

Important Code Reference: script.js

A focused code/control-file explanation covering purpose, owner layer, APIs, variables, implementation signals, source excerpts, and debugging rules.

Item	Explanation
File	script.js
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	158,556 bytes
Extension	.js
MIME type	application/javascript
Text lines	3,710

Why this file exists

- Public website JavaScript for navigation, portfolio rendering, search, AI UI, and interactive project windows.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

API endpoints mentioned or called

Item	Explanation
/api/portfolio-ai	Line 1492. Handles AI assistant questions.
/api/portfolio-ai	Line 1493. Handles AI assistant questions.

Important implementation signals

Item	Explanation
Browser DOM at line 1	Builder or website interface behavior in the browser/Electron renderer. Evidence: const grid = document.querySelector("#project-grid");
Browser DOM at line 2	Builder or website interface behavior in the browser/Electron renderer. Evidence: const searchInput = document.querySelector("#project-search");
Browser DOM at line 3	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectFilters = document.querySelector("#project-filters");
Browser DOM at line 4	Builder or website interface behavior in the browser/Electron renderer. Evidence: let filterButtons = [...document.querySelectorAll(".filter-button")];
Browser DOM at line 5	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCount = document.querySelector("#project-count");
Browser DOM at line 6	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackCount = document.querySelector("#project-track-count");
Browser DOM at line 7	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackLabels = document.querySelector("#project-track-labels");
Browser DOM at line 8	Builder or website interface behavior in the browser/Electron renderer. Evidence: const year = document.querySelector("#year");

Item	Explanation
Browser DOM at line 9	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const funFactsCallout = document.querySelector("#fun-facts-callout");</code>
Browser DOM at line 10	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroEyebrow = document.querySelector(".hero-content .eyebrow");</code>
Browser DOM at line 11	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroTitle = document.querySelector("#hero-title");</code>
Browser DOM at line 12	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroCopy = document.querySelector(".hero-copy");</code>
Browser DOM at line 13	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandName = document.querySelector(".brand strong");</code>
Browser DOM at line 14	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandSubtitle = document.querySelector(".brand small");</code>
Browser DOM at line 15	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandIcon = document.querySelector(".brand-icon");</code>
Browser DOM at line 16	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandText = document.querySelector(".omb-engraving");</code>
Browser DOM at line 17	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatar = document.querySelector(".header-avatar");</code>
Browser DOM at line 18	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatarImage = document.querySelector(".header-avatar img");</code>
Browser DOM at line 19	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroImage = document.querySelector(".hero-image");</code>
Browser DOM at line 20	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const resumeSection = document.querySelector("#resume");</code>
Browser DOM at line 21	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactBand = document.querySelector("#contact");</code>
Browser DOM at line 22	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePhoto = document.querySelector(".profile-photo");</code>
Browser DOM at line 23	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactTitle = document.querySelector("#contact-title");</code>
Browser DOM at line 24	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactIntro = document.querySelector(".contact-intro");</code>
Browser DOM at line 25	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactDetails = document.querySelector(".contact-details");</code>
Browser DOM at line 26	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactLinks = document.querySelector(".contact-links");</code>
Browser DOM at line 27	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const footerOwner = document.querySelector(".site-footer span");</code>
Browser DOM at line 29	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantForm = document.querySelector("#ai-assistant-form");</code>
Browser DOM at line 30	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantPanel = document.querySelector("#ask-ai");</code>
Browser DOM at line 31	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantInput = document.querySelector("#ai-assistant-input");</code>
Browser DOM at line 32	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantLog = document.querySelector("#ai-assistant-log");</code>
Browser DOM at line 33	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantStatus = document.querySelector("#ai-assistant-status");</code>
Cloudflare or AI at line 34	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const aiClearChatButton = document.querySelector("#ai-clear-chat");</code>
Cloudflare or AI at line 57	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>let assistantChatHistory = [];</code>
Compiler or simulator at line 73	Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "verilog", label: "Verilog", aliases: ["verilog", "v"] },</code>

Item	Explanation
Compiler or simulator at line 74	Part of the Compile Code workspace or language/tool detection. Evidence: { id: "systemverilog", label: "SystemVerilog", aliases: ["systemverilog", "system verilog", "sv"] },
Compiler or simulator at line 76	Part of the Compile Code workspace or language/tool detection. Evidence: { id: "java", label: "Java", aliases: ["java"] },
Compiler or simulator at line 78	Part of the Compile Code workspace or language/tool detection. Evidence: { id: "python", label: "Python", aliases: ["python", "py"] },
Compiler or simulator at line 104	Part of the Compile Code workspace or language/tool detection. Evidence: "embedded-wireless-node": "appearance-deep-navy-cyan-click",
Browser DOM at line 175	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-draggable-dialog");
Browser DOM at line 184	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (dialog.querySelector("[data-section-dialog-resize-handle]")) return;
Browser DOM at line 186	Builder or website interface behavior in the browser/Electron renderer. Evidence: const handle = document.createElement("div");
Browser DOM at line 188	Builder or website interface behavior in the browser/Electron renderer. Evidence: handle.dataset.sectionDialogResizeHandle = direction;
Browser DOM at line 197	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-resized-dialog");
Browser DOM at line 255	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-dragging-dialog");
Browser DOM at line 273	Builder or website interface behavior in the browser/Electron renderer. Evidence: activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");
Browser DOM at line 281	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (!handle !dialog?.open dialog.classList.contains("is-minimized-dialog")) return;
Browser DOM at line 285	Builder or website interface behavior in the browser/Electron renderer. Evidence: direction: handle.dataset.sectionDialogResizeHandle,
Browser DOM at line 293	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-resizing-dialog");
Browser DOM at line 309	Builder or website interface behavior in the browser/Electron renderer. Evidence: activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");
Browser DOM at line 314	Builder or website interface behavior in the browser/Electron renderer. Evidence: const minimized = dialog.classList.contains("is-minimized-dialog");
Browser DOM at line 315	Builder or website interface behavior in the browser/Electron renderer. Evidence: const button = dialog.querySelector(".section-view-minimize");
Browser DOM at line 324	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.toggle("is-minimized-dialog");
Browser DOM at line 331	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerdown", beginSectionDialogResize);
Browser DOM at line 332	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerdown", beginSectionDialogDrag);
Browser DOM at line 333	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointermove", moveSectionDialogResize);
Browser DOM at line 334	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointermove", moveSectionDialogDrag);
Browser DOM at line 335	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerup", endSectionDialogResize);
Browser DOM at line 336	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerup", endSectionDialogDrag);
Browser DOM at line 337	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointercancel", endSectionDialogResize);
Browser DOM at line 338	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointercancel", endSectionDialogDrag);
Browser DOM at line 339	Builder or website interface behavior in the browser/Electron renderer. Evidence: window.addEventListener("resize", () => {

Item	Explanation
Browser DOM at line 340	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) => {</code>
Rich text at line 370	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>function normalizeCodePasteMode(value = "") {</code>
Rich text at line 374	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>function normalizeCodeText(value = "", pasteMode = "source") {</code>
Rich text at line 376	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/t/g, " ").replace(/s+\$/g, "");</code>
Compiler or simulator at line 400	Part of the Compile Code workspace or language/tool detection. Evidence: <code>if (/^(\b(always_ff always_comb always_latch logic interface covergroup assert s+property typedef s+enum class s+ w+)\b/.test(code)) return "systemverilog";</code>
Compiler or simulator at line 401	Part of the Compile Code workspace or language/tool detection. Evidence: <code>if (/^(\b(module endmodule always assign reg wire initial posedge negedge)\b/.test(code)) return "verilog";</code>
Cloudflare or AI at line 402	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>if (/^s*\.(?!(tran ac dc op model subckt ends param)\b/im.test(code) \bV w+ s+ w+ s+ w+ s+(?:DC SIN PULSE)?/i.test(code)) return "Itspice";</code>
Compiler or simulator at line 403	Part of the Compile Code workspace or language/tool detection. Evidence: <code>if (/^(\b(def elif import s+ w+ from s+ w+ s+import self None True False)\b/.test(code)) return "python";</code>
Compiler or simulator at line 404	Part of the Compile Code workspace or language/tool detection. Evidence: <code>if (/^(\b(public s+class private s+ protected s+ static s+void s+main System\.out)\b/.test(code)) return "java";</code>
Security or auth at line 412	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>function tokenizedCodeHtml(code = "", language = "javascript") {</code>
Security or auth at line 414	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>const tokens = [];</code>
Security or auth at line 417	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>const token = `@@CODE_TOKEN_\${tokens.length}@`;</code>
Security or auth at line 418	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>tokens.push(`<span class="\${className}">\${match}</span>`);</code>
Security or auth at line 419	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>return token;</code>
Security or auth at line 424	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>protect(/&lt;!-[s\S]*?--&gt;/g, "code-token-comment");</code>
Security or auth at line 425	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>protect(/&lt;\/?)([a-zA-Z0-9:-]+)([s\S]*?)(\/?&gt;)/g, "code-token-tag");</code>
Compiler or simulator at line 427	Part of the Compile Code workspace or language/tool detection. Evidence: <code>const commentPattern = ["python", "Itspice"].includes(normalizeCodeLanguage(language))</code>
Security or auth at line 430	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>protect(commentPattern, "code-token-comment");</code>

Important constants and variables

Item	Explanation
grid	Line 1: <code>const grid = document.querySelector("#project-grid");</code>
searchInput	Line 2: <code>const searchInput = document.querySelector("#project-search");</code>
projectFilters	Line 3: <code>const projectFilters = document.querySelector("#project-filters");</code>
filterButtons	Line 4: <code>let filterButtons = [...document.querySelectorAll(".filter-button")];</code>
projectCount	Line 5: <code>const projectCount = document.querySelector("#project-count");</code>
projectTrackCount	Line 6: <code>const projectTrackCount = document.querySelector("#project-track-count");</code>
projectTrackLabels	Line 7: <code>const projectTrackLabels = document.querySelector("#project-track-labels");</code>
year	Line 8: <code>const year = document.querySelector("#year");</code>
funFactsCallout	Line 9: <code>const funFactsCallout = document.querySelector("#fun-facts-callout");</code>
heroEyebrow	Line 10: <code>const heroEyebrow = document.querySelector(".hero-content .eyebrow");</code>
heroTitle	Line 11: <code>const heroTitle = document.querySelector("#hero-title");</code>

Item	Explanation
heroCopy	Line 12: const heroCopy = document.querySelector(".hero-copy");
brandName	Line 13: const brandName = document.querySelector(".brand strong");
brandSubtitle	Line 14: const brandSubtitle = document.querySelector(".brand small");
brandIcon	Line 15: const brandIcon = document.querySelector(".brand-icon");
brandText	Line 16: const brandText = document.querySelector(".omb-engraving");
headerAvatar	Line 17: const headerAvatar = document.querySelector(".header-avatar");
headerAvatarImage	Line 18: const headerAvatarImage = document.querySelector(".header-avatar img");
heroImage	Line 19: const heroImage = document.querySelector(".hero-image");
resumeSection	Line 20: const resumeSection = document.querySelector("#resume");
contactBand	Line 21: const contactBand = document.querySelector("#contact");
profilePhoto	Line 22: const profilePhoto = document.querySelector(".profile-photo");
contactTitle	Line 23: const contactTitle = document.querySelector("#contact-title");
contactIntro	Line 24: const contactIntro = document.querySelector(".contact-intro");
contactDetails	Line 25: const contactDetails = document.querySelector(".contact-details");
contactLinks	Line 26: const contactLinks = document.querySelector(".contact-links");
footerOwner	Line 27: const footerOwner = document.querySelector(".site-footer span");
searchWrap	Line 28: const searchWrap = searchInput?.closest(".search-wrap");
aiAssistantForm	Line 29: const aiAssistantForm = document.querySelector("#ai-assistant-form");
aiAssistantPanel	Line 30: const aiAssistantPanel = document.querySelector("#ask-ai");
aiAssistantInput	Line 31: const aiAssistantInput = document.querySelector("#ai-assistant-input");
aiAssistantLog	Line 32: const aiAssistantLog = document.querySelector("#ai-assistant-log");
aiAssistantStatus	Line 33: const aiAssistantStatus = document.querySelector("#ai-assistant-status");
aiClearChatButton	Line 34: const aiClearChatButton = document.querySelector("#ai-clear-chat");
categories	Line 36: let categories = [];
projects	Line 37: let projects = [];
siteSections	Line 38: let siteSections = [];
funFacts	Line 39: let funFacts = [];
funFactsRich	Line 40: let funFactsRich = null;
fieldStyles	Line 41: let fieldStyles = {};
siteContent	Line 42: let siteContent = null;
siteContentRich	Line 43: let siteContentRich = {};
profile	Line 44: let profile = null;
profileRich	Line 45: let profileRich = {};
activeFilter	Line 46: let activeFilter = "all";
activeSectionDialogDrag	Line 47: let activeSectionDialogDrag = null;
activeSectionDialogResize	Line 48: let activeSectionDialogResize = null;
sectionDialogDragEnabled	Line 49: let sectionDialogDragEnabled = false;
isApplyingSectionRoute	Line 50: let isApplyingSectionRoute = false;
searchPanel	Line 51: let searchPanel = null;
searchStatus	Line 52: let searchStatus = null;
searchLimitSelect	Line 53: let searchLimitSelect = null;

Item	Explanation
currentSearchResults	Line 54: let currentSearchResults = [];
assistantSourceCounter	Line 55: let assistantSourceCounter = 0;
assistantSourceMap	Line 56: let assistantSourceMap = new Map();
assistantChatHistory	Line 57: let assistantChatHistory = [];
searchableEntries	Line 58: let searchableEntries = [];
searchResultLimit	Line 59: let searchResultLimit = 10;
searchHighlightTimer	Line 60: let searchHighlightTimer = 0;
searchHighlightName	Line 61: const searchHighlightName = "portfolio-search-highlight";
sectionRouteKey	Line 63: const sectionRouteKey = "portfolio-section";
sectionRouteParams	Line 64: const sectionRouteParams = {
supportedCodeLanguages	Line 70: const supportedCodeLanguages = [
legacyTemplateSkins	Line 82: const legacyTemplateSkins = {
defaultSiteContent	Line 108: const defaultSiteContent = {
defaultProfile	Line 114: const defaultProfile = {
items	Line 137: const items = Array.isArray(value) ? value : String(value "").split(/\r?\n/);
facts	Line 146: const facts = normalizeFunFacts(funFacts);
hasRichFacts	Line 147: const hasRichFacts = Boolean(funFactsRich?.blocks?.length);
rect	Line 162: const rect = dialog.getBoundingClientRect();
margin	Line 163: const margin = 12;
maxLeft	Line 164: const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
maxTop	Line 165: const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
rect	Line 173: const rect = dialog.getBoundingClientRect();
position	Line 174: const position = clampSectionDialogPosition(rect.left, rect.top, dialog);
handle	Line 186: const handle = document.createElement("div");
rect	Line 196: const rect = dialog.getBoundingClientRect();
margin	Line 204: const margin = 12;
minWidth	Line 205: const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
minHeight	Line 206: const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));

JSON/YAML-style keys detected

Item	Explanation
skin-light-blue	Line 83: "skin-light-blue": "appearance-light-blue-red-click",
skin-clean-white	Line 84: "skin-clean-white": "appearance-white-blue-click",
skin-deep-navy	Line 85: "skin-deep-navy": "appearance-deep-navy-cyan-click",
skin-red-warm	Line 86: "skin-red-warm": "appearance-warm-red-pale-click",
skin-emerald-instrument	Line 87: "skin-emerald-instrument": "appearance-emerald-mint-click",
skin-amber-power	Line 88: "skin-amber-power": "appearance-amber-cream-click",
skin-violet-mixed	Line 89: "skin-violet-mixed": "appearance-violet-lilac-click",
skin-graphite-asic	Line 90: "skin-graphite-asic": "appearance-graphite-white-click",
analog-opamp-topology	Line 91: "analog-opamp-topology": "appearance-light-blue-red-click",
analog-power-charger	Line 92: "analog-power-charger": "appearance-amber-cream-click",

Item	Explanation
analog-filter-front-end	Line 93: "analog-filter-front-end": "appearance-light-blue-red-click",
analog-sensor-interface	Line 94: "analog-sensor-interface": "appearance-emerald-mint-click",
analog-mixed-signal-timing	Line 95: "analog-mixed-signal-timing": "appearance-violet-lilac-click",
digital-fpga-pipeline	Line 96: "digital-fpga-pipeline": "appearance-deep-navy-cyan-click",
digital-asic-block	Line 97: "digital-asic-block": "appearance-graphite-white-click",
digital-verification-suite	Line 98: "digital-verification-suite": "appearance-graphite-white-click",
digital-interface-controller	Line 99: "digital-interface-controller": "appearance-deep-navy-cyan-click",
digital-signal-processing	Line 100: "digital-signal-processing": "appearance-violet-lilac-click",
embedded-stm32-sensor	Line 101: "embedded-stm32-sensor": "appearance-emerald-mint-click",
embedded-rtos-control	Line 102: "embedded-rtos-control": "appearance-graphite-white-click",
embedded-power-monitor	Line 103: "embedded-power-monitor": "appearance-amber-cream-click",
embedded-wireless-node	Line 104: "embedded-wireless-node": "appearance-deep-navy-cyan-click",
embedded-motor-control	Line 105: "embedded-motor-control": "appearance-violet-lilac-click"
Brief	Line 792: "Brief": "Overview",
Project Brief	Line 793: "Project Brief": "Overview",
Design Brief	Line 794: "Design Brief": "Design Overview",
Simulation Brief	Line 795: "Simulation Brief": "Simulation Overview"
--responsive-section-card-min	Line 1029: "--responsive-section-card-min": profile.sectionCardMin,
--responsive-content-max	Line 1030: "--responsive-content-max": profile.contentMax,
--responsive-media-max	Line 1031: "--responsive-media-max": profile.mediaMax,
--responsive-touch-target	Line 1032: "--responsive-touch-target": profile.touchTarget
--accent	Line 1047: "--accent": accent,
--template-bg	Line 1048: "--template-bg": visual.background,
--template-panel	Line 1049: "--template-panel": visual.panel,
--template-accent	Line 1050: "--template-accent": visual.accent,
--template-hover	Line 1051: "--template-hover": visual.hover,
--template-click	Line 1052: "--template-click": visual.click,
--template-click-text	Line 1053: "--template-click-text": visual.clickText,
--template-line	Line 1054: "--template-line": visual.line,
--template-text	Line 1055: "--template-text": visual.text,
--accent	Line 1080: "--accent": accent,
--template-bg	Line 1081: "--template-bg": visual.background,
--template-panel	Line 1082: "--template-panel": visual.panel,
--template-accent	Line 1083: "--template-accent": visual.accent,
--template-hover	Line 1084: "--template-hover": visual.hover,
--template-click	Line 1085: "--template-click": visual.click,
--template-click-text	Line 1086: "--template-click-text": visual.clickText,
--template-line	Line 1087: "--template-line": visual.line,
--template-text	Line 1088: "--template-text": visual.text,

Event handlers and user interactions

Item	Explanation
Line 331	<code>document.addEventListener("pointerdown", beginSectionDialogResize);</code>
Line 332	<code>document.addEventListener("pointerdown", beginSectionDialogDrag);</code>
Line 333	<code>document.addEventListener("pointermove", moveSectionDialogResize);</code>
Line 334	<code>document.addEventListener("pointermove", moveSectionDialogDrag);</code>
Line 335	<code>document.addEventListener("pointerup", endSectionDialogResize);</code>
Line 336	<code>document.addEventListener("pointerup", endSectionDialogDrag);</code>
Line 337	<code>document.addEventListener("pointercancel", endSectionDialogResize);</code>
Line 338	<code>document.addEventListener("pointercancel", endSectionDialogDrag);</code>
Line 339	<code>window.addEventListener("resize", () => {</code>
Line 1214	<code>button.addEventListener("click", () => {</code>
Line 3272	<code>searchLimitSelect.addEventListener("change", () => {</code>
Line 3276	<code>searchPanel.addEventListener("mousedown", (event) => event.preventDefault());</code>
Line 3277	<code>searchPanel.addEventListener("click", (event) => {</code>
Line 3434	<code>dialog.querySelector(".section-view-close").addEventListener("click", () => closeSectionDialog(dialog));</code>
Line 3435	<code>dialog.addEventListener("cancel", () => clearSectionRouteInHistory());</code>
Line 3436	<code>dialog.addEventListener("close", () => {</code>
Line 3445	<code>dialog.querySelector(".section-view-back").addEventListener("click", () => {</code>
Line 3457	<code>dialog.querySelector(".section-view-forward").addEventListener("click", () => {</code>
Line 3464	<code>dialog.querySelector("#section-view-content").addEventListener("click", (event) => {</code>
Line 3562	<code>window.addEventListener("popstate", (event) => {</code>
Line 3567	<code>searchInput.addEventListener("input", handleSearchInput);</code>
Line 3568	<code>searchInput.addEventListener("focus", showSearchPanelIfNeeded);</code>
Line 3569	<code>searchInput.addEventListener("keydown", (event) => {</code>
Line 3580	<code>document.addEventListener("click", (event) => {</code>
Line 3586	<code>document.addEventListener("click", (event) => {</code>
Line 3599	<code>aiAssistantForm?.addEventListener("submit", (event) => {</code>
Line 3606	<code>aiClearChatButton?.addEventListener("click", clearAssistantChat);</code>
Line 3607	<code>window.addEventListener("resize", updateAssistantPanelGrowth);</code>
Line 3609	<code>aiAssistantLog?.addEventListener("click", (event) => {</code>
Line 3616	<code>grid.addEventListener("click", (event) => {</code>

Function-by-function detail

normalize - lines 132-134

Item	Explanation
Source location	Lines 132-134 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	<code>function normalize(value) {</code>
Touches	<code>filesystem</code>
Calls detected	<code>toLowerCase</code>
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

132: function normalize(value) {
133:   return String(value || "").toLowerCase();
134: }

```

normalizeFunFacts - lines 136-142

Item	Explanation
Source location	Lines 136-142 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFunFacts(value) {
Touches	filesystem
Calls detected	isArray, split, map, replace, trim, filter, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

136: function normalizeFunFacts(value) {
137:   const items = Array.isArray(value) ? value : String(value || "").split(/\r?\n/);
138:   return items
139:     .map((item) => String(item || "").replace(/\s+/g, " ").trim())
140:     .filter(Boolean)
141:     .slice(0, 10);
142: }

```

renderFunFacts - lines 144-159

Item	Explanation
Source location	Lines 144-159 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderFunFacts() {
Touches	filesystem
Calls detected	normalizeFunFacts, renderRichContent, slice, join, map, renderInlineMath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

144: function renderFunFacts() {
145:   if (!funFactsCallout) return;
146:   const facts = normalizeFunFacts(funFacts);
147:   const hasRichFacts = Boolean(funFactsRich?.blocks?.length);
148:   funFactsCallout.hidden = !facts.length && !hasRichFacts;
149:   funFactsCallout.innerHTML = facts.length || hasRichFacts ? `
150:     <div class="fun-facts-window">
151:       <span class="fun-facts-label">Fun fact:</span>
152:       <div class="fun-facts-lines">

```

```

153:         ${hasRichFacts
154:           ? renderRichContent({ blocks: funFactsRich.blocks.slice(0, 3) }, facts.join("\n"))
155:           : facts.map((fact) => `
```

clampSectionDialogPosition - lines 161-170

Item	Explanation
Source location	Lines 161-170 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function clampSectionDialogPosition(left, top, dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	getBoundingClientRect, max, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

161: function clampSectionDialogPosition(left, top, dialog) {
162:   const rect = dialog.getBoundingClientRect();
163:   const margin = 12;
164:   const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
165:   const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
166:   return {
167:     left: Math.min(Math.max(left, margin), maxLeft),
168:     top: Math.min(Math.max(top, margin), maxTop)
169:   };
170: }

```

anchorSectionDialog - lines 172-181

Item	Explanation
Source location	Lines 172-181 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function anchorSectionDialog(dialog) {
Touches	browser/Electron DOM
Calls detected	getBoundingClientRect, clampSectionDialogPosition, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

172: function anchorSectionDialog(dialog) {
173:   const rect = dialog.getBoundingClientRect();
174:   const position = clampSectionDialogPosition(rect.left, rect.top, dialog);
175:   dialog.classList.add("is-draggable-dialog");
176:   dialog.style.left = `${position.left}px`;
177:   dialog.style.top = `${position.top}px`;
178:   dialog.style.right = "auto";
179:   dialog.style.bottom = "auto";
180:   dialog.style.margin = "0";
181: }

```

ensureSectionResizeHandles - lines 183-192

Item	Explanation
Source location	Lines 183-192 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function ensureSectionResizeHandles(dialog) {
Touches	browser/Electron DOM
Calls detected	querySelector, forEach, createElement, setAttribute, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

183: function ensureSectionResizeHandles(dialog) {
184:   if (dialog.querySelector("[data-section-dialog-resize-handle]")) return;
185:   ["n", "e", "s", "w", "ne", "nw", "se", "sw"].forEach((direction) => {
186:     const handle = document.createElement("div");
187:     handle.className = `dialog-resize-handle resize-${direction}`;
188:     handle.dataset.sectionDialogResizeHandle = direction;
189:     handle.setAttribute("aria-hidden", "true");
190:     dialog.append(handle);
191:   });
192: }

```

anchorSectionDialogForResize - lines 194-201

Item	Explanation
Source location	Lines 194-201 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function anchorSectionDialogForResize(dialog) {
Touches	browser/Electron DOM
Calls detected	anchorSectionDialog, getBoundingClientRect, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

194: function anchorSectionDialogForResize(dialog) {
195:   anchorSectionDialog(dialog);
196:   const rect = dialog.getBoundingClientRect();
197:   dialog.classList.add("is-resized-dialog");
198:   dialog.style.width = `${rect.width}px`;
199:   dialog.style.height = `${rect.height}px`;
200:   dialog.style.maxHeight = "none";
201: }

```

sectionDialogResizeBounds - lines 203-230

Item	Explanation
Source location	Lines 203-230 (28 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionDialogResizeBounds(state, event) {
Touches	Mostly local calculations or local UI state.
Calls detected	min, max, includes

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

203: function sectionDialogResizeBounds(state, event) {
204:   const margin = 12;
205:   const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
206:   const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));
207:   const direction = state.direction;
208:   let { left, top, width, height } = state.rect;
209:   const dx = event.clientX - state.startX;
210:   const dy = event.clientY - state.startY;
211:
212:   if (direction.includes("e")) {
213:     width = Math.min(Math.max(state.rect.width + dx, minWidth), window.innerWidth - left - margin);
214:   }
215:   if (direction.includes("s")) {
216:     height = Math.min(Math.max(state.rect.height + dy, minHeight), window.innerHeight - top - margin);
217:   }
218:   if (direction.includes("w")) {
219:     const right = state.rect.left + state.rect.width;
220:     left = Math.min(Math.max(state.rect.left + dx, margin), right - minWidth);
221:     width = right - left;
222:   }
223:   if (direction.includes("n")) {
224:     const bottom = state.rect.top + state.rect.height;
225:     top = Math.min(Math.max(state.rect.top + dy, margin), bottom - minHeight);
226:     height = bottom - top;
227:   }
228:
229:   return { height, left, top, width };
230: }

```

sectionDialogFromDragEvent - lines 232-239

Item	Explanation
Source location	Lines 232-239 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionDialogFromDragEvent(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

232: function sectionDialogFromDragEvent(event) {
233:   if (event.target.closest("[data-section-dialog-resize-handle]")) return null;
234:   const handle = event.target.closest("[data-section-dialog-drag='true']");
235:   if (!handle) return null;
236:   if (event.target.closest("button, a, input, textarea, select, label, summary")) return null;
237:   const dialog = handle.closest("dialog");
238:   return dialog?.open ? dialog : null;
239: }

```

beginSectionDialogDrag - lines 241-257

Item	Explanation
Source location	Lines 241-257 (17 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function beginSectionDialogDrag(event) {
Touches	browser/Electron DOM
Calls detected	sectionDialogFromDragEvent, anchorSectionDialog, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

241: function beginSectionDialogDrag(event) {
242:   if (event.button !== 0) return;
243:   const dialog = sectionDialogFromDragEvent(event);
244:   if (!dialog) return;
245:   anchorSectionDialog(dialog);
246:   const rect = dialog.getBoundingClientRect();
247:   activeSectionDialogDrag = {
248:     dialog,
249:     offsetX: event.clientX - rect.left,
250:     offsetY: event.clientY - rect.top,
251:     pointerId: event.pointerId,
252:     pointerTarget: event.target
253:   };
254:   event.target.setPointerCapture?.(event.pointerId);
255:   dialog.classList.add("is-dragging-dialog");
256:   event.preventDefault();
257: }

```

moveSectionDialogDrag - lines 259-268

Item	Explanation
Source location	Lines 259-268 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function moveSectionDialogDrag(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	clampSectionDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

259: function moveSectionDialogDrag(event) {
260:   if (!activeSectionDialogDrag) return;
261:   const next = clampSectionDialogPosition(
262:     event.clientX - activeSectionDialogDrag.offsetX,
263:     event.clientY - activeSectionDialogDrag.offsetY,
264:     activeSectionDialogDrag.dialog
265:   );
266:   activeSectionDialogDrag.dialog.style.left = `${next.left}px`;
267:   activeSectionDialogDrag.dialog.style.top = `${next.top}px`;
268: }

```

endSectionDialogDrag - lines 270-275

Item	Explanation
Source location	Lines 270-275 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function endSectionDialogDrag() {

Item	Explanation
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

270: function endSectionDialogDrag() {
271:   if (!activeSectionDialogDrag) return;
272:   activeSectionDialogDrag.pointerTarget?.releasePointerCapture?.(activeSectionDialogDrag.pointerId);
273:   activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");
274:   activeSectionDialogDrag = null;
275: }

```

beginSectionDialogResize - lines 277-295

Item	Explanation
Source location	Lines 277-295 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function beginSectionDialogResize(event) {
Touches	browser/Electron DOM
Calls detected	closest, contains, anchorSectionDialogForResize, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

277: function beginSectionDialogResize(event) {
278:   if (event.button !== 0) return;
279:   const handle = event.target.closest("[data-section-dialog-resize-handle]");
280:   const dialog = handle?.closest("dialog");
281:   if (!handle || !dialog?.open || dialog.classList.contains("is-minimized-dialog")) return;
282:   anchorSectionDialogForResize(dialog);
283:   activeSectionDialogResize = {
284:     dialog,
285:     direction: handle.dataset.sectionDialogResizeHandle,
286:     pointerId: event.pointerId,
287:     pointerTarget: event.target,
288:     rect: dialog.getBoundingClientRect(),
289:     startX: event.clientX,
290:     startY: event.clientY
291:   };
292:   event.target.setPointerCapture?.(event.pointerId);
293:   dialog.classList.add("is-resizing-dialog");
294:   event.preventDefault();
295: }

```

moveSectionDialogResize - lines 297-304

Item	Explanation
Source location	Lines 297-304 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function moveSectionDialogResize(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionDialogResizeBounds

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

297: function moveSectionDialogResize(event) {
298:   if (!activeSectionDialogResize) return;
299:   const next = sectionDialogResizeBounds(activeSectionDialogResize, event);
300:   activeSectionDialogResize.dialog.style.left = `${next.left}px`;
301:   activeSectionDialogResize.dialog.style.top = `${next.top}px`;
302:   activeSectionDialogResize.dialog.style.width = `${next.width}px`;
303:   activeSectionDialogResize.dialog.style.height = `${next.height}px`;
304: }

```

endSectionDialogResize - lines 306-311

Item	Explanation
Source location	Lines 306-311 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function endSectionDialogResize() {
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

306: function endSectionDialogResize() {
307:   if (!activeSectionDialogResize) return;
308:   activeSectionDialogResize.pointerTarget?.releasePointerCapture?.(activeSectionDialogResize.pointerId);
309:   activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");
310:   activeSectionDialogResize = null;
311: }

```

updateSectionDialogMinimize - lines 313-320

Item	Explanation
Source location	Lines 313-320 (8 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSectionDialogMinimize(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, querySelector, setAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

313: function updateSectionDialogMinimize(dialog) {
314:   const minimized = dialog.classList.contains("is-minimized-dialog");
315:   const button = dialog.querySelector(".section-view-minimize");

```

```

316:   if (!button) return;
317:   button.textContent = minimized ? "+" : "-";
318:   button.title = minimized ? "Restore window" : "Minimize window";
319:   button.setAttribute("aria-label", minimized ? "Restore window" : "Minimize window");
320: }

```

toggleSectionDialogMinimized - lines 322-326

Item	Explanation
Source location	Lines 322-326 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function toggleSectionDialogMinimized(dialog) {
Touches	browser/Electron DOM
Calls detected	anchorSectionDialog, toggle, updateSectionDialogMinimize
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

322: function toggleSectionDialogMinimized(dialog) {
323:   anchorSectionDialog(dialog);
324:   dialog.classList.toggle("is-minimized-dialog");
325:   updateSectionDialogMinimize(dialog);
326: }

```

enableSectionDialogDrag - lines 328-347

Item	Explanation
Source location	Lines 328-347 (20 source lines).
Purpose	Part of the public website runtime.
Declaration	function enableSectionDialogDrag() {
Touches	browser/Electron DOM, filesystem
Calls detected	addEventListener, querySelectorAll, forEach, getBoundingClientRect, clampSectionDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

328: function enableSectionDialogDrag() {
329:   if (sectionDialogDragEnabled) return;
330:   sectionDialogDragEnabled = true;
331:   document.addEventListener("pointerdown", beginSectionDialogResize);
332:   document.addEventListener("pointerdown", beginSectionDialogDrag);
333:   document.addEventListener("pointermove", moveSectionDialogResize);
334:   document.addEventListener("pointermove", moveSectionDialogDrag);
335:   document.addEventListener("pointerup", endSectionDialogResize);
336:   document.addEventListener("pointerup", endSectionDialogDrag);
337:   document.addEventListener("pointercancel", endSectionDialogResize);
338:   document.addEventListener("pointercancel", endSectionDialogDrag);
339:   window.addEventListener("resize", () => {
340:     document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) => {
341:       const rect = dialog.getBoundingClientRect();
342:       const next = clampSectionDialogPosition(rect.left, rect.top, dialog);
343:       dialog.style.left = `${next.left}px`;
344:       dialog.style.top = `${next.top}px`;
345:     });
346:   });
347: }

```


escapeHtml - lines 349-356

Item	Explanation
Source location	Lines 349-356 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function escapeHtml(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
349: function escapeHtml(value) {
350:   return String(value || "")
351:     .replaceAll("&", "&amp;")
352:     .replaceAll("<", "&lt;")
353:     .replaceAll(">", "&gt;")
354:     .replaceAll("'", "&quot;")
355:     .replaceAll('"', "&#039;");
356: }
```

normalizeCodeLanguage - lines 358-364

Item	Explanation
Source location	Lines 358-364 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeLanguage(value = "") {
Touches	filesystem
Calls detected	trim, toLowerCase, replace, find, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
358: function normalizeCodeLanguage(value = "") {
359:   const clean = String(value || "").trim().toLowerCase().replace(/[_-]+/g, " ");
360:   const match = supportedCodeLanguages.find((language) =>
361:     language.id === clean || language.aliases.includes(clean)
362:   );
363:   return match?.id || "javascript";
364: }
```

codeLanguageLabel - lines 366-368

Item	Explanation
Source location	Lines 366-368 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function codeLanguageLabel(value = "") {
Touches	filesystem
Calls detected	find, normalizeCodeLanguage

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

366: function codeLanguageLabel(value = "") {
367:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value)?.label
|| "Code";
368: }

```

normalizeCodePasteMode - lines 370-372

Item	Explanation
Source location	Lines 370-372 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodePasteMode(value = "") {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

370: function normalizeCodePasteMode(value = "") {
371:   return value === "source" ? "source" : "basic";
372: }

```

normalizeCodeText - lines 374-383

Item	Explanation
Source location	Lines 374-383 (10 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeText(value = "", pasteMode = "source") {
Touches	filesystem
Calls detected	replace, normalizeCodePasteMode, split, map, trimEnd, join, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

374: function normalizeCodeText(value = "", pasteMode = "source") {
375:   const normalized = String(value || "").replace(/\r\n?/g, "\n").replace(/\u00a0/g, " ");
376:   if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/\t/g, "
").replace(/\s+$/g, "");
377:   return normalized
378:     .split("\n")
379:     .map((line) => line.trimEnd())
380:     .join("\n")
381:     .replace(/\n{4,}/g, "\n\n\n")
382:     .trim();

```

```
383: }
```

sourceLooksCpp - lines 385-389

Item	Explanation
Source location	Lines 385-389 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sourceLooksCpp(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
385: function sourceLooksCpp(source = "") {
386:   const text = String(source || "");
387:   return
  /#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
  ||
  388:    /\b(?:std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w+|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
389: }
```

sourceLooksC - lines 391-395

Item	Explanation
Source location	Lines 391-395 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sourceLooksC(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
391: function sourceLooksC(source = "") {
392:   const text = String(source || "");
393:   return /#include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
394:    /\b(?:printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
395: }
```

detectCodeLanguage - lines 397-410

Item	Explanation
Source location	Lines 397-410 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function detectCodeLanguage(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b, sourceLooksCpp, sourceLooksC


```

440:      c: "_Alignas|_Alignof|_Atomic|_Bool|_Complex|_Generic|_Imaginary|_Noreturn|_Static_assert|_Thread_loc
al|auto|break|case|char|const|continue|default|do|double|else|enum|extern|float|for|goto|if|inline|int|long|regi
ster|restrict|return|short|signed|sizeof|static|struct|switch|typedef|union|unsigned|void|volatile|while",
441:      cpp: "alignas|alignof|and|and_eq|asm|auto|bitand|bitor|bool|break|case|catch|char|char8_t|char16_t|ch
ar32_t|class|compl|concept|const|consteval|constexpr|constinit|const_cast|continue|co_await|co_return|co_yield|d
ecltype|default|delete|do|double|dynamic_cast|else|enum|explicit|export|extern|false|float|for|friend|if|inline|
int|long|mutable|namespace|new|noexcept|not|not_eq|nullptr|operator|or|or_eq|private|protected|public|register|r
einterpret_cast|requires|return|short|signed|sizeof|static|static_assert|static_cast|struct|switch|template|this
|thread_local|throw|true|try|typedef|typeid|typename|union|unsigned|using|virtual|void|volatile|wchar_t|while|xor
|xor_eq",
442:      verilog: "always|and|assign|begin|buf|case|casex|casez|deassign|default|defparam|disable|edge|else|en
d|endcase|endfunction|endmodule|endprimitive|endspecify|endtable|endtask|event|for|force|forever|fork|function|g
enerate|genvar|if|initial|inout|input|integer|join|module|nand|negedge|nor|not|or|output|parameter|posedge|primi
tive|reg|release|repeat|signed|specify|supply0|supply1|table|task|tri|wand|while|wire|wor|xnor|xor",
443:      systemverilog: "always|always_comb|always_ff|always_latch|assign|automatic|begin|bit|case|class|clock
ing|covergroup|default|disable|do|else|end|endcase|endclass|endclocking|endfunction|endmodule|endpackage|endtask
|enum|for|forever|function|generate|genvar|if|initial|input|int|interface|logic|module|negedge|output|package|pa
rameter|posedge|reg|return|signed|task|typedef|wire",
444:      ltspice: "ac|dc|end|ends|four|func|global|ic|include|lib|meas|model|nodeset|op|options|param|plot|pro
be|save|step|subckt|temp|tran",
445:      java: "abstract|assert|boolean|break|byte|case|catch|char|class|const|continue|default|do|double|else
|enum|extends|final|finally|float|for|if|implements|import|instanceof|int|interface|long|native|new|null|package
|private|protected|public|return|short|static|strictfp|super|switch|synchronized|this|throw|throws|transient|try
|void|volatile|while",
446:      javascript: "async|await|break|case|catch|class|const|continue|debugger|default|delete|do|else|export
|extends|false|finally|for|from|function|if|import|in|instanceof|let|new|null|of|return|static|super|switch|this
|throw|true|try|typeof|undefined|var|void|while|yield",
447:      python: "and|as|assert|async|await|break|class|continue|def|del|elif|else|except|False|finally|for|fr
om|global|if|import|in|is|lambda|None|nonlocal|not|or|pass|raise|return|True|try|while|with|yield",
448:      html: "html|head|body|main|section|article|div|span|script|style|link|meta|title|header|footer|nav|bu
tton|input|form|label|img|a|p|pre|code"
449:  };
450:  const keywords = keywordMap[normalizeCodeLanguage(language)] || keywordMap.javascript;
451:  html = html.replace(new RegExp(`\\b(${keywords})\\b`, "g"), `

```

protect - lines 415-421

Item	Explanation
Source location	Lines 415-421 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>const protect = (pattern, className) => {</code>
Touches	authentication/security data
Calls detected	replace, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

415:  const protect = (pattern, className) => {
416:    html = html.replace(pattern, (match) => {
417:      const token = `@@CODE_TOKEN_${tokens.length}@@`;
418:      tokens.push(`<span class="${className}">${match}</span>`);
419:      return token;
420:    });
421:  };

```

renderRichCodeBlock - lines 456-465

Item	Explanation
Source location	Lines 456-465 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	<code>function renderRichCodeBlock(block = {}) {</code>
Touches	filesystem, authentication/security data

Item	Explanation
Calls detected	normalizeCodeLanguage, detectCodeLanguage, normalizeCodeText, escapeHtml, codeLanguageLabel, tokenizedCodeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

456: function renderRichCodeBlock(block = {}) {
457:   const language = normalizeCodeLanguage(block.language || detectCodeLanguage(block.code || ""));
458:   const code = normalizeCodeText(block.code || "", block.pasteMode || "source");
459:   return `
460:     <figure class="rich-code-block rich-code-language-${language}">
461:       <figcaption><span>&lt;/span></span> ${escapeHtml(codeLanguageLabel(language))}</figcaption>
462:       <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
463:     </figure>
464:   `;
465: }
```

renderPlainMultiline - lines 467-469

Item	Explanation
Source location	Lines 467-469 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPlainMultiline(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

467: function renderPlainMultiline(value = "") {
468:   return escapeHtml(value).replace(/\r\n?|\n/g, "<br>");
469: }
```

normalizeSiteContent - lines 471-477

Item	Explanation
Source location	Lines 471-477 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeSiteContent(content = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

471: function normalizeSiteContent(content = {}) {
```

```

472:   return {
473:     heroCopy: String(content.heroCopy || "").trim() || defaultSiteContent.heroCopy,
474:     heroEyebrow: String(content.heroEyebrow || "").trim() || defaultSiteContent.heroEyebrow,
475:     heroTitle: String(content.heroTitle || "").trim() || defaultSiteContent.heroTitle
476:   };
477: }

```

normalizeProfile - lines 479-495

Item	Explanation
Source location	Lines 479-495 (17 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeProfile(profileValue = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

479: function normalizeProfile(profileValue = {}) {
480:   return {
481:     brandImage: String(profileValue.brandImage || "").trim(),
482:     brandText: String(profileValue.brandText || "").trim() || String(profileValue.displayName ||
483:   "").trim() || defaultProfile.brandText,
484:     contactIntro: String(profileValue.contactIntro || "").trim(),
485:     displayName: String(profileValue.displayName || "").trim(),
486:     email: String(profileValue.email || "").trim(),
487:     githubUrl: String(profileValue.githubUrl || "").trim(),
488:     heroImage: String(profileValue.heroImage || "").trim(),
489:     linkedinUrl: String(profileValue.linkedinUrl || "").trim(),
490:     phone: String(profileValue.phone || "").trim(),
491:     portfolioLabel: String(profileValue.portfolioLabel || "").trim() || defaultProfile.portfolioLabel,
492:     profileImage: String(profileValue.profileImage || "").trim(),
493:     resumeUrl: String(profileValue.resumeUrl || "").trim(),
494:     websiteUrl: String(profileValue.websiteUrl || "").trim()
495:   };
496: }

```

mailComposeLink - lines 497-500

Item	Explanation
Source location	Lines 497-500 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function mailComposeLink(email = "") {
Touches	network/API requests
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

497: function mailComposeLink(email = "") {
498:   const clean = String(email || "").trim();
499:   return clean ? `https://mail.google.com/mail/?view=cm&fs=1&to=${encodeURIComponent(clean)}` : "";
500: }

```

phoneLink - lines 502-505

Item	Explanation
Source location	Lines 502-505 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function phoneLink(phone = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
502: function phoneLink(phone = "") {
503:   const clean = String(phone || "").trim();
504:   return clean ? `tel:${clean.replace(/^[^d+]/g, "")}` : "";
505: }
```

renderSiteContent - lines 507-515

Item	Explanation
Source location	Lines 507-515 (9 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteContent() {
Touches	filesystem
Calls detected	normalizeSiteContent, renderRichFieldContent, applyPlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
507: function renderSiteContent() {
508:   const content = normalizeSiteContent(siteContent || {});
509:   if (heroEyebrow) heroEyebrow.innerHTML = renderRichFieldContent(siteContentRich.heroEyebrow,
content.heroEyebrow);
510:   if (heroTitle) heroTitle.innerHTML = renderRichFieldContent(siteContentRich.heroTitle,
content.heroTitle);
511:   if (heroCopy) heroCopy.innerHTML = renderRichFieldContent(siteContentRich.heroCopy, content.heroCopy);
512:   applyPlainFieldStyle(heroEyebrow, "site-hero-eyebrow");
513:   applyPlainFieldStyle(heroTitle, "site-hero-title");
514:   applyPlainFieldStyle(heroCopy, "site-hero-copy");
515: }
```

renderProfile - lines 517-627

Item	Explanation
Source location	Lines 517-627 (111 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProfile() {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeProfile, mailComposeLink, phoneLink, normalizeLinkTarget, filter, renderRichFieldContent, applyPlainFieldStyle, slice, toUpperCase, setAttribute, querySelectorAll, forEach
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

517: function renderProfile() {
518:   const current = normalizeProfile(profile || {});
519:   const displayName = current.displayName || "Portfolio";
520:   const label = current.portfolioLabel || "Portfolio";
521:   const emailLink = mailComposeLink(current.email);
522:   const callLink = phoneLink(current.phone);
523:   const links = [
524:     current.email ? { label: "Email", url: emailLink, external: true } : null,
525:     current.phone ? { label: "Call", url: callLink } : null,
526:     current.githubUrl ? { label: "GitHub", url: normalizeLinkTarget(current.githubUrl, { assumeWeb: true
527:   })), external: true } : null,
528:     current.linkedinUrl ? { label: "LinkedIn", url: normalizeLinkTarget(current.linkedinUrl, { assumeWeb:
529:   true })), external: true } : null,
530:     current.websiteUrl ? { label: "Website", url: normalizeLinkTarget(current.websiteUrl, { assumeWeb:
531:   true })), external: true } : null,
532:     current.resumeUrl ? { label: "Resume", url: current.resumeUrl, external: false } : null
533:   ].filter(Boolean);
534:   document.title = `${displayName} | ${label}`;
535:   if (brandName) brandName.innerHTML = renderRichFieldContent(profileRich.displayName, displayName);
536:   if (brandSubtitle) brandSubtitle.innerHTML = renderRichFieldContent(profileRich.portfolioLabel, label);
537:   applyPlainFieldStyle(brandName, "profile-display-name");
538:   applyPlainFieldStyle(brandSubtitle, "profile-portfolio-label");
539:   if (brandText) brandText.textContent = current.brandText || displayName.slice(0, 3).toUpperCase() ||
"PORT";
540:   if (brandIcon) {
541:     if (current.brandImage) {
542:       brandIcon.src = current.brandImage;
543:       brandIcon.hidden = false;
544:     } else {
545:       brandIcon.hidden = true;
546:     }
547:   }
548:   if (headerAvatar && headerAvatarImage) {
549:     headerAvatar.hidden = !current.profileImage;
550:     if (current.profileImage) {
551:       headerAvatarImage.src = current.profileImage;
552:       headerAvatarImage.alt = displayName;
553:       headerAvatar.setAttribute("aria-label", `Go to ${displayName} contact information`);
554:     }
555:   }
556:   if (heroImage) {
557:     if (current.heroImage) {
558:       heroImage.hidden = false;
559:       heroImage.src = current.heroImage;
560:       heroImage.alt = `${displayName} portfolio background`;
561:     } else {
562:       heroImage.hidden = true;
563:     }
564:   }
565:   document.querySelectorAll('a[href="#resume"]').forEach((link) => {
566:     link.hidden = !current.resumeUrl;
567:   });
568:   const heroGithubLink = [...document.querySelectorAll(".hero-actions a")]
569:     .find((link) => /github/i.test(link.textContent || ""));
570:   if (heroGithubLink) {
571:     heroGithubLink.hidden = !current.githubUrl;
572:     if (current.githubUrl) heroGithubLink.href = normalizeLinkTarget(current.githubUrl, { assumeWeb: true
573:   });
574:   }
575:   if (resumeSection) {
576:     resumeSection.hidden = !current.resumeUrl;
577:     resumeSection.querySelectorAll("a").forEach((link) => {
578:       link.href = current.resumeUrl || "#";
579:     });
580:     const resumeObject = resumeSection.querySelector("object");
581:     if (resumeObject) resumeObject.data = current.resumeUrl || "";
582:     const fallbackLink = resumeSection.querySelector("object a");
583:     if (fallbackLink) fallbackLink.href = current.resumeUrl || "#";
584:   }
585:   if (profilePhoto) {
586:     profilePhoto.hidden = !current.profileImage;
587:     if (current.profileImage) {
588:       profilePhoto.src = current.profileImage;
589:     }
590:   }

```

```

591:     profilePhoto.alt = `Portrait of ${displayName}`;
592:   }
593: }
594: if (contactTitle) contactTitle.innerHTML = renderRichFieldContent(profileRich.displayName,
displayName);
595: applyPlainFieldStyle(contactTitle, "profile-display-name");
596: if (contactIntro) {
597:   contactIntro.innerHTML = renderRichFieldContent(
598:     profileRich.contactIntro,
599:     current.contactIntro || (current.displayName ? `${displayName}'s contact information and public
links.` : "Add contact details in the builder.")
600:   );
601: }
602: applyPlainFieldStyle(contactIntro, "profile-contact-intro");
603: if (contactDetails) {
604:   contactDetails.innerHTML = [
605:     current.email ? `` : "",
606:     current.phone ? `` : ""
607:   \\].filter\\(Boolean\\).join\\(""\\);
608: }
609: if \\(contactLinks\\) {
610:   contactLinks.innerHTML = links.map\\(\\(link\\) => {
611:     const styleId = link.label === "GitHub"
612:       ? "profile-github"
613:       : link.label === "LinkedIn"
614:       ? "profile-linkedin"
615:       : link.label === "Website"
616:       ? "profile-website"
617:       : link.label === "Email"
618:       ? "profile-email"
619:       : link.label === "Call"
620:       ? "profile-phone"
621:       : "";
622:     return `

```

textBlocksFromPlainText - lines 629-637

Item	Explanation
Source location	Lines 629-637 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function textBlocksFromPlainText(text) {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

629: function textBlocksFromPlainText(text) {
630:   return [{
631:     align: "left",
632:     fontFamily: "Arial",
633:     fontSize: "normal",
634:     text: String(text || ""),
635:     type: "paragraph"
636:   }];
637: }

```

unwrapFormula - lines 639-652

Item	Explanation
Source location	Lines 639-652 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function unwrapFormula(value) {
Touches	filesystem
Calls detected	trim, match
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

639: function unwrapFormula(value) {
640:   const text = String(value || "").trim();
641:   const wrappers = [
642:     [/^\$\$([\s\S]+\)\$\$\$/, 1],
643:     [/^\$\$([\s\S]+\)\$\$/, 1],
644:     [/^\$\$([\s\S]+\)\$/, 1],
645:     [/^\$([\s\S]+\)\$/, 1]
646:   ];
647:   for (const [pattern, group] of wrappers) {
648:     const match = text.match(pattern);
649:     if (match) return match[group].trim();
650:   }
651:   return text;
652: }

```

renderInlineMath - lines 654-663

Item	Explanation
Source location	Lines 654-663 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderInlineMath(text) {
Touches	filesystem
Calls detected	escapeHtml, replace, b, match, slice, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

654: function renderInlineMath(text) {
655:   const escaped = escapeHtml(text);
656:   const withMath = escaped.replace(/\$([\^$]+\)\$/g, '<span class="rich-inline-formula">$1</span>');
657:   return withMath.replace(/\b(https?:\/\/[\^s<]+|www\.[\^s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[\^s<]+\))/gi, (match) => {
658:     const trailing = match.match(/[.,;:!?]+\$/)?.[0] || "";
659:     const clean = trailing ? match.slice(0, -trailing.length) : match;
660:     const href = normalizeLinkTarget(clean, { assumeWeb: true });
661:     return `<a href="${href}" target="_blank" rel="noreferrer">${clean}</a>${trailing}`;
662:   });
663: }

```

renderMultilineInlineText - lines 665-667

Item	Explanation
Source location	Lines 665-667 (3 source lines).
Purpose	Renders UI, markup, or display output.

Item	Explanation
Declaration	function renderMultilineInlineText(text = "") {
Touches	filesystem
Calls detected	renderInlineMath, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

665: function renderMultilineInlineText(text = "") {
666:   return renderInlineMath(text).replace(/\r\n?|\n/g, "<br>");
667: }

```

looksLikeBareWebAddress - lines 669-676

Item	Explanation
Source location	Lines 669-676 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function looksLikeBareWebAddress(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, split, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

669: function looksLikeBareWebAddress(value = "") {
670:   const clean = String(value || "").trim();
671:   if (!clean || /\s/.test(clean) || /^(\.?\.\?\/|#|mailto:|tel:)/i.test(clean)) return false;
672:   if (/^[a-z][a-z0-9+.-]*$/i.test(clean)) return false;
673:   const host = clean.split(/[/?#]/)[0].toLowerCase();
674:   if (!/^[a-z0-9-]+\.(.[a-z0-9-]+)+$/i.test(host)) return false;
675:   return !/\.(pdf|docx?|xlsx?|pptx?|zip|7z|rar|png|jpe?g|gif|svg|webp|txt|md|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|l?spice|cir|net|asc|sch|kicad_sch|kicad_pcb)$/i.test(host);
676: }

```

looksLikeWebOrContactText - lines 678-686

Item	Explanation
Source location	Lines 678-686 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function looksLikeWebOrContactText(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, b, looksLikeBareWebAddress, split, some, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
678: function looksLikeWebOrContactText(value = "") {
679:   const clean = String(value || "").trim();
680:   if (!clean) return false;
681:   if (/^b(?:https?:\/\/|www\.)[^s<>'"]+/i.test(clean)) return true;
682:   if (/^b[a-z0-9._%+~]@[a-z0-9.-]+\.[a-z]{2,}\b/i.test(clean)) return true;
683:   if (/^b(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[^s<>'"]+/i.test(clean)) return true;
684:   if (looksLikeBareWebAddress(clean)) return true;
685:   return clean.split(/\s+/).some((word) => looksLikeBareWebAddress(word.replace(/^[('"]+|[)]'";:;!]?$/g,
  "")));
686: }
```

normalizeLinkTarget - lines 688-695

Item	Explanation
Source location	Lines 688-695 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeLinkTarget(target = "", options = {}) {
Touches	filesystem, network/API requests
Calls detected	trim, looksLikeBareWebAddress
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
688: function normalizeLinkTarget(target = "", options = {}) {
689:   const clean = String(target || "").trim();
690:   if (!clean) return "";
691:   if (/^\/\//.test(clean)) return `https:${clean}`;
692:   if (/^www\./i.test(clean)) return `https://${clean}`;
693:   if (options.assumeWeb && looksLikeBareWebAddress(clean)) return `https://${clean}`;
694:   return clean;
695: }
```

isWebsiteLinkItem - lines 697-705

Item	Explanation
Source location	Lines 697-705 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function isWebsiteLinkItem(item = {}, target = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, filter, join, toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
697: function isWebsiteLinkItem(item = {}, target = "") {
698:   const clean = String(target || "").trim();
699:   if (/^(https?:)?\/\//i.test(clean) || /^www\./i.test(clean)) return true;
700:   const typeText = [item.kind, item.type, item.status, item.meta, item.label, item.title]
701:     .filter(Boolean)
702:     .join(" ")
703:     .toLowerCase();
704:   return /^b(link|website|web link|url|external|github|linkedin|drive|social|profile)\b/i.test(typeText);
705: }
```

linkifyRichTextNodes - lines 707-742

Item	Explanation
Source location	Lines 707-742 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	function linkifyRichTextNodes(root) {
Touches	browser/Electron DOM, filesystem
Calls detected	createTreeWalker, acceptNode, closest, nextNode, push, forEach, createDocumentFragment, replace, b, append, createTextNode, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
707: function linkifyRichTextNodes(root) {
708:   const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
709:     acceptNode(node) {
710:       if (!node.textContent ||
!/(https?:\/\/|www\.|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/)/i.test(node.textContent)) return
NodeFilter.FILTER_REJECT;
711:       return node.parentElement?.closest("a") ? NodeFilter.FILTER_REJECT : NodeFilter.FILTER_ACCEPT;
712:     }
713:   });
714:   const nodes = [];
715:   let node = walker.nextNode();
716:   while (node) {
717:     nodes.push(node);
718:     node = walker.nextNode();
719:   }
720:
721:   nodes.forEach((textNode) => {
722:     const text = textNode.textContent || "";
723:     const fragment = document.createDocumentFragment();
724:     let cursor = 0;
725:     text.replace(/(https?:\/\/|www\.|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com
\/|\/|s<]+)\/gi, (match, _url, offset) => {
726:       if (offset > cursor) fragment.appendChild(document.createTextNode(text.slice(cursor, offset)));
727:       const trailing = match.match(/[,.;:!?]+$/)?.[0] || "";
728:       const clean = trailing ? match.slice(0, -trailing.length) : match;
729:       const link = document.createElement("a");
730:       link.href = normalizeLinkTarget(clean, { assumeWeb: true });
731:       link.target = "_blank";
732:       link.rel = "noreferrer";
733:       link.textContent = clean;
734:       fragment.appendChild(link);
735:       if (trailing) fragment.appendChild(document.createTextNode(trailing));
736:       cursor = offset + match.length;
737:       return match;
738:     });
739:     if (cursor < text.length) fragment.appendChild(document.createTextNode(text.slice(cursor)));
740:     textNode.replaceWith(fragment);
741:   });
742: }
```

sanitizeRichInlineHtml - lines 744-787

Item	Explanation
Source location	Lines 744-787 (44 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function sanitizeRichInlineHtml(value = "") {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, Set, querySelectorAll, forEach, has, replaceWith, getAttribute, trim, normalizeLinkTarget, cleanFontFamily, normalizeFontPx, normalizeTextColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

744: function sanitizeRichInlineHtml(value = "") {
745:   const template = document.createElement("template");
746:   template.innerHTML = String(value || "");
747:   const allowedTags = new Set(["A", "B", "BR", "EM", "I", "SPAN", "STRONG", "U"]);
748:
749:   [...template.content.querySelectorAll("*")].forEach((node) => {
750:     if (!allowedTags.has(node.tagName)) {
751:       node.replaceWith(...node.childNodes);
752:       return;
753:     }
754:
755:     const href = node.tagName === "A" ? String(node.getAttribute("href") || "").trim() : "";
756:     const normalizedHref = normalizeLinkTarget(href, { assumeWeb: true });
757:     const safeHref = /^(https?:|mailto:|tel:|#|\/)/i.test(normalizedHref) ? normalizedHref : "";
758:     const fontFamily = cleanFontFamily(node.style.fontFamily || "");
759:     const fontPx = normalizeFontPx(parseFloat(node.style.fontSize || ""));
760:     const color = normalizeTextColor(node.style.color || "");
761:     const rawFontWeight = node.style.fontWeight || "";
762:     const fontWeight = /^(bold|[6-9]00)$/i.test(rawFontWeight)
763:       ? "700"
764:       : /^(normal|[1-5]00)$/i.test(rawFontWeight)
765:       ? "400"
766:       : "";
767:     const fontStyle = node.style.fontStyle === "italic" ? "italic" : "";
768:     const textDecoration = node.style.textDecoration.includes("underline") ? "underline" : "";
769:
770:     [...node.attributes].forEach((attribute) => node.removeAttribute(attribute.name));
771:
772:     if (node.tagName === "A" && safeHref) {
773:       node.setAttribute("href", safeHref);
774:       node.setAttribute("target", "_blank");
775:       node.setAttribute("rel", "noreferrer");
776:     }
777:     if (fontFamily) node.style.fontFamily = fontFamily;
778:     if (fontPx) node.style.fontSize = `${fontPx}px`;
779:     if (color) node.style.color = color;
780:     if (fontWeight) node.style.fontWeight = fontWeight;
781:     if (fontStyle) node.style.fontStyle = fontStyle;
782:     if (textDecoration) node.style.textDecoration = textDecoration;
783:   });
784:
785:   linkifyRichTextNodes(template.content);
786:   return template.innerHTML;
787: }

```

displayTitle - lines 789-798

Item	Explanation
Source location	Lines 789-798 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function displayTitle(value, fallback = "Untitled item") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

789: function displayTitle(value, fallback = "Untitled item") {
790:   const title = String(value || fallback).trim();
791:   const replacements = {
792:     "Brief": "Overview",
793:     "Project Brief": "Overview",
794:     "Design Brief": "Design Overview",
795:     "Simulation Brief": "Simulation Overview"

```

```

796:   };
797:   return replacements[title] || title;
798: }

```

cleanFontFamily - lines 800-807

Item	Explanation
Source location	Lines 800-807 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function cleanFontFamily(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, replace, trim, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

800: function cleanFontFamily(value = "") {
801:   return String(value)
802:     .split(",")
803:     .map((item) => item.replace(/^[^w\s-]/g, "").trim())
804:     .filter(Boolean)
805:     .slice(0, 3)
806:     .join(", ");
807: }

```

normalizeFontPx - lines 809-812

Item	Explanation
Source location	Lines 809-812 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFontPx(value) {
Touches	filesystem
Calls detected	isFinite
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

809: function normalizeFontPx(value) {
810:   const size = Number(value);
811:   return Number.isFinite(size) && size >= 8 && size <= 24 ? String(size) : "";
812: }

```

normalizeTextColor - lines 813-820

Item	Explanation
Source location	Lines 813-820 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeTextColor(value = "") {
Touches	filesystem
Calls detected	trim

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

813: function normalizeTextColor(value = "") {
814:     const color = String(value || "").trim();
815:     return /^(#[0-9a-f]{3}|[0-9a-f]{6})$/i.test(color) ||
816:         /^rgba?\(/i.test(color) ||
817:         /^hsla?\(/i.test(color)
818:         ? color
819:         : "";
820: }

```

rgbColorToHex - lines 822-830

Item	Explanation
Source location	Lines 822-830 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function rgbColorToHex(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, toLowerCase, match, map, max, min, toString, padStart, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

822: function rgbColorToHex(value = "") {
823:     const color = String(value || "").trim();
824:     if (/^[0-9a-f]{6}$/i.test(color)) return color.toLowerCase();
825:     const shortHex = color.match(/^[0-9a-f]([0-9a-f])([0-9a-f])$/i);
826:     if (shortHex) return
`#${shortHex[1]}${shortHex[1]}${shortHex[2]}${shortHex[2]}${shortHex[3]}${shortHex[3]}`.toLowerCase();
827:     const rgb = color.match(/^rgba?\((\d+),\s*(\d+),\s*(\d+)/i);
828:     if (!rgb) return color;
829:     return `#${[rgb[1], rgb[2], rgb[3]].map((part) => Math.max(0, Math.min(255,
Number(part))).toString(16).padStart(2, "0")).join("")}`;
830: }

```

normalizePlainFieldStyle - lines 832-844

Item	Explanation
Source location	Lines 832-844 (13 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizePlainFieldStyle(style = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
832: function normalizePlainFieldStyle(style = {}) {
833:   const normalized = {};
834:   const fontFamily = cleanFontFamily(style.fontFamily || "");
835:   const fontPx = normalizeFontPx(style.fontPx || style.fontSize || "");
836:   const color = normalizeTextColor(style.color || "");
837:   if (fontFamily) normalized.fontFamily = fontFamily;
838:   if (fontPx) normalized.fontPx = fontPx;
839:   if (color) normalized.color = rgbColorToHex(color);
840:   if (style.bold === true || style.bold === "true") normalized.bold = true;
841:   if (style.italic === true || style.italic === "true") normalized.italic = true;
842:   if (style.underline === true || style.underline === "true") normalized.underline = true;
843:   return normalized;
844: }
```

normalizeFieldStyles - lines 846-852

Item	Explanation
Source location	Lines 846-852 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFieldStyles(styles = {}) {
Touches	filesystem
Calls detected	fromEntries, entries, map, normalizePlainFieldStyle, filter, keys
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
846: function normalizeFieldStyles(styles = {}) {
847:   return Object.fromEntries(
848:     Object.entries(styles || {})
849:       .map(([fieldId, style]) => [fieldId, normalizePlainFieldStyle(style)])
850:       .filter(([ , style]) => Object.keys(style).length)
851:   );
852: }
```

plainFieldStyleToCss - lines 854-864

Item	Explanation
Source location	Lines 854-864 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function plainFieldStyleToCss(style = {}) {
Touches	filesystem
Calls detected	normalizePlainFieldStyle, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
854: function plainFieldStyleToCss(style = {}) {
855:   const normalized = normalizePlainFieldStyle(style);
856:   const rules = [];
857:   if (normalized.fontFamily) rules.push(`font-family: ${normalized.fontFamily}`);
858:   if (normalized.fontPx) rules.push(`font-size: ${normalized.fontPx}px`);
859:   if (normalized.color) rules.push(`color: ${normalized.color}`);
860:   if (normalized.bold) rules.push("font-weight: 700");
861:   if (normalized.italic) rules.push("font-style: italic");
862:   if (normalized.underline) rules.push("text-decoration: underline");
863:   return rules.join(" ");
}
```

864: }

applyPlainFieldStyle - lines 866-875

Item	Explanation
Source location	Lines 866-875 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function applyPlainFieldStyle(element, fieldId = "") {
Touches	filesystem
Calls detected	normalizePlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
866: function applyPlainFieldStyle(element, fieldId = "") {
867:   if (!element) return;
868:   const style = normalizePlainFieldStyle(fieldStyles[fieldId] || {});
869:   element.style.fontFamily = style.fontFamily || "";
870:   element.style.fontSize = style.fontPx ? `${style.fontPx}px` : "";
871:   element.style.color = style.color || "";
872:   element.style.fontWeight = style.bold ? "700" : "";
873:   element.style.fontStyle = style.italic ? "italic" : "";
874:   element.style.textDecoration = style.underline ? "underline" : "";
875: }
```

plainFieldStyleAttribute - lines 877-880

Item	Explanation
Source location	Lines 877-880 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function plainFieldStyleAttribute(fieldId = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	plainFieldStyleToCss, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
877: function plainFieldStyleAttribute(fieldId = "") {
878:   const css = plainFieldStyleToCss(fieldStyles[fieldId] || {});
879:   return css ? `style="${escapeHtml(css)}"` : "";
880: }
```

richTextStyle - lines 882-896

Item	Explanation
Source location	Lines 882-896 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function richTextStyle(block = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, push, escapeHtml, join

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

882: function richTextStyle(block = {}) {
883:     const styles = [];
884:     const fontFamily = cleanFontFamily(block.fontFamily || "Arial") || "Arial";
885:     const fontPx = normalizeFontPx(block.fontPx);
886:     const color = normalizeTextColor(block.color || "");
887:
888:     if (fontFamily) styles.push(`font-family: ${fontFamily}`);
889:     if (fontPx) styles.push(`font-size: ${fontPx}px`);
890:     if (color) styles.push(`color: ${color}`);
891:     if (block.bold) styles.push("font-weight: 700");
892:     if (block.italic) styles.push("font-style: italic");
893:     if (block.underline) styles.push("text-decoration: underline");
894:
895:     return styles.length ? ` style="${escapeHtml(styles.join("; "))}"` : "";
896: }

```

normalizeCropAspect - lines 898-900

Item	Explanation
Source location	Lines 898-900 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropAspect(value = "") {
Touches	filesystem
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

898: function normalizeCropAspect(value = "") {
899:     return ["1 / 1", "4 / 3", "16 / 9", "3 / 4"].includes(value) ? value : "original";
900: }

```

normalizeCropZoom - lines 902-905

Item	Explanation
Source location	Lines 902-905 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropZoom(value = 1) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

902: function normalizeCropZoom(value = 1) {
903:     const zoom = Number(value);
904:     return Number.isFinite(zoom) ? Math.min(3, Math.max(1, zoom)) : 1;
905: }

```

normalizeCropPosition - lines 907-910

Item	Explanation
Source location	Lines 907-910 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropPosition(value = 50) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

907: function normalizeCropPosition(value = 50) {
908:     const position = Number(value);
909:     return Number.isFinite(position) ? Math.min(100, Math.max(0, position)) : 50;
910: }

```

richImageCropStyle - lines 912-921

Item	Explanation
Source location	Lines 912-921 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function richImageCropStyle(block = {}) {
Touches	filesystem
Calls detected	normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, push, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

912: function richImageCropStyle(block = {}) {
913:     const aspect = normalizeCropAspect(block.cropAspect);
914:     const styles = [
915:         `--crop-zoom: ${normalizeCropZoom(block.cropZoom)}`,
916:         `--crop-x: ${normalizeCropPosition(block.cropX)}%`,
917:         `--crop-y: ${normalizeCropPosition(block.cropY)}%`
918:     ];
919:     if (aspect !== "original") styles.push(`--crop-aspect: ${aspect}`);
920:     return ` style="${escapeHtml(styles.join("; "))}"`;
921: }

```

richImageDownloadLink - lines 923-940

Item	Explanation
Source location	Lines 923-940 (18 source lines).
Purpose	Part of the public website runtime.
Declaration	function richImageDownloadLink(block = {}) {

Item	Explanation
Touches	filesystem
Calls detected	escapeHtml, cleanRichImageTitle, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

923: function richImageDownloadLink(block = {}) {
924:   const label = escapeHtml(cleanRichImageTitle(block) || block.caption || "Image file");
925:   const url = block.url || "#";
926:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(block, url) });
927:
928:   return `
929:   <p class="rich-download-only">
930:     <a
931:       class="resource-link"
932:       href="${escapeHtml(target)}"
933:       ${linkAttributes(target, block)}
934:       ${downloadAttribute(target, block)}
935:     >
936:       ${label}
937:     </a>
938:   </p>
939:   `;
940: }

```

cleanRichImageTitle - lines 942-945

Item	Explanation
Source location	Lines 942-945 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function cleanRichImageTitle(block = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

942: function cleanRichImageTitle(block = {}) {
943:   const title = String(block.title || "").trim();
944:   return /^pasted image\b/i.test(title) ? "" : title;
945: }

```

renderRichContent - lines 947-982

Item	Explanation
Source location	Lines 947-982 (36 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichContent(rich, fallbackText = "") {
Touches	filesystem
Calls detected	textBlocksFromPlainText, map, includes, richImageDownloadLink, cleanRichImageTitle, normalizeLinkTarget, normalizeCropAspect, richImageCropStyle, escapeHtml, unwrapFormula, looksLikeWebOrContactText, renderInlineMath

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

947: function renderRichContent(rich, fallbackText = "") {
948:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
949:   return `
950:     <div class="rich-content">
951:       ${blocks.map((block) => {
952:         const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
953:         if (block.type === "image") {
954:           if (block.display === "download") return richImageDownloadLink(block);
955:           const title = cleanRichImageTitle(block);
956:           const imageSrc = normalizeLinkTarget(block.url, { assumeWeb: true });
957:           return `
958:             <figure class="rich-image justify-${align}">
959:               <span class="rich-image-viewport crop-${normalizeCropAspect(block.cropAspect)} ===
"original" ? "original" : "active"${richImageCropStyle(block)}>
960:                 
961:               </span>
962:               ${title || block.caption} ? `<figcaption>${title} ?
`<strong>${escapeHtml(title)}</strong>` : ""}${block.caption ? `<span>${escapeHtml(block.caption)}</span>` :
""}</figcaption>` : ""}
963:             </figure>
964:           `;
965:         }
966:         if (block.type === "formula") {
967:           const formula = unwrapFormula(block.formula);
968:           if (looksLikeWebOrContactText(formula)) {
969:             return `<p class="rich-paragraph rich-text-normal
text-${align}">${renderInlineMath(formula)}</p>`;
970:           }
971:           return `<div class="rich-formula justify-${align}">${escapeHtml(formula)}</div>`;
972:         }
973:         if (block.type === "code") return renderRichCodeBlock(block);
974:         const size = ["small", "normal", "large"].includes(block.fontSize) ? block.fontSize : "normal";
975:         const content = block.html
976:           ? sanitizeRichInlineHtml(block.html)
977:           : renderInlineMath(block.text || "");
978:         return `<p class="rich-paragraph rich-text-${size}
text-${align}"${richTextStyle(block)}>${content}</p>`;
979:       }).join("")}
980:     </div>
981:   `;
982: }

```

renderRichFieldContent - lines 984-994

Item	Explanation
Source location	Lines 984-994 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichFieldContent(rich, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	textBlocksFromPlainText, map, includes, sanitizeRichInlineHtml, renderInlineMath, richTextStyle, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

984: function renderRichFieldContent(rich, fallbackText = "") {
985:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
986:   return blocks.map((block) => {
987:     if (block.type !== "paragraph") return "";
988:     const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
989:     const content = block.html

```

```

990:         ? sanitizeRichInlineHtml(block.html)
991:         : renderInlineMath(block.text || "");
992:     return `<span class="rich-field-line text-${align}"${richTextStyle(block)}>${content}</span>`;
993:   }).filter(Boolean).join("");
994: }

```

slugLabel - lines 996-999

Item	Explanation
Source location	Lines 996-999 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function slugLabel(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

996: function slugLabel(value) {
997:   const category = categories.find((item) => item.id === value);
998:   return category ? category.label : value;
999: }

```

canonicalTemplateId - lines 1001-1003

Item	Explanation
Source location	Lines 1001-1003 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function canonicalTemplateId(id) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1001: function canonicalTemplateId(id) {
1002:   return legacyTemplateSkins[id] || id || "";
1003: }

```

projectTemplateId - lines 1005-1007

Item	Explanation
Source location	Lines 1005-1007 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateId(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	canonicalTemplateId
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1005: function projectTemplateId(project) {
1006:   return canonicalTemplateId(project?.portfolioView?.template?.id || project?.templateId || "");
1007: }

```

projectTemplateClass - lines 1009-1012

Item	Explanation
Source location	Lines 1009-1012 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1009: function projectTemplateClass(project) {
1010:   const templateId = projectTemplateId(project);
1011:   return templateId ? `project-template project-template-${templateId}` : "project-template-white";
1012: }

```

responsiveClassName - lines 1014-1019

Item	Explanation
Source location	Lines 1014-1019 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveClassName(value, fallback) {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1014: function responsiveClassName(value, fallback) {
1015:   return String(value || fallback || "")
1016:     .toLowerCase()
1017:     .replace(/[^\a-z0-9]+/g, "-")
1018:     .replace(/^-|-$/g, "") || fallback;
1019: }

```

projectResponsiveClass - lines 1021-1024

Item	Explanation
Source location	Lines 1021-1024 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectResponsiveClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile, responsiveClassName
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1021: function projectResponsiveClass(project) {
1022:   const profile = projectResponsiveProfile(project);
1023:   return `responsive-project responsive-density-${responsiveClassName(profile.density, "balanced")}
responsive-card-${responsiveClassName(profile.cardLayout, "single")}`;
1024: }

```

responsiveStyleValues - lines 1026-1034

Item	Explanation
Source location	Lines 1026-1034 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveStyleValues(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1026: function responsiveStyleValues(project) {
1027:   const profile = projectResponsiveProfile(project);
1028:   return {
1029:     "--responsive-section-card-min": profile.sectionCardMin,
1030:     "--responsive-content-max": profile.contentMax,
1031:     "--responsive-media-max": profile.mediaMax,
1032:     "--responsive-touch-target": profile.touchTarget
1033:   };
1034: }

```

hasPublicTemplate - lines 1036-1038

Item	Explanation
Source location	Lines 1036-1038 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function hasPublicTemplate(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1036: function hasPublicTemplate(project) {
1037:   return Boolean(projectTemplateId(project));
1038: }

```

projectTemplateVisual - lines 1040-1042

Item	Explanation
Source location	Lines 1040-1042 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateVisual(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1040: function projectTemplateVisual(project) {
1041:   return project?.portfolioView?.template?.visual || project?.templateVisual || null;
1042: }

```

projectTemplateStyle - lines 1044-1062

Item	Explanation
Source location	Lines 1044-1062 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateStyle(project, accent) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateVisual, responsiveStyleValues, entries, filter, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1044: function projectTemplateStyle(project, accent) {
1045:   const visual = projectTemplateVisual(project) || {};
1046:   const values = {
1047:     "--accent": accent,
1048:     "--template-bg": visual.background,
1049:     "--template-panel": visual.panel,
1050:     "--template-accent": visual.accent,
1051:     "--template-hover": visual.hover,
1052:     "--template-click": visual.click,
1053:     "--template-click-text": visual.clickText,
1054:     "--template-line": visual.line,
1055:     "--template-text": visual.text,
1056:     ...responsiveStyleValues(project)
1057:   };
1058:   return Object.entries(values)

```

```

1059:     .filter(([, value]) => value)
1060:     .map(([key, value]) => `${key}: ${value}`)
1061:     .join("; ");
1062: }

```

applyProjectTemplateToElement - lines 1064-1098

Item	Explanation
Source location	Lines 1064-1098 (35 source lines).
Purpose	Part of the public website runtime.
Declaration	function applyProjectTemplateToElement(element, project, accent) {
Touches	browser/Electron DOM
Calls detected	filter, startsWith, forEach, remove, projectTemplateClass, split, add, projectResponsiveClass, projectTemplateVisual, responsiveStyleValues, entries, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1064: function applyProjectTemplateToElement(element, project, accent) {
1065:   if (!element) return;
1066:   [...element.classList]
1067:   .filter((className) =>
1068:     className === "project-template" ||
1069:     className === "project-template-white" ||
1070:     className === "responsive-project" ||
1071:     className.startsWith("project-template-") ||
1072:     className.startsWith("responsive-density-") ||
1073:     className.startsWith("responsive-card-")
1074:   )
1075:   .forEach((className) => element.classList.remove(className));
1076:   projectTemplateClass(project).split(" ").forEach((className) => element.classList.add(className));
1077:   projectResponsiveClass(project).split(" ").forEach((className) => element.classList.add(className));
1078:   const visual = projectTemplateVisual(project) || {};
1079:   const values = {
1080:     "--accent": accent,
1081:     "--template-bg": visual.background,
1082:     "--template-panel": visual.panel,
1083:     "--template-accent": visual.accent,
1084:     "--template-hover": visual.hover,
1085:     "--template-click": visual.click,
1086:     "--template-click-text": visual.clickText,
1087:     "--template-line": visual.line,
1088:     "--template-text": visual.text,
1089:     ...responsiveStyleValues(project)
1090:   };
1091:   Object.entries(values).forEach(([key, value]) => {
1092:     if (value) {
1093:       element.style.setProperty(key, value);
1094:     } else {
1095:       element.style.removeProperty(key);
1096:     }
1097:   });
1098: }

```

parsedItemTerms - lines 1100-1110

Item	Explanation
Source location	Lines 1100-1110 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedItemTerms(item) {
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1100: function parsedItemTerms(item) {
1101:   return [
1102:     item?.title,
1103:     item?.description,
1104:     item?.meta,
1105:     item?.url,
1106:     item?.kind,
1107:     ...(item?.rich?.blocks || []).flatMap((block) => [block.text, block.formula, block.title,
block.caption]),
1108:     ...(item?.children || []).flatMap(parsedItemTerms)
1109:   ];
1110: }

```

flattenProject - lines 1112-1165

Item	Explanation
Source location	Lines 1112-1165 (54 source lines).
Purpose	Part of the public website runtime.
Declaration	function flattenProject(project) {
Touches	filesystem
Calls detected	slugLabel, electronicsSearchKeywords, flatMap, map, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1112: function flattenProject(project) {
1113:   const categoryLabel = slugLabel(project.category);
1114:   const electronicsKeywords = window.electronicsSearchKeywords
1115:     ? window.electronicsSearchKeywords(project, categoryLabel)
1116:     : [];
1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};
1127:   const designTerms = [
1128:     design.brief?.summary,
1129:     ...(design.brief?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1130:     design.documentation?.summary,
1131:     ...(design.documentation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1132:     ...(design.documentation?.references || []).flatMap((item) => [item.title, item.description,
item.url]),
1133:     ...(design.documentation?.mathAnalysis || []).flatMap((item) => [item.title, item.description]),
1134:     design.simulation?.summary,
1135:     ...(design.simulation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1136:     ...(design.simulation?.results || []).flatMap((item) => [item.title, item.description, item.url])
1137:   ];
1138:   return [
1139:     project.id,
1140:     project.title,
1141:     categoryLabel,
1142:     project.status,
1143:     project.summary,
1144:     ...richSummaryTerms,
1145:     ...parsedChildTerms,
1146:     ...designTerms,
1147:     ...(project.focus || []),
1148:     ...(project.highlights || []),
1149:     ...(project.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title,

```

```

item.label, item.description].filter(Boolean).join(" "),
1150:   ...(project.languages || []),
1151:   ...(project.documents || []).flatMap((item) => [item.title, item.type, item.status, item.url]),
1152:   ...(project.tests || []).flatMap((item) => [item.name, item.method, item.status, item.result,
item.artifact]),
1153:   ...(project.pcbbs || []).flatMap((item) => [item.name, item.revision, item.status, item.artifact]),
1154:   ...(project.media || []).flatMap((item) => [item.title, item.caption, item.url]),
1155:   ...(project.sections || []).flatMap((section) => [
1156:     section.title,
1157:     section.description,
1158:     ...(section.items || []).flatMap((item) => [item.title, item.description, item.type, item.status,
item.url])
1159:   ]),
1160:   ...(project.links || []).flatMap((item) => [item.label, item.url]),
1161:   ...electronicsKeywords
1162: ]
1163:   .map(normalize)
1164:   .join(" ");
1165: }

```

richSummaryTerms - lines 1117-1126

Item	Explanation
Source location	Lines 1117-1126 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	const richSummaryTerms = (project.summaryRich?.blocks []).flatMap((block) => [
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};

```

parsedChildTerms - lines 1123-1126

Item	Explanation
Source location	Lines 1123-1126 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	const parsedChildTerms = (project.portfolioView?.sections []).flatMap((section) =>
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );

```

```
1126:   const design = project.design || {};
```

projectMatches - lines 1167-1169

Item	Explanation
Source location	Lines 1167-1169 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectMatches(project, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	flattenProject, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1167: function projectMatches(project, query) {  
1168:   return !query || flattenProject(project).includes(query);  
1169: }
```

projectVisible - lines 1171-1174

Item	Explanation
Source location	Lines 1171-1174 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectVisible(project, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectMatches
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1171: function projectVisible(project, query) {  
1172:   const categoryMatch = activeFilter === "all" || project.category === activeFilter;  
1173:   return categoryMatch && projectMatches(project, query);  
1174: }
```

normalizeCategory - lines 1176-1184

Item	Explanation
Source location	Lines 1176-1184 (9 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCategory(category = {}) {
Touches	filesystem
Calls detected	trim, normalizeTextColor, toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1176: function normalizeCategory(category = {}) {
1177:   const label = String(category.label || category.title || category.id || "Project category").trim() ||
"Project category";
1178:   return {
1179:     accent: normalizeTextColor(category.accent || "" || "#1677a8",
1180:     description: String(category.description || "").trim(),
1181:     id: String(category.id || label.toLowerCase().replace(/^[a-z0-9]+/g, "-").replace(/^-|-$/g, "")) ||
"category"),
1182:     label
1183:   };
1184: }

```

hydrateProjectCategories - lines 1186-1201

Item	Explanation
Source location	Lines 1186-1201 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function hydrateProjectCategories(rawCategories = [], rawProjects = []) {
Touches	filesystem
Calls detected	map, Set, forEach, trim, has, split, filter, charAt, toUpperCase, slice, join, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1186: function hydrateProjectCategories(rawCategories = [], rawProjects = []) {
1187:   const normalized = rawCategories.map(normalizeCategory);
1188:   const knownIds = new Set(normalized.map((category) => category.id));
1189:   rawProjects.forEach((project) => {
1190:     const id = String(project?.category || "").trim();
1191:     if (!id || knownIds.has(id)) return;
1192:     const label = id
1193:       .split(/[-_\s]+/)
1194:       .filter(Boolean)
1195:       .map((part) => part.charAt(0).toUpperCase() + part.slice(1))
1196:       .join(" ") || "Project category";
1197:     normalized.push(normalizeCategory({ id, label, description: "Project category" }));
1198:     knownIds.add(id);
1199:   });
1200:   return normalized;
1201: }

```

setActiveFilter - lines 1203-1209

Item	Explanation
Source location	Lines 1203-1209 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function setActiveFilter(filter = "all") {
Touches	browser/Electron DOM
Calls detected	some, forEach, toggle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1203: function setActiveFilter(filter = "all") {
1204:   activeFilter = filter;
1205:   if (activeFilter !== "all" && !categories.some((category) => category.id === activeFilter)) {
1206:     activeFilter = "all";
1207:   }
1208:   filterButtons.forEach((item) => item.classList.toggle("active", item.dataset.filter === activeFilter));
1209: }

```

bindFilterButtons - lines 1211-1220

Item	Explanation
Source location	Lines 1211-1220 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function bindFilterButtons() {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, addEventListener, setActiveFilter, renderProjects, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1211: function bindFilterButtons() {
1212:   filterButtons = [...document.querySelectorAll(".filter-button")];
1213:   filterButtons.forEach((button) => {
1214:     button.addEventListener("click", () => {
1215:       setActiveFilter(button.dataset.filter || "all");
1216:       renderProjects();
1217:       updateSearchDropdown();
1218:     });
1219:   });
1220: }

```

renderCategoryFilters - lines 1222-1241

Item	Explanation
Source location	Lines 1222-1241 (20 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCategoryFilters() {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join, escapeHtml, bindFilterButtons, setActiveFilter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1222: function renderCategoryFilters() {
1223:   if (projectTrackCount) {
1224:     projectTrackCount.textContent = `${categories.length} Track${categories.length === 1 ? "" : "s"}`;
1225:   }
1226:   if (projectTrackLabels) {
1227:     projectTrackLabels.textContent = categories.length

```

```

1228:         ? categories.map((category) => category.label).join(", ")
1229:         : "project categories";
1230:     }
1231:     if (projectFilters) {
1232:         projectFilters.innerHTML = `
1233:         <button class="filter-button active" type="button" data-filter="all">All</button>
1234:         ${categories.map((category) => `
1235:         <button class="filter-button" type="button" data-
filter="${escapeHtml(category.id)}">${escapeHtml(category.label)}</button>
1236:         `).join("")}
1237:         `;
1238:     }
1239:     bindFilterButtons();
1240:     setActiveFilter(activeFilter);
1241: }

```

flattenSearchText - lines 1243-1251

Item	Explanation
Source location	Lines 1243-1251 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function flattenSearchText(values = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	flat, filter, map, stringify, join, replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1243: function flattenSearchText(values = []) {
1244:     return values
1245:         .flat(Infinity)
1246:         .filter((value) => value !== null && value !== undefined)
1247:         .map((value) => typeof value === "object" ? JSON.stringify(value) : String(value))
1248:         .join(" ")
1249:         .replace(/\s+/g, " ")
1250:         .trim();
1251: }

```

richTextTerms - lines 1253-1264

Item	Explanation
Source location	Lines 1253-1264 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function richTextTerms(rich) {
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1253: function richTextTerms(rich) {
1254:     return (rich?.blocks || []).flatMap((block) => [
1255:         block.text,
1256:         block.html,
1257:         block.formula,
1258:         block.code,
1259:         block.language,
1260:         block.title,

```

```

1261:     block.caption,
1262:     block.url
1263:   });
1264: }

```

addSearchEntry - lines 1266-1276

Item	Explanation
Source location	Lines 1266-1276 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function addSearchEntry(entries, entry) {
Touches	filesystem
Calls detected	flattenSearchText, normalize, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1266: function addSearchEntry(entries, entry) {
1267:   const text = flattenSearchText([entry.title, entry.type, entry.context, entry.text, entry.url]);
1268:   if (!entry.title && !text) return null;
1269:   const storedEntry = {
1270:     ...entry,
1271:     normalizedText: normalize(text),
1272:     text
1273:   };
1274:   entries.push(storedEntry);
1275:   return storedEntry;
1276: }

```

searchSnippet - lines 1278-1286

Item	Explanation
Source location	Lines 1278-1286 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function searchSnippet(text = "", query = "") {
Touches	filesystem
Calls detected	replace, trim, normalize, indexOf, slice, max, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1278: function searchSnippet(text = "", query = "") {
1279:   const clean = String(text || "").replace(/\s+/g, " ").trim();
1280:   if (!clean) return "";
1281:   const index = normalize(clean).indexOf(normalize(query));
1282:   if (index < 0) return clean.slice(0, 150);
1283:   const start = Math.max(0, index - 54);
1284:   const end = Math.min(clean.length, index + query.length + 80);
1285:   return `${start ? "... " : ""}${clean.slice(start, end)}${end < clean.length ? "... " : ""}`;
1286: }

```

entryScore - lines 1288-1314

Item	Explanation
Source location	Lines 1288-1314 (27 source lines).
Purpose	Part of the public website runtime.
Declaration	function entryScore(entry, query) {
Touches	filesystem
Calls detected	normalize, trim, split, filter, startsWith, includes, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1288: function entryScore(entry, query) {
1289:   const normalizedQuery = normalize(query).trim();
1290:   if (!normalizedQuery) return 0;
1291:   const title = normalize(entry.title);
1292:   const type = normalize(entry.type);
1293:   const context = normalize(entry.context);
1294:   const text = entry.normalizedText || "";
1295:   const words = normalizedQuery.split(/\s+/).filter(Boolean);
1296:   let score = 0;
1297:
1298:   if (title === normalizedQuery) score += 160;
1299:   if (title.startsWith(normalizedQuery)) score += 120;
1300:   if (title.includes(normalizedQuery)) score += 90;
1301:   if (context.includes(normalizedQuery)) score += 55;
1302:   if (type.includes(normalizedQuery)) score += 35;
1303:   if (text.includes(normalizedQuery)) score += 45;
1304:   words.forEach((word) => {
1305:     if (title.includes(word)) score += 16;
1306:     if (context.includes(word)) score += 10;
1307:     if (text.includes(word)) score += 6;
1308:   });
1309:
1310:   if (score > 0 && entry.kind === "file") score += 18;
1311:   if (score > 0 && entry.kind === "section") score += 14;
1312:   if (score > 0 && entry.kind === "project") score += 12;
1313:   return score;
1314: }

```

entryMatches - lines 1316-1318

Item	Explanation
Source location	Lines 1316-1318 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function entryMatches(entry, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	entryScore
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1316: function entryMatches(entry, query) {
1317:   return entryScore(entry, query) > 0;
1318: }

```

fileIsBrowserSearchable - lines 1320-1323

Item	Explanation
Source location	Lines 1320-1323 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function fileIsBrowserSearchable(url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1320: function fileIsBrowserSearchable(url = "") {
1321:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1322:   return /\.(txt|md|markdown|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|?|spice|cir|net|asc|sch|k
icad_sch|kicad_pcb)$/i.test(clean);
1323: }

```

indexSearchableFileText - lines 1325-1338

Item	Explanation
Source location	Lines 1325-1338 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	async function indexSearchableFileText(entry) {
Touches	filesystem, network/API requests
Calls detected	fileIsBrowserSearchable, startsWith, fetch, text, flattenSearchText, slice, normalize, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1325: async function indexSearchableFileText(entry) {
1326:   if (!entry.url || !fileIsBrowserSearchable(entry.url)) return;
1327:   if (/^(https?:)?\/\//i.test(entry.url) && !entry.url.startsWith(window.location.origin)) return;
1328:   try {
1329:     const response = await fetch(entry.url);
1330:     if (!response.ok) return;
1331:     const text = await response.text();
1332:     entry.text = flattenSearchText([entry.text, text.slice(0, 60000)]);
1333:     entry.normalizedText = normalize(entry.text);
1334:     updateSearchDropdown();
1335:   } catch {
1336:     // Files remain searchable by title, path, caption, and catalog text if direct text fetch is
unavailable.
1337:   }
1338: }

```

addProjectSearchEntries - lines 1340-1414

Item	Explanation
Source location	Lines 1340-1414 (75 source lines).
Purpose	Part of the public website runtime.
Declaration	function addProjectSearchEntries(entries, project) {
Touches	filesystem

Item	Explanation
Calls detected	slugLabel, richTextTerms, electronicsSearchKeywords, addSearchEntry, filter, sectionHasRenderableContent, forEach, displayTitle, parsedItemTerms, fileNameFromUrl, itemUrl, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1340: function addProjectSearchEntries(entries, project) {
1341:   const categoryLabel = slugLabel(project.category);
1342:   const baseContext = `${categoryLabel} / ${project.title}`;
1343:   const projectTerms = [
1344:     project.id,
1345:     project.summary,
1346:     richTextTerms(project.summaryRich),
1347:     project.focus,
1348:     project.highlights,
1349:     project.tools,
1350:     project.languages,
1351:     project.links,
1352:     window.electronicsSearchKeywords ? window.electronicsSearchKeywords(project, categoryLabel) : []
1353:   ];
1354:
1355:   addSearchEntry(entries, {
1356:     context: categoryLabel,
1357:     kind: "project",
1358:     projectId: project.id,
1359:     title: project.title,
1360:     type: "Project",
1361:     text: projectTerms
1362:   });
1363:
1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });
1400:
1401:   uniqueDownloads(collectDownloadsFromValue(project, "Project file", [])).forEach((download) => {
1402:     const entry = {
1403:       context: baseContext,
1404:       kind: "file",
1405:       projectId: project.id,
1406:       title: download.title || fileNameFromUrl(download.url),
1407:       type: download.type || "Uploaded file",
1408:       url: download.url,
1409:       text: [download.description, download.status, download.url]

```

```

1410:     };
1411:     const storedEntry = addSearchEntry(entries, entry);
1412:     if (storedEntry) indexSearchableFileText(storedEntry);
1413:   });
1414: }

```

sections - lines 1364-1399

Item	Explanation
Source location	Lines 1364-1399 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>const sections = (project.portfolioView?.sections []).filter((section) => section.id !== "brief" && sectionHasRenderableContent(section));</code>
Touches	filesystem
Calls detected	filter, sectionHasRenderableContent, forEach, addSearchEntry, displayTitle, richTextTerms, parsedItemTerms, fileNameFromUrl, itemUrl, join, pathToString, indexSearchableFileText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });

```

addSiteSectionSearchEntries - lines 1416-1463

Item	Explanation
Source location	Lines 1416-1463 (48 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function addSiteSectionSearchEntries(entries) {</code>

Item	Explanation
Touches	filesystem
Calls detected	normalizeProfile, splice, forEach, addSearchEntry, indexSearchableFileText, filter, richTextTerms
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1416: function addSiteSectionSearchEntries(entries) {
1417:   const current = normalizeProfile(profile || {});
1418:   const pageSections = [
1419:     { id: "top", title: heroTitle?.textContent || "Front page", type: "Page section", text:
[heroEyebrow?.textContent, heroCopy?.textContent] },
1420:     { id: "projects", title: "Engineering Projects", type: "Directory", text: "project directory
categories files sections" },
1421:     { id: "process", title: "Process", type: "Page section", text: "engineering process design testing
documentation" },
1422:     { id: "contact", title: "Contact", type: "Page section", text: [current.displayName, current.email,
current.phone, current.githubUrl, current.linkedinUrl, current.websiteUrl, current.contactIntro] }
1423:   ];
1424:   if (current.resumeUrl) {
1425:     pageSections.splice(2, 0, { id: "resume", title: "Resume", type: "Page section", text: "resume
professional profile document" });
1426:   }
1427:   pageSections.forEach((section) => addSearchEntry(entries, {
1428:     domTarget: section.id,
1429:     kind: "page",
1430:     title: section.title,
1431:     type: section.type,
1432:     text: section.text
1433:   }));
1434:
1435:   if (current.resumeUrl) {
1436:     const resumeTextEntry = addSearchEntry(entries, {
1437:       context: "Professional Profile / Resume",
1438:       kind: "file",
1439:       title: "Resume",
1440:       type: "Resume",
1441:       url: current.resumeUrl,
1442:       text: [current.displayName, current.portfolioLabel, current.contactIntro, current.resumeUrl]
1443:     });
1444:     if (resumeTextEntry) indexSearchableFileText(resumeTextEntry);
1445:   }
1446:
1447:   (siteSections || []).filter(siteSectionRenderable).forEach((section) => {
1448:     addSearchEntry(entries, {
1449:       domTarget: section.id,
1450:       kind: "page",
1451:       title: section.title || "Portfolio section",
1452:       type: "Portfolio section",
1453:       text: [section.description, richTextTerms(section.richDescription), section.links, section.assets]
1454:     });
1455:     (section.subsections || []).forEach((item) => addSearchEntry(entries, {
1456:       domTarget: section.id,
1457:       kind: "page",
1458:       title: item.title || "Portfolio subsection",
1459:       type: "Portfolio subsection",
1460:       text: [item.description, richTextTerms(item.richDescription), item.links]
1461:     }));
1462:   });
1463: }

```

rebuildSearchIndex - lines 1465-1470

Item	Explanation
Source location	Lines 1465-1470 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function rebuildSearchIndex() {
Touches	Mostly local calculations or local UI state.
Calls detected	addSiteSectionSearchEntries, forEach, addProjectSearchEntries
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1465: function rebuildSearchIndex() {
1466:   const entries = [];
1467:   addSiteSectionSearchEntries(entries);
1468:   projects.forEach((project) => addProjectSearchEntries(entries, project));
1469:   searchableEntries = entries;
1470: }

```

searchResultsFor - lines 1472-1486

Item	Explanation
Source location	Lines 1472-1486 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function searchResultsFor(query) {
Touches	filesystem
Calls detected	normalize, trim, Set, map, entryScore, filter, sort, localeCompare, join, has, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1472: function searchResultsFor(query) {
1473:   const normalizedQuery = normalize(query).trim();
1474:   if (!normalizedQuery) return [];
1475:   const seen = new Set();
1476:   return searchableEntries
1477:     .map((entry) => ({ ...entry, score: entryScore(entry, normalizedQuery) }))
1478:     .filter((entry) => entry.score > 0)
1479:     .sort((a, b) => b.score - a.score || a.title.localeCompare(b.title))
1480:     .filter((entry) => {
1481:       const key = [entry.kind, entry.projectId, entry.sectionIndex, entry.resourcePath, entry.domTarget,
entry.url, entry.title].join("|");
1482:       if (seen.has(key)) return false;
1483:       seen.add(key);
1484:       return true;
1485:     });
1486: }

```

assistantEndpoint - lines 1488-1494

Item	Explanation
Source location	Lines 1488-1494 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantEndpoint() {
Touches	browser/Electron DOM, network/API requests
Calls detected	trim, querySelector, includes
API endpoints	/api/portfolio-ai
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1488: function assistantEndpoint() {
1489:   if (typeof window.OMB_AI_ENDPOINT === "string") return window.OMB_AI_ENDPOINT.trim();
1490:   const metaEndpoint = document.querySelector('meta[name="portfolio-ai-endpoint"]')?.content || "";
1491:   if (metaEndpoint.trim()) return String(metaEndpoint).trim();
1492:   if (["localhost", "127.0.0.1"].includes(window.location.hostname)) return "/api/portfolio-ai";
1493:   return "/api/portfolio-ai";
1494: }
```

assistantSourceLabel - lines 1496-1505

Item	Explanation
Source location	Lines 1496-1505 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourceLabel(result = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, trim, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1496: function assistantSourceLabel(result = {}) {
1497:   const title = result.title || "Portfolio item";
1498:   if (result.kind === "project" || result.kind === "page") return title;
1499:   const contextParts = String(result.context || "")
1500:     .split("/")
1501:     .map((item) => item.trim())
1502:     .filter(Boolean);
1503:   const specificContext = contextParts.slice(-2).join(" / ");
1504:   return [title, specificContext].filter(Boolean).join(" - ");
1505: }
```

assistantProjectSummary - lines 1507-1522

Item	Explanation
Source location	Lines 1507-1522 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectSummary(project = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	slugLabel, filter, sectionHasRenderableContent, map, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1507: function assistantProjectSummary(project = {}) {
1508:   const categoryLabel = slugLabel(project.category);
1509:   const sections = (project.portfolioView?.sections || [])
1510:     .filter((section) => section.id !== "brief" && sectionHasRenderableContent(section))
1511:     .map((section) => displayTitle(section.title, "Section"));
1512:   return {
1513:     category: categoryLabel,
1514:     focus: project.focus || [],
1515:     id: project.id,
1516:     links: project.links || [],
1517:     sections,
1518:     summary: project.summary || "",
1519:     title: project.portfolioView?.title || project.title,
```

```

1520:     tools: project.tools || []
1521:   };
1522: }

```

assistantProjectTitleAcronyms - lines 1524-1535

Item	Explanation
Source location	Lines 1524-1535 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectTitleAcronyms(title = "") {
Touches	filesystem
Calls detected	normalize, split, filter, Set, min, add, slice, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1524: function assistantProjectTitleAcronyms(title = "") {
1525:   const words = normalize(title)
1526:   .split(/\s+/)
1527:   .filter((word) => word.length > 2);
1528:   const acronyms = new Set();
1529:   for (let start = 0; start < words.length; start += 1) {
1530:     for (let length = 2; length <= Math.min(5, words.length - start); length += 1) {
1531:       acronyms.add(words.slice(start, start + length).map((word) => word[0]).join(""));
1532:     }
1533:   }
1534:   return [...acronyms].filter((item) => item.length >= 2);
1535: }

```

assistantProjectIsNamedInQuestion - lines 1537-1553

Item	Explanation
Source location	Lines 1537-1553 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectIsNamedInQuestion(question = "", project = {}) {
Touches	filesystem
Calls detected	normalize, Set, includes, split, assistantProjectTitleAcronyms, some, has, filter, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1537: function assistantProjectIsNamedInQuestion(question = "", project = {}) {
1538:   const clean = normalize(question);
1539:   if (!clean) return false;
1540:   const fillerWords = new Set(["with", "from", "into", "using", "project", "design", "system",
"systems"]);
1541:   const rawTitle = project.portfolioView?.title || project.title || project.id;
1542:   const title = normalize(rawTitle);
1543:   if (!title) return false;
1544:   if (clean.includes(title)) return true;
1545:   const cleanWords = new Set(clean.split(/\s+/));
1546:   const acronymMatch = assistantProjectTitleAcronyms(rawTitle)
1547:   .some((acronym) => cleanWords.has(acronym));
1548:   if (acronymMatch) return true;
1549:   const words = title.split(/\s+/).filter((word) => word.length > 3 && !fillerWords.has(word));
1550:   if (!words.length) return false;
1551:   const matches = words.filter((word) => clean.includes(word)).length;
1552:   return words.length <= 2 ? matches === words.length : matches >= Math.min(3, words.length);

```

```
1553: }
```

assistantNamedProjectIds - lines 1555-1559

Item	Explanation
Source location	Lines 1555-1559 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function assistantNamedProjectIds(question = "") {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>filter</code> , <code>assistantProjectIsNamedInQuestion</code> , <code>map</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1555: function assistantNamedProjectIds(question = "") {
1556:   return projects
1557:     .filter((project) => assistantProjectIsNamedInQuestion(question, project))
1558:     .map((project) => project.id);
1559: }
```

assistantProjectTitleMatches - lines 1561-1563

Item	Explanation
Source location	Lines 1561-1563 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function assistantProjectTitleMatches(question = "") {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>assistantNamedProjectIds</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1561: function assistantProjectTitleMatches(question = "") {
1562:   return assistantNamedProjectIds(question).length > 0;
1563: }
```

assistantUrlKind - lines 1565-1577

Item	Explanation
Source location	Lines 1565-1577 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function assistantUrlKind(url = "") {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>split</code> , <code>toLowerCase</code> , <code>fileIsBrowserSearchable</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1565: function assistantUrlKind(url = "") {
1566:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1567:   if (!clean) return "unknown";
1568:   if (/github\.com|raw\.githubusercontent\.com|gist\.githubusercontent\.com/i.test(clean)) return
"github";
1569:   if (/linkedin\.com/i.test(clean)) return "linkedin";
1570:   if (/\. (png|jpe?g|gif|webp|svg)$/i.test(clean)) return "image";
1571:   if (/\.pdf$/i.test(clean)) return "pdf";
1572:   if (/\. (zip|7z|rar)$/i.test(clean)) return "archive";
1573:   if (/\. (xlsx?|csv)$/i.test(clean)) return "spreadsheet";
1574:   if (fileIsBrowserSearchable(clean)) return "text";
1575:   if (/^https?:\/\/\//i.test(clean)) return "webpage";
1576:   return "file";
1577: }

```

assistantAbsoluteUrl - lines 1579-1658

Item	Explanation
Source location	Lines 1579-1658 (80 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantAbsoluteUrl(url = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget, looksLikeBareWebAddress, URL, assistantLinkRecordsFromValue, uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantUrlsFromTextValue, flattenSearchText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1579: function assistantAbsoluteUrl(url = "") {
1580:   const target = normalizeLinkTarget(url, { assumeWeb: looksLikeBareWebAddress(url) ||
/^https?:\/\/\//i.test(url) });
1581:   if (!target) return "";
1582:   try {
1583:     return new URL(target, window.location.href).href;
1584:   } catch {
1585:     return target;
1586:   }
1587: }
1588:
1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:   const records = [];
1591:   const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:   downloads.forEach((download) => {
1593:     const rawUrl = download.url || download.href || download.path || "";
1594:     if (!rawUrl) return;
1595:     records.push({
1596:       context,
1597:       description: download.description || "",
1598:       kind: download.kind || assistantUrlKind(rawUrl),
1599:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:       url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602:   });
1603:   return records;
1604: }
1605:
1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/\|www\.)[^s<>"']+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[(),.;:!?]+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,

```

```

1613:         description: "",
1614:         kind: assistantUrlKind(url),
1615:         label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:         type: "Public link",
1617:         url: assistantAbsoluteUrl(url)
1618:     }));
1619: }
1620:
1621: function assistantPublicProfileLinks() {
1622:     const records = [];
1623:     const add = (item = {}, context = "") => {
1624:         const rawUrl = item.url || item.href || item.path || "";
1625:         if (!rawUrl) return;
1626:         const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:         records.push({
1628:             context,
1629:             kind: item.kind || assistantUrlKind(rawUrl),
1630:             label,
1631:             text: item.description || "",
1632:             url: assistantAbsoluteUrl(rawUrl)
1633:         });
1634:     };
1635:
1636:     const current = normalizeProfile(profile || {});
1637:     if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:     if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:     if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:     if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:     (siteSections || []).forEach((section) => {
1643:         (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:     });
1645:
1646:     return records.filter((record, index, array) => (
1647:         record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:     ));
1649: }
1650:
1651: function assistantProjectEvidence(project = {}) {
1652:     const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:     const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:         const rawUrl = download.url || "";
1655:         return {
1656:             description: download.description || "",
1657:             kind: assistantUrlKind(rawUrl),
1658:             label: download.title || download.label || fileNameFromUrl(rawUrl),

```

assistantLinkRecordsFromValue - lines 1589-1604

Item	Explanation
Source location	Lines 1589-1604 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantLinkRecordsFromValue(value, context = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:     const records = [];
1591:     const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:     downloads.forEach((download) => {
1593:         const rawUrl = download.url || download.href || download.path || "";
1594:         if (!rawUrl) return;
1595:         records.push({
1596:             context,
1597:             description: download.description || "",
1598:             kind: download.kind || assistantUrlKind(rawUrl),

```

```

1599:         label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:         url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602: });
1603: return records;
1604: }

```

assistantUrlsFromTextValue - lines 1606-1619

Item	Explanation
Source location	Lines 1606-1619 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantUrlsFromTextValue(value, context = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	flattenSearchText, match, b, Set, map, replace, slice, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/|www\.)[^s<>"])+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[.,:!?]+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,
1613:       description: "",
1614:       kind: assistantUrlKind(url),
1615:       label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:       type: "Public link",
1617:       url: assistantAbsoluteUrl(url)
1618:     }));
1619: }

```

assistantPublicProfileLinks - lines 1621-1649

Item	Explanation
Source location	Lines 1621-1649 (29 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantPublicProfileLinks() {
Touches	filesystem
Calls detected	fileNameFromUrl, push, assistantUrlKind, assistantAbsoluteUrl, normalizeProfile, add, forEach, filter, findIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1621: function assistantPublicProfileLinks() {
1622:   const records = [];
1623:   const add = (item = {}, context = "") => {
1624:     const rawUrl = item.url || item.href || item.path || "";
1625:     if (!rawUrl) return;
1626:     const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:     records.push({
1628:       context,
1629:       kind: item.kind || assistantUrlKind(rawUrl),
1630:       label,
1631:       text: item.description || "",
1632:       url: assistantAbsoluteUrl(rawUrl)

```

```

1633:     });
1634:   });
1635:
1636:   const current = normalizeProfile(profile || {});
1637:   if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:   if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:   if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:   if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:   (siteSections || []).forEach((section) => {
1643:     (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:   });
1645:
1646:   return records.filter((record, index, array) => (
1647:     record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:   ));
1649: }

```

assistantProjectEvidence - lines 1651-1667

Item	Explanation
Source location	Lines 1651-1667 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectEvidence(project = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, slice, map, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl, assistantUrlsFromTextValue, filter, findIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1651: function assistantProjectEvidence(project = {}) {
1652:   const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:   const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:     const rawUrl = download.url || "";
1655:     return {
1656:       description: download.description || "",
1657:       kind: assistantUrlKind(rawUrl),
1658:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1659:       type: download.type || "",
1660:       url: assistantAbsoluteUrl(rawUrl)
1661:     };
1662:   });
1663:   const textLinkRecords = assistantUrlsFromTextValue([project.summary, project.links,
project.portfolioView, project.portfolioView?.title || project.title];
1664:   return [...downloadRecords, ...textLinkRecords].filter((record, index, array) => (
1665:     record.url && array.findIndex((item) => item.url === record.url) === index
1666:   ));
1667: }

```

assistantQuestionIsCasual - lines 1669-1675

Item	Explanation
Source location	Lines 1669-1675 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionIsCasual(question = "") {
Touches	filesystem
Calls detected	normalize
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1669: function assistantQuestionIsCasual(question = "") {
1670:   const clean = normalize(question);
1671:   return
1672:   /^(hi|hello|hey|good\s+(morning|afternoon|evening)|yo|sup|thanks|thank\s+you|ok|okay)\b/.test(clean)
1673:   || /\^(how\s+are\s+you|what\s+can\s+you\s+help.*|what\s+do\s+you\s+do|who\s+are\s+you|help|help\s+me|c
1674:   an\s+you\s+help|nice\s+to\s+meet\s+you)\??$/ .test(clean)
1675:   || /\^(what'\s+is|do\s+you\s+know)\s+my\s+name\??$/ .test(clean)
1676:   || /\^who\s+am\s+i\??$/ .test(clean);
1677: }
```

assistantQuestionHasPortfolioIntent - lines 1677-1682

Item	Explanation
Source location	Lines 1677-1682 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionHasPortfolioIntent(question = "") {
Touches	filesystem
Calls detected	normalize, assistantProjectTitleMatches, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1677: function assistantQuestionHasPortfolioIntent(question = "") {
1678:   const clean = normalize(question);
1679:   return assistantProjectTitleMatches(question)
1680:   || /\b(maurice|otieno|portfolio|resume|github|linkedin|contact|email|phone|project|projects|repo|repo
1681:   sitory|repositories|file|files|document|documents|artifact|artifacts|link|links|download|open|show|where|built|b
1682:   uild|created|designed|implemented)\b/.test(clean)
1683:   || /\b(your|his|maurice's)\s+(project|projects|resume|github|linkedin|portfolio|work|email|phone|cont
1684:   act|repo|repository|files?|documents?|links?)\b/.test(clean);
1685: }
```

assistantConversationSuggestsPortfolioContext - lines 1684-1688

Item	Explanation
Source location	Lines 1684-1688 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantConversationSuggestsPortfolioContext(history = []) {
Touches	filesystem
Calls detected	slice, some, b, normalize
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1684: function assistantConversationSuggestsPortfolioContext(history = []) {
```

```

1685:   return history
1686:     .slice(-4)
1687:     .some((item) => /\b(maurice|otieno|portfolio|project|resume|github|linkedin|electronics|engineering)\
b/.test(normalize(item?.content)));
1688: }

```

assistantQuestionAllowsPublicLookup - lines 1690-1695

Item	Explanation
Source location	Lines 1690-1695 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalize, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1690: function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {
1691:   const clean = normalize(question);
1692:   return intent === "general_engineering"
1693:     || intent === "general_knowledge"
1694:     || /\b(github|linkedin|repo|repository|repositories|public\s+link|website|web\s+page|profile|resume|u
ploadeds+file|document|source\s+code)\b/.test(clean);
1695: }

```

assistantQuestionLooksConceptual - lines 1697-1702

Item	Explanation
Source location	Lines 1697-1702 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionLooksConceptual(question = "") {
Touches	filesystem
Calls detected	normalize, assistantQuestionIsEngineeringRelated, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1697: function assistantQuestionLooksConceptual(question = "") {
1698:   const clean = normalize(question);
1699:   if (!assistantQuestionIsEngineeringRelated(question)) return false;
1700:   return
/^(what|what\s+is|what\s+are|what's|define|explain|describe|how|why|compare|differentiate)\b/.test(clean)
1701:     || /\b(definition|meaning|basics|overview|introduction|difference\s+between|how\s+does|how\s+do|why\s
+does|why\s+do)\b/.test(clean);
1702: }

```

assistantQuestionIntent - lines 1704-1717

Item	Explanation
Source location	Lines 1704-1717 (14 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function assistantQuestionIntent(question = "", history = assistantChatHistory) {
Touches	filesystem
Calls detected	assistantQuestionIsCasual, normalize, assistantQuestionHasPortfolioIntent, assistantConversationSuggestsPortfolioContext, b, assistantQuestionIsEngineeringRelated, assistantQuestionLooksConceptual
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1704: function assistantQuestionIntent(question = "", history = assistantChatHistory) {
1705:   if (assistantQuestionIsCasual(question)) return "general_conversation";
1706:   if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(normalize(question)) ||
/who\s+am\s+i\??$/i.test(normalize(question))) {
1707:     return "general_conversation";
1708:   }
1709:   const hasPortfolioIntent = assistantQuestionHasPortfolioIntent(question);
1710:   if (!hasPortfolioIntent && assistantConversationSuggestsPortfolioContext(history) &&
/b(name|you|your|he|his|h|him|that|this|it)\b/i.test(normalize(question))) {
1711:     return "general_conversation";
1712:   }
1713:   if (!hasPortfolioIntent && assistantQuestionIsEngineeringRelated(question)) return
"general_engineering";
1714:   if (!hasPortfolioIntent && assistantQuestionLooksConceptual(question)) return "general_engineering";
1715:   if (hasPortfolioIntent) return "portfolio_specific";
1716:   return "general_knowledge";
1717: }

```

assistantQuestionTokens - lines 1719-1730

Item	Explanation
Source location	Lines 1719-1730 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionTokens(question = "") {
Touches	filesystem, authentication/security data
Calls detected	Set, normalize, split, map, trim, filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1719: function assistantQuestionTokens(question = "") {
1720:   const stopWords = new Set([
1721:     "about", "again", "also", "and", "are", "can", "could", "does", "for", "from", "give", "have", "his",
1722:     "into", "link", "links", "maurice", "me", "my", "open", "otieno", "please", "portfolio", "project",
1723:     "projects", "show", "tell", "that", "the", "their", "this", "what", "where", "which", "who", "with",
1724:     "work", "your"
1725:   ]);
1726:   return normalize(question)
1727:     .split(/[\a-z0-9+#.]+/i)
1728:     .map((token) => token.trim())
1729:     .filter((token) => token.length > 2 && !stopWords.has(token));
1730: }

```

assistantSpecificSourceTokens - lines 1732-1739

Item	Explanation
Source location	Lines 1732-1739 (8 source lines).

Item	Explanation
Purpose	Part of the public website runtime.
Declaration	function assistantSpecificSourceTokens(question = "") {
Touces	authentication/security data
Calls detected	Set, assistantQuestionTokens, filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1732: function assistantSpecificSourceTokens(question = "") {
1733:   const genericTokens = new Set([
1734:     "built", "build", "created", "designed", "did", "done", "engineering", "file", "files", "hardware",
1735:     "item", "items", "section", "sections", "system", "systems", "thing", "things", "tool", "tools",
1736:     "use", "used", "using", "website"
1737:   ]);
1738:   return assistantQuestionTokens(question).filter((token) => !genericTokens.has(token));
1739: }

```

assistantPromptTargetsProjectLanding - lines 1741-1746

Item	Explanation
Source location	Lines 1741-1746 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantPromptTargetsProjectLanding(question = "") {
Touces	filesystem
Calls detected	normalize, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1741: function assistantPromptTargetsProjectLanding(question = "") {
1742:   const clean = normalize(question);
1743:   return /\b(open|show|view|go\s+to|take\s+me\s+to)\b/.test(clean)
1744:     && /\b(project|overview|page)\b/.test(clean)
1745:     && /\b(code|source|repo|repository|github|file|files|document|documents|test|tests|result|results|pc
b|schematic|simulation|tool|tools|resume|linkedin)\b/.test(clean);
1746: }

```

assistantEntryRelationScore - lines 1748-1762

Item	Explanation
Source location	Lines 1748-1762 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantEntryRelationScore(result = {}, question = "") {
Touces	filesystem, authentication/security data
Calls detected	assistantSpecificSourceTokens, normalize, filter, join, includes, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1748: function assistantEntryRelationScore(result = {}, question = "") {
1749:   const tokens = assistantSpecificSourceTokens(question);
1750:   const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1751:   const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1752:   let relation = tokenHits * 30;
1753:
1754:   if (tokens.length && tokenHits === tokens.length) relation += 25;
1755:   if (Number(result.score || 0) >= 120) relation += 15;
1756:   if (Number(result.score || 0) >= 80) relation += 8;
1757:   if (result.kind === "project") relation += 12;
1758:   if (result.kind === "file" && /\b(file|files|download|document|documents|code|source|github|repo|repository|test|tests|result|results|pcb|schematic|simulation)\b/.test(normalize(question))) {
1759:     relation += 16;
1760:   }
1761:   return relation;
1762: }

```

assistantNamedProjectTopicTokens - lines 1764-1776

Item	Explanation
Source location	Lines 1764-1776 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {
Touches	filesystem, authentication/security data
Calls detected	assistantNamedProjectIds, Set, map, find, filter, forEach, normalize, split, add, assistantProjectTitleAcronyms, assistantSpecificSourceTokens, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1764: function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {
1765:   const projectVocabulary = new Set();
1766:   namedIds
1767:     .map((id) => projects.find((project) => project.id === id))
1768:     .filter(Boolean)
1769:     .forEach((project) => {
1770:       const title = project.portfolioView?.title || project.title || project.id;
1771:       normalize(title).split(/\s+/).filter((word) => word.length > 2).forEach((word) =>
projectVocabulary.add(word));
1772:       assistantProjectTitleAcronyms(title).forEach((acronym) => projectVocabulary.add(acronym));
1773:       projectVocabulary.add(normalize(project.id));
1774:     });
1775:   return assistantSpecificSourceTokens(question).filter((token) => !projectVocabulary.has(token));
1776: }

```

assistantSourcesForDisplay - lines 1778-1813

Item	Explanation
Source location	Lines 1778-1813 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourcesForDisplay(question = "", results = [], intent = assistantQuestionIntent(question)) {
Touches	filesystem, authentication/security data
Calls detected	assistantQuestionIntent, assistantSpecificSourceTokens, assistantNamedProjectIds, Set, assistantNamedProjectTopicTokens, b, normalize, assistantPromptTargetsProjectLanding, map, assistantEntryRelationScore, filter, has

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 14 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1778: function assistantSourcesForDisplay(question = "", results = [], intent =
assistantQuestionIntent(question)) {
1779:   if (intent !== "portfolio_specific") return [];
1780:   const tokens = assistantSpecificSourceTokens(question);
1781:   const namedProjectIdList = assistantNamedProjectIds(question);
1782:   const namedProjectIds = new Set(namedProjectIdList);
1783:   const namedProjectTopicTokens = assistantNamedProjectTopicTokens(question, namedProjectIdList);
1784:   const asksForLinks = /\b(open|show|where|link|links|github|linkedin|resume|download|file|files|repo|rep
ository|source\s+code)\b/.test(normalize(question));
1785:   const asksForProjectLanding = assistantPromptTargetsProjectLanding(question);
1786:   const pageLinkIntent = /\b(resume|github|linkedin|contact|email|phone)\b/.test(normalize(question));
1787:   const asksForSpecificArtifact = /\b(code|source|repo|repository|github|file|files|document|documents|te
st|tests|result|results|pcb|schematic|simulation|tool|tools)\b/.test(normalize(question));
1788:
1789:   if (!namedProjectIds.size && !pageLinkIntent && tokens.length === 0) return [];
1790:
1791:   const filtered = results
1792:     .map((result) => ({ ...result, relation: assistantEntryRelationScore(result, question) })))
1793:     .filter((result) => {
1794:       if (!result) return false;
1795:       if (namedProjectIds.size && !namedProjectIds.has(result.projectId)) return false;
1796:       if (asksForProjectLanding) return result.kind === "project";
1797:       const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1798:       const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1799:       const topicHits = namedProjectTopicTokens.filter((token) => haystack.includes(token)).length;
1800:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind !== "project" &&
topicHits === 0) return false;
1801:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind === "project" &&
asksForSpecificArtifact) return false;
1802:       if (namedProjectIds.size && result.kind === "project") return true;
1803:       if (pageLinkIntent && tokenHits >= 1) return true;
1804:       if (!tokens.length) return false;
1805:       if (tokenHits >= Math.min(2, tokens.length)) return true;
1806:       if (asksForLinks && tokenHits >= 1 && result.relation >= 35) return true;
1807:       return result.relation >= 55;
1808:     });
1809:
1810:   return filtered
1811:     .sort((a, b) => b.relation - a.relation || Number(b.score || 0) - Number(a.score || 0))
1812:     .slice(0, asksForLinks ? 4 : 2);
1813: }

```

assistantContextForQuestion - lines 1815-1890

Item	Explanation
Source location	Lines 1815-1890 (76 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question), knownResults = null) {
Touches	filesystem
Calls detected	assistantQuestionIntent, includes, searchResultsFor, slice, Set, map, filter, assistantNamedProjectIds, find, b, normalize, assistantPublicProfileLinks
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1815: function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question),
knownResults = null) {
1816:   const directResults = ["general_engineering", "general_knowledge",
"general_conversation"].includes(intent)

```

```

1817:     ? []
1818:     : (knownResults || searchResultsFor(question)).slice(0, 8);
1819: const projectIds = [...new Set([
1820:   ...directResults.map((item) => item.projectId).filter(Boolean),
1821:   ...assistantNamedProjectIds(question)
1822: ])]);
1823: const matchedProjects = projectIds
1824:   .map((id) => projects.find((project) => project.id === id))
1825:   .filter(Boolean)
1826:   .slice(0, 5);
1827: const includePortfolioLinkContext = intent === "portfolio_specific";
1828: const wantsPublicCode = /\b(github|repo|repository|repositories|source\s|code|code|snippet|firmware|ver
1829: iolog|systemverilog|python|javascript)\b/.test(normalize(question));
1829: const publicProfileFetches = includePortfolioLinkContext
1830:   ? assistantPublicProfileLinks()
1831:   .map((item) => ({ ...item, question }));
1832:   .sort((a, b) => {
1833:     if (!wantsPublicCode) return 0;
1834:     const aGithub = /github/i.test(`${a.kind || ""} ${a.url || ""}`) ? 1 : 0;
1835:     const bGithub = /github/i.test(`${b.kind || ""} ${b.url || ""}`) ? 1 : 0;
1836:     return bGithub - aGithub;
1837:   })
1838:   : [];
1839: const resultFetches = directResults.map((result) => ({
1840:   context: result.context,
1841:   kind: assistantUrlKind(result.url),
1842:   title: result.title,
1843:   type: result.type,
1844:   url: assistantAbsoluteUrl(result.url || ""),
1845:   question
1846: }));
1847: const projectFetches = matchedProjects.flatMap((project) => assistantProjectEvidence(project)
1848:   .filter((item) => ["text", "webpage", "github", "linkedin", "resume_text"].includes(item.kind))
1849:   .slice(0, 6)
1850:   .map((item) => ({ ...item, question })));
1851: const sourceFetches = [
1852:   ...(wantsPublicCode ? publicProfileFetches : []),
1853:   ...resultFetches,
1854:   ...(wantsPublicCode ? [] : publicProfileFetches),
1855:   ...projectFetches
1856: ].filter((item) => item.url);
1857: const uniqueSourceFetches = sourceFetches.filter((item, index, array) => (
1858:   array.findIndex((candidate) => candidate.url === item.url) === index
1859: )).slice(0, 14);
1860: return {
1861:   categories: categories.map((category) => ({ id: category.id, label: category.label })),
1862:   intent,
1863:   profile: normalizeProfile(profile || {}),
1864:   knowledgeManifest: {
1865:     publicProfiles: includePortfolioLinkContext ? assistantPublicProfileLinks() : [],
1866:     publicSourcePolicy: "Only public profile and project links shown in the portfolio are used. GitHub
repository links can be expanded into repository metadata, README text, and selected public source files when
the visitor asks for code.",
1867:     matchedProjectEvidence: matchedProjects.map((project) => ({
1868:       evidence: assistantProjectEvidence(project),
1869:       projectId: project.id,
1870:       title: project.portfolioView?.title || project.title
1871:     })),
1872:     note: "Image records include filenames, captions, descriptions, and neighboring portfolio text. Use
a vision-capable backend before claiming visual details not present in captions or text."
1873:   },
1874:   portfolioContextPolicy: intent !== "portfolio_specific"
1875:     ? "Answer from general knowledge first. Do not lead with the portfolio owner's project context
unless the visitor explicitly asks to connect the concept to the portfolio."
1876:     : "Answer from the configured portfolio context first. Do not invent project details outside the
supplied context.",
1877:   projects: matchedProjects.map(assistantProjectSummary),
1878:   question,
1879:   results: directResults.map((result) => ({
1880:     context: result.context,
1881:     kind: result.kind,
1882:     text: searchSnippet(result.text, question),
1883:     title: result.title,
1884:     type: result.type,
1885:     url: result.url
1886: })),
1887:   sourceFetches: uniqueSourceFetches,
1888:   siteSections: siteSections.filter(siteSectionRenderable).map((section) => ({ id: section.id, title:
section.title })))
1889:   };
1890: }

```

assistantRemember - lines 1892-1901

Item	Explanation
Source location	Lines 1892-1901 (10 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function assistantRemember(role, content = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, push, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1892: function assistantRemember(role, content = "") {
1893:   const cleanRole = role === "assistant" ? "assistant" : "user";
1894:   const cleanContent = String(content || "").trim();
1895:   if (!cleanContent) return;
1896:   assistantChatHistory.push({
1897:     role: cleanRole,
1898:     content: cleanContent.slice(0, 1400)
1899:   });
1900:   assistantChatHistory = assistantChatHistory.slice(-10);
1901: }

```

assistantConversationForRequest - lines 1903-1912

Item	Explanation
Source location	Lines 1903-1912 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice, map, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1903: function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory)
{
1904:   const history = priorConversation.slice(-8).map((item) => ({
1905:     role: item.role === "assistant" ? "assistant" : "user",
1906:     content: String(item.content || "").slice(0, 1200)
1907:   }));
1908:   if (currentQuestion) {
1909:     history.push({ role: "user", content: String(currentQuestion).slice(0, 1200) });
1910:   }
1911:   return history;
1912: }

```

assistantFallbackResults - lines 1914-1972

Item	Explanation
Source location	Lines 1914-1972 (59 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalize, trim, assistantNamedProjectIds, searchResultsFor, filter, includes, slice, map, find, slugLabel, flattenSearchText

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1914: function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {
1915:   if (intent !== "portfolio_specific") return [];
1916:   const clean = normalize(question).trim();
1917:   if (!clean) return [];
1918:   const namedProjectIds = assistantNamedProjectIds(question);
1919:   let results = searchResultsFor(clean);
1920:   if (namedProjectIds.length && results.length) {
1921:     const focusedResults = results.filter((result) => namedProjectIds.includes(result.projectId));
1922:     if (focusedResults.length) return focusedResults.slice(0, 6);
1923:   }
1924:   if (results.length) return results.slice(0, 6);
1925:
1926:   if (namedProjectIds.length) {
1927:     return namedProjectIds
1928:       .map((id) => projects.find((project) => project.id === id))
1929:       .filter(Boolean)
1930:       .map((project) => ({
1931:         context: slugLabel(project.category),
1932:         kind: "project",
1933:         projectId: project.id,
1934:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1935:         title: project.portfolioView?.title || project.title,
1936:         type: "Project"
1937:       })))
1938:     .slice(0, 6);
1939:   }
1940:
1941:   const category = categories.find((item) => clean.includes(normalize(item.label)) ||
clean.includes(normalize(item.id)));
1942:   if (category) {
1943:     return projects
1944:       .filter((project) => project.category === category.id)
1945:       .map((project) => ({
1946:         context: category.label,
1947:         kind: "project",
1948:         projectId: project.id,
1949:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1950:         title: project.portfolioView?.title || project.title,
1951:         type: "Project"
1952:       })))
1953:     .slice(0, 6);
1954:   }
1955:
1956:   if (/resume|professional|contact|email|phone|github|linkedin|profile/.test(clean)) {
1957:     return searchResultsFor("resume contact github professional profile").slice(0, 6);
1958:   }
1959:
1960:   if (/project|work|portfolio|hardware|design/.test(clean)) {
1961:     return projects.slice(0, 6).map((project) => ({
1962:       context: slugLabel(project.category),
1963:       kind: "project",
1964:       projectId: project.id,
1965:       text: project.summary || "",
1966:       title: project.portfolioView?.title || project.title,
1967:       type: "Project"
1968:     }));
1969:   }
1970:
1971:   return [];
1972: }

```

assistantQuestionIsEngineeringRelated - lines 1974-1977

Item	Explanation
Source location	Lines 1974-1977 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionIsEngineeringRelated(question = "") {
Touches	filesystem
Calls detected	normalize, b

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1974: function assistantQuestionIsEngineeringRelated(question = "") {
1975:   const clean = normalize(question);
1976:   return /\b(op\s*amp|amplifier|analog|mixed\s*signal|adc|dac|filter|vco|oscillator|pwm|charger|rectifier
|regulator|buck|boost|ldo|mosfet|bjt|transistor|diode|pcb|schematic|layout|ground|noise|frequency|gain|phase|lts
pice|kicad|vivado|verilog|vhdl|fpga|asic|stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart|sensor|co
ntrol|signal|circuit|electronics|hardware|power)\b/.test(clean);
1977: }

```

assistantGeneralEngineeringAnswer - lines 1979-2091

Item	Explanation
Source location	Lines 1979-2091 (113 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantGeneralEngineeringAnswer(question = "") {
Touches	filesystem
Calls detected	normalize, assistantQuestionIsEngineeringRelated, join, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1979: function assistantGeneralEngineeringAnswer(question = "") {
1980:   const clean = normalize(question);
1981:   if (!assistantQuestionIsEngineeringRelated(question)) return "";
1982:
1983:   if (/bembedded\s+systems?b/.test(clean)) {
1984:     return [
1985:       "Embedded systems is a field of engineering focused on computers that are built into larger
products, machines, instruments, or control systems.",
1986:       "",
1987:       "Unlike a laptop or desktop computer, an embedded system is usually designed for a specific job. It
combines hardware, firmware, sensors, actuators, communication interfaces, and timing constraints so the
product can make measurements, control outputs, or respond to events reliably.",
1988:       "",
1989:       "Common examples include:",
1990:       "- A microcontroller controlling a motor drive.",
1991:       "- A sensor node collecting temperature, pressure, or motion data.",
1992:       "- A medical device running real-time measurement firmware.",
1993:       "- An automotive controller reading sensors and driving actuators.",
1994:       "",
1995:       "The important engineering ideas are timing, interrupts, power use, peripheral interfaces,
reliability, and how the firmware proves the hardware is behaving correctly."
1996:     ].join("\n");
1997:   }
1998:
1999:   if (/b(op\s*amp|amplifier|fully differential|differential)b/.test(clean)) {
2000:     return [
2001:       "A useful way to think about an op amp is as a high-gain error-correcting block. With negative
feedback, the circuit around it sets the useful behavior: gain, filtering, buffering, summing, integration, or
differential conversion.",
2002:       "",
2003:       "For a fully differential op amp:",
2004:       "- The signal is carried on two opposite-phase outputs instead of one output referenced to
ground.",
2005:       "- The differential output improves noise immunity and dynamic range.",
2006:       "- A common-mode feedback loop is normally needed to hold the average output voltage at the desired
common-mode level.",
2007:       "",
2008:       "In portfolio terms, this connects most naturally to analog and mixed-signal design, especially
filters, VCO control paths, ADC front ends, and precision measurement circuits."
2009:     ].join("\n");

```

```

2010: }
2011:
2012: if (/b(pwm|vco|oscillator|frequency)\b/.test(clean)) {
2013:     return [
2014:         "PWM-driven VCO work is about converting a digital timing signal into an analog control quantity
and then into a frequency.",
2015:         "",
2016:         "The usual signal chain is:",
2017:         "- PWM duty cycle encodes the control value.",
2018:         "- A low-pass filter turns the PWM waveform into an average voltage.",
2019:         "- The VCO maps that voltage to oscillation frequency.",
2020:         "- Tests compare duty cycle, control voltage, frequency range, ripple, and stability.",
2021:         "",
2022:         "That is a mixed-signal problem because digital timing, analog filtering, oscillator behavior, and
measurement all interact."
2023:     ].join("\n");
2024: }
2025:
2026: if (/b(fpga|asic|verilog|vhdl|vivado|digital)\b/.test(clean)) {
2027:     return [
2028:         "FPGA and ASIC work both implement digital hardware, but the engineering tradeoffs are different.",
2029:         "",
2030:         "- FPGA design is reconfigurable, fast to test, and useful for prototyping or deployment when
flexibility matters.",
2031:         "- ASIC design targets a fixed silicon implementation, so verification, timing closure, power,
area, and design-for-test become much more permanent decisions.",
2032:         "- Verilog or VHDL describes hardware structure and behavior; it is not software running line by
line.",
2033:         "",
2034:         "For a recruiter, strong evidence includes block diagrams, RTL, simulation waveforms, timing
reports, verification plans, and notes explaining design tradeoffs."
2035:     ].join("\n");
2036: }
2037:
2038: if (/b(stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart)\b/.test(clean)) {
2039:     return [
2040:         "Embedded work is strongest when the firmware is tied clearly to hardware behavior.",
2041:         "",
2042:         "A good explanation usually covers:",
2043:         "- The MCU role in the system.",
2044:         "- The peripherals used, such as ADC, PWM, timers, I2C, SPI, UART, or GPIO.",
2045:         "- How timing, interrupts, sampling, power, and fault handling were managed.",
2046:         "- What was measured to prove the system worked.",
2047:         "",
2048:         "STM32CubeIDE is useful because it helps configure peripherals, generate startup code, debug
firmware, and manage the build flow around STM32 devices."
2049:     ].join("\n");
2050: }
2051:
2052: if (/b(pcb|layout|kicad|schematic|ground|noise)\b/.test(clean)) {
2053:     return [
2054:         "PCB design is where the schematic becomes a physical electrical system.",
2055:         "",
2056:         "The important ideas are:",
2057:         "- Current returns matter as much as current paths.",
2058:         "- Grounding, decoupling, trace width, component placement, and loop area affect noise and
stability.",
2059:         "- Analog, power, and switching sections should be arranged so noisy currents do not corrupt
sensitive measurements.",
2060:         "- A good portfolio PCB section should show schematic intent, layout decisions, board files, bring-
up notes, and test results.",
2061:         "",
2062:         "KiCad evidence is strongest when the design files are paired with clear measurements or bring-up
photos."
2063:     ].join("\n");
2064: }
2065:
2066: if (/b(charger|rectifier|ac\s*\to\s*\dc|regulator|buck|boost|ldo|power)\b/.test(clean)) {
2067:     return [
2068:         "An AC-to-DC charger or power supply usually has a few major stages.",
2069:         "",
2070:         "- Input protection and filtering handle transients and noise.",
2071:         "- Rectification converts AC into pulsating DC.",
2072:         "- Bulk capacitance smooths the rectified waveform.",
2073:         "- Regulation sets the desired output voltage or current.",
2074:         "- Load testing checks ripple, thermal behavior, regulation, and fault response.",
2075:         "",
2076:         "For portfolio presentation, the best proof is a clear schematic, expected waveforms, measured
output behavior, and a short explanation of component choices."
2077:     ].join("\n");
2078: }
2079:
2080: return [
2081:     "I can answer this as a general electronics or hardware-engineering question, even if it is not
directly tied to one saved project yet.",
2082:     "",
2083:     "The best way to approach it is:",
2084:     "- Define the signal or energy flow.",
2085:     "- Identify the active devices, passive networks, and control points.",
2086:     "- State what should be measured to prove the design works.",
2087:     "- Connect the explanation back to a project artifact: schematic, code, test data, PCB, or
simulation."

```

```

2088:     "",
2089:     "If you ask about a specific circuit, tool, waveform, or failure mode, I can make the explanation
more concrete."
2090:   ].join("\n");
2091: }

```

assistantLocalAnswer - lines 2093-2188

Item	Explanation
Source location	Lines 2093-2188 (96 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalizeProfile, normalize, b, join, assistantGeneralEngineeringAnswer, replace, Set, map, filter, find, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 11 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2093: function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {
2094:   const ownerName = normalizeProfile(profile || {}).displayName || "the portfolio owner";
2095:   if (intent === "general_conversation") {
2096:     const clean = normalize(question);
2097:     if (/^(hi|hello|hey|yo|sup)\b/.test(clean)) {
2098:       return "Hi, what can I do for you?";
2099:     }
2100:     if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(clean) ||
/who\s+am\s+i\??$/i.test(clean)) {
2101:       return `I am an AI agent for ${ownerName}'s portfolio. I know the portfolio owner is ${ownerName},
but I do not know a visitor's personal name unless they tell me.`;
2102:     }
2103:     if (/^(who are you|what are you)\b/.test(clean)) {
2104:       return `I am an AI agent for ${ownerName}'s portfolio. I can help with projects, resume links,
public links, files, and related engineering questions.`;
2105:     }
2106:     return [
2107:       `I am here with you. I can help explore ${ownerName}'s portfolio, explain projects, summarize
files, open relevant sections, or answer related engineering questions.`,
2108:       "",
2109:       "You can ask things like:",
2110:       "- What is embedded systems?",
2111:       "- Show me a project.",
2112:       "- What tools were used?",
2113:       "- Explain the difference between FPGA and ASIC design."
2114:     ].join("\n");
2115:   }
2116:   if (intent === "general_engineering") {
2117:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2118:     if (generalAnswer) return generalAnswer;
2119:     return [
2120:       "I can answer that as a general engineering question, but I need a little more detail to make the
answer useful.",
2121:       "",
2122:       "Try asking about a specific circuit, device, tool, protocol, waveform, or failure mode."
2123:     ].join("\n");
2124:   }
2125:   if (intent === "general_knowledge") {
2126:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2127:     if (generalAnswer) return generalAnswer;
2128:     return [
2129:       "I can answer that as a general question from the local fallback, but the live model will usually
give a deeper answer when it is reachable.",
2130:       "",
2131:       "Short answer:",
2132:       `~ ${question.replace(/\?+$/, "")} is a topic I can discuss by breaking it into definition,
purpose, main parts, and practical examples.`
2133:     ].join("\n");
2134:     if (question.toLowerCase().includes(ownerName.toLowerCase())) {
2135:       return `If the question connects to Maurice's portfolio, I will use the saved project context and links
as evidence.`;
2136:     }
2137:     return `If it is broader than the portfolio, I will answer it as a general engineering or technology
concept first.`;
2138:   }
2139: }

```

```

2140:   if (!projects.length) return "The project catalog is still loading. Try again in a moment.";
2141:   if (!results.length) {
2142:     return [
2143:       `I could not find a strong portfolio match for "${question}".`,
2144:       "",
2145:       "Try asking about:",
2146:       "- A project name or category.",
2147:       "- A tool such as LTspice, KiCad, Vivado, STM32CubeIDE, or GitHub.",
2148:       "- A hardware topic such as op amps, PCB testing, VCOs, embedded firmware, FPGA, ASIC, or power
circuits."
2149:     ].join("\n");
2150:   }
2151:
2152:   const projectIds = [...new Set(results.map((item) => item.projectId).filter(Boolean))];
2153:   const matchedProjects = projectIds
2154:     .map((id) => projects.find((project) => project.id === id))
2155:     .filter(Boolean);
2156:   const projectNames = matchedProjects.map((project) => project.portfolioView?.title ||
project.title).slice(0, 3);
2157:   const fileCount = results.filter((item) => item.kind === "file").length;
2158:   const sectionCount = results.filter((item) => item.kind === "section" || item.kind ===
"subsection").length;
2159:   const pageCount = results.filter((item) => item.kind === "page").length;
2160:   const parts = [];
2161:
2162:   if (projectNames.length) {
2163:     parts.push(`The strongest portfolio match is ${projectNames.join(projectNames.length > 2 ? ", " : "
and ")}.`);
2164:   } else if (pageCount) {
2165:     parts.push("The strongest matches are in the profile, resume, contact, or portfolio information
sections.");
2166:   } else {
2167:     parts.push(`I found ${results.length} relevant portfolio item${results.length === 1 ? "" : "s"}.`);
2168:   }
2169:
2170:   parts.push("");
2171:   parts.push("What I found:");
2172:   if (sectionCount || fileCount) {
2173:     parts.push(`- ${sectionCount} section match${sectionCount === 1 ? "" : "es"}.`);
2174:     if (fileCount) parts.push(`- ${fileCount} file or link match${fileCount === 1 ? "" : "es"}.`);
2175:   }
2176:   if (pageCount) parts.push(`- ${pageCount} page or profile match${pageCount === 1 ? "" : "es"}.`);
2177:   if (projectNames.length) {
2178:     parts.push(`- Best project candidate${projectNames.length === 1 ? "" : "s"}: ${projectNames.join(",
")}.`);
2179:   }
2180:
2181:   const firstSnippet = searchSnippet(results[0].text, question);
2182:   if (firstSnippet) {
2183:     parts.push("");
2184:     parts.push("Why it matched:");
2185:     parts.push(firstSnippet);
2186:   }
2187:   return parts.join("\n");
2188: }

```

assistantSourcesMarkup - lines 2190-2208

Item	Explanation
Source location	Lines 2190-2208 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourcesMarkup(results = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice, map, set, escapeHtml, assistantSourceLabel, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2190: function assistantSourcesMarkup(results = []) {
2191:   const sources = results.slice(0, 5);
2192:   if (!sources.length) return "";
2193:   const sourceButtons = sources.map((source) => {
2194:     const id = String(assistantSourceCounter++);
2195:     assistantSourceMap.set(id, source);
2196:     return `

```

```

2197:         <button type="button" data-ai-source-
id="${id}">${escapeHtml(assistantSourceLabel(source))}</button>
2198:     `;
2199:   }).join("");
2200:   return `
2201:     <div class="ai-source-panel">
2202:       <p class="ai-source-heading">Relevant portfolio links</p>
2203:       <div class="ai-source-list" aria-label="Assistant sources">
2204:         ${sourceButtons}
2205:       </div>
2206:     </div>
2207:   `;
2208: }

```

setAssistantStatus - lines 2210-2214

Item	Explanation
Source location	Lines 2210-2214 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setAssistantStatus(message, state = "ready") {
Touches	browser/Electron DOM
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2210: function setAssistantStatus(message, state = "ready") {
2211:   if (!aiAssistantStatus) return;
2212:   aiAssistantStatus.textContent = message;
2213:   aiAssistantStatus.dataset.state = state;
2214: }

```

assistantDesktopDockedHeight - lines 2216-2221

Item	Explanation
Source location	Lines 2216-2221 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantDesktopDockedHeight() {
Touches	Mostly local calculations or local UI state.
Calls detected	getComputedStyle, getPropertyValue, trim, endsWith, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2216: function assistantDesktopDockedHeight() {
2217:   if (!aiAssistantPanel) return 360;
2218:   const value = getComputedStyle(aiAssistantPanel).getPropertyValue("--ai-console-rest-height").trim();
2219:   if (value.endsWith("px")) return Number.parseFloat(value) || 360;
2220:   return Math.min(Math.max(window.innerHeight * 0.38, 340), 390);
2221: }

```

updateAssistantPanelGrowth - lines 2223-2246

Item	Explanation
Source location	Lines 2223-2246 (24 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateAssistantPanelGrowth() {
Touches	browser/Electron DOM, filesystem
Calls detected	matchMedia, removeProperty, remove, assistantDesktopDockedHeight, getBoundingClientRect, querySelector, querySelectorAll, reduce, min, round, max, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2223: function updateAssistantPanelGrowth() {
2224:   if (!aiAssistantPanel || !aiAssistantLog || !aiAssistantForm) return;
2225:   const isDesktop = window.matchMedia("(min-width: 901px)").matches;
2226:   if (!isDesktop) {
2227:     aiAssistantPanel.style.removeProperty("--ai-console-live-height");
2228:     aiAssistantPanel.classList.remove("ai-assistant-panel-expanded");
2229:     return;
2230:   }
2231:
2232:   const dockedHeight = assistantDesktopDockedHeight();
2233:   const formHeight = aiAssistantForm.getBoundingClientRect().height || 46;
2234:   const clearHeight = aiClearChatButton?.getBoundingClientRect().height || 38;
2235:   const titleHeight = document.querySelector("#ai-assistant-title")?.getBoundingClientRect().height ||
34;
2236:   const chromeHeight = titleHeight + formHeight + clearHeight + 92;
2237:   const messages = [...aiAssistantLog.querySelectorAll(".ai-message")];
2238:   const messageGap = messages.length > 1 ? (messages.length - 1) * 10 : 0;
2239:   const messageHeight = messages.reduce((sum, message) => sum + message.getBoundingClientRect().height,
0) + messageGap;
2240:   const contentHeight = messageHeight + chromeHeight;
2241:   const maxHeight = Math.min(window.innerHeight * 0.76, 720);
2242:   const nextHeight = Math.round(Math.max(dockedHeight, Math.min(contentHeight, maxHeight)));
2243:
2244:   aiAssistantPanel.style.setProperty("--ai-console-live-height", `${nextHeight}px`);
2245:   aiAssistantPanel.classList.toggle("ai-assistant-panel-expanded", nextHeight > dockedHeight + 12);
2246: }

```

renderAssistantInline - lines 2248-2250

Item	Explanation
Source location	Lines 2248-2250 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAssistantInline(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	renderInlineMath, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2248: function renderAssistantInline(value = "") {
2249:   return renderInlineMath(value).replace(/\^*([\^*]+)\^*/g, "<strong>$1</strong>");
2250: }

```

renderAssistantAnswerContent - lines 2252-2304

Item	Explanation
Source location	Lines 2252-2304 (53 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAssistantAnswerContent(content = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace, split, push, map, renderAssistantInline, join, escapeHtml, forEach, trim, match, flushCode, flushBullets
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2252: function renderAssistantAnswerContent(content = "") {
2253:   const lines = String(content || "").replace(/\r\n?/g, "\n").split("\n");
2254:   const html = [];
2255:   let bulletItems = [];
2256:   let codeBlock = null;
2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };
2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}" ` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };
2268:
2269:   lines.forEach((line) => {
2270:     const trimmed = line.trim();
2271:     const fence = trimmed.match(/^````([a-z0-9_+.#-]*)\s*$/i);
2272:     if (fence) {
2273:       if (codeBlock) {
2274:         flushCode();
2275:       } else {
2276:         flushBullets();
2277:         codeBlock = { language: fence[1] || "", lines: [] };
2278:       }
2279:       return;
2280:     }
2281:     if (codeBlock) {
2282:       codeBlock.lines.push(line);
2283:       return;
2284:     }
2285:     if (!trimmed) {
2286:       flushBullets();
2287:       return;
2288:     }
2289:     const bullet = trimmed.match(/^[-*\u2022]\s+(.+)/u);
2290:     if (bullet) {
2291:       bulletItems.push(bullet[1]);
2292:       return;
2293:     }
2294:     flushBullets();
2295:     if (/^[^!?]{2,64}:$/.test(trimmed)) {
2296:       html.push(`<p><strong>${renderAssistantInline(trimmed.slice(0, -1))}</strong></p>`);
2297:     } else {
2298:       html.push(`<p>${renderAssistantInline(trimmed)}</p>`);
2299:     }
2300:   });
2301:   flushCode();
2302:   flushBullets();
2303:   return `<div class="ai-answer-content">${html.join("")} || "<p>I am ready.</p>"</div>`;
2304: }

```

flushBullets - lines 2257-2261

Item	Explanation
Source location	Lines 2257-2261 (5 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	const flushBullets = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	push, map, renderAssistantInline, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };

```

flushCode - lines 2262-2267

Item	Explanation
Source location	Lines 2262-2267 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	const flushCode = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}"` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };

```

appendAssistantMessage - lines 2306-2317

Item	Explanation
Source location	Lines 2306-2317 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function appendAssistantMessage(role, content, sources = []) {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, renderAssistantAnswerContent, assistantSourcesMarkup, renderMultilineInlineText, append, updateAssistantPanelGrowth
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2306: function appendAssistantMessage(role, content, sources = []) {
2307:   if (!aiAssistantLog) return;
2308:   const article = document.createElement("article");
2309:   article.className = `ai-message ai-message-${role}`;
2310:   article.innerHTML = role === "assistant"
2311:     ? `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`
2312:     : `<p>${renderMultilineInlineText(content)}</p>`;
2313:   aiAssistantLog.append(article);
2314:   updateAssistantPanelGrowth();
2315:   aiAssistantLog.scrollTop = aiAssistantLog.scrollHeight;
2316:   return article;
2317: }

```

scrollAssistantAnswerToStart - lines 2319-2329

Item	Explanation
Source location	Lines 2319-2329 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function scrollAssistantAnswerToStart(article) {
Touches	Mostly local calculations or local UI state.
Calls detected	max, requestAnimationFrame
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2319: function scrollAssistantAnswerToStart(article) {
2320:   if (!aiAssistantLog || !article) return;
2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };
2324:   if (typeof window.requestAnimationFrame === "function") {
2325:     window.requestAnimationFrame(scrollToAnswer);
2326:   } else {
2327:     window.setTimeout(scrollToAnswer, 0);
2328:   }
2329: }

```

scrollToAnswer - lines 2321-2323

Item	Explanation
Source location	Lines 2321-2323 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	const scrollToAnswer = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };

```

appendAssistantPendingMessage - lines 2331-2335

Item	Explanation
Source location	Lines 2331-2335 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function appendAssistantPendingMessage() {
Touches	browser/Electron DOM
Calls detected	appendAssistantMessage, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2331: function appendAssistantPendingMessage() {
2332:   const article = appendAssistantMessage("assistant", "Thinking", []);
2333:   article?.classList.add("ai-message-pending");
2334:   return article;
2335: }

```

replaceAssistantMessage - lines 2337-2346

Item	Explanation
Source location	Lines 2337-2346 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function replaceAssistantMessage(article, content, sources = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	appendAssistantMessage, renderAssistantAnswerContent, assistantSourcesMarkup, updateAssistantPanelGrowth, scrollAssistantAnswerToStart
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2337: function replaceAssistantMessage(article, content, sources = []) {
2338:   if (!article) {
2339:     appendAssistantMessage("assistant", content, sources);
2340:     return;
2341:   }
2342:   article.className = "ai-message ai-message-assistant";
2343:   article.innerHTML = `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`;
2344:   updateAssistantPanelGrowth();
2345:   scrollAssistantAnswerToStart(article);
2346: }

```

setAssistantBusy - lines 2348-2352

Item	Explanation
Source location	Lines 2348-2352 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setAssistantBusy(isBusy) {

Item	Explanation
Touches	browser/Electron DOM, filesystem
Calls detected	querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2348: function setAssistantBusy(isBusy) {
2349:   const submitButton = aiAssistantForm?.querySelector('button[type="submit"]');
2350:   if (aiAssistantInput) aiAssistantInput.disabled = isBusy;
2351:   if (submitButton) submitButton.disabled = isBusy;
2352: }

```

clearAssistantChat - lines 2354-2363

Item	Explanation
Source location	Lines 2354-2363 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearAssistantChat() {
Touches	Mostly local calculations or local UI state.
Calls detected	Map, setAssistantStatus, updateAssistantPanelGrowth, focus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2354: function clearAssistantChat() {
2355:   if (!aiAssistantLog) return;
2356:   assistantSourceCounter = 0;
2357:   assistantSourceMap = new Map();
2358:   assistantChatHistory = [];
2359:   aiAssistantLog.innerHTML = "";
2360:   setAssistantStatus("Chat cleared");
2361:   updateAssistantPanelGrowth();
2362:   aiAssistantInput?.focus();
2363: }

```

assistantRemoteTimeoutMs - lines 2365-2374

Item	Explanation
Source location	Lines 2365-2374 (10 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {
Touches	filesystem
Calls detected	normalize, isArray, some, toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2365: function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {
2366:   const clean = normalize(question);
2367:   const sourceFetches = Array.isArray(context.sourceFetches) ? context.sourceFetches : [];
2368:   const usesGitHub = sourceFetches.some((source) => String(source.kind || "").toLowerCase() === "github"
|| /github\.com/i.test(source.url || ""));
2369:   if (usesGitHub || /\b(github|repo|repository|repositories|source\s+code|code|snippet|firmware|verilog|s
ystemverilog|python|javascript|circuit\s+file)\b/.test(clean)) {
2370:     return 45000;
2371:   }
2372:   if (options.allowWebSearch) return 30000;
2373:   return 18000;
2374: }
```

askRemoteAssistant - lines 2376-2411

Item	Explanation
Source location	Lines 2376-2411 (36 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function askRemoteAssistant(question, context, options = {}) {
Touches	network/API requests
Calls detected	assistantEndpoint, AbortController, abort, assistantRemoteTimeoutMs, fetch, stringify, Error, json, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2376: async function askRemoteAssistant(question, context, options = {}) {
2377:   const endpoint = assistantEndpoint();
2378:   if (!endpoint) return null;
2379:   const controller = new AbortController();
2380:   let didTimeout = false;
2381:   const timer = setTimeout(() => {
2382:     didTimeout = true;
2383:     controller.abort();
2384:   }, assistantRemoteTimeoutMs(question, context, options));
2385:   let response;
2386:   try {
2387:     response = await fetch(endpoint, {
2388:       method: "POST",
2389:       headers: { "Content-Type": "application/json" },
2390:       signal: controller.signal,
2391:       body: JSON.stringify({
2392:         context,
2393:         conversation: options.conversation || [],
2394:         intent: options.intent || context.intent || "portfolio_specific",
2395:         allowWebSearch: Boolean(options.allowWebSearch),
2396:         question,
2397:         mode: "portfolio-plus-general-knowledge",
2398:         responseType: "chatgpt-like-answer-first-links-second"
2399:       })
2400:     });
2401:   } catch (error) {
2402:     if (didTimeout) throw new Error("AI request timed out while reading public sources.");
2403:     throw error;
2404:   } finally {
2405:     clearTimeout(timer);
2406:   }
2407:   if (!response.ok) throw new Error("AI service unavailable");
2408:   const data = await response.json();
2409:   const answer = String(data.answer || data.message || "").trim();
2410:   return answer ? { answer, model: data.model || "", usedWebSearch: Boolean(data.usedWebSearch) } : null;
2411: }
```

answerAssistantQuestion - lines 2413-2458

Item	Explanation
Source location	Lines 2413-2458 (46 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	async function answerAssistantQuestion(question = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, slice, appendAssistantMessage, assistantRemember, appendAssistantPendingMessage, setAssistantBusy, setAssistantStatus, assistantEndpoint, assistantQuestionIntent, assistantFallbackResults, assistantSourcesForDisplay, assistantContextForQuestion
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2413: async function answerAssistantQuestion(question = "") {
2414:   const cleanQuestion = String(question || "").trim();
2415:   if (!cleanQuestion) return;
2416:   const priorConversation = assistantChatHistory.slice();
2417:   appendAssistantMessage("user", cleanQuestion);
2418:   assistantRemember("user", cleanQuestion);
2419:   const pendingMessage = appendAssistantPendingMessage();
2420:   setAssistantBusy(true);
2421:   try {
2422:     setAssistantStatus(assistantEndpoint() ? "Composing AI answer" : "Composing local answer",
"working");
2423:
2424:     const intent = assistantQuestionIntent(cleanQuestion, priorConversation);
2425:     const rawSources = assistantFallbackResults(cleanQuestion, intent);
2426:     const sources = assistantSourcesForDisplay(cleanQuestion, rawSources, intent);
2427:     const context = assistantContextForQuestion(cleanQuestion, intent, rawSources);
2428:     const conversation = assistantConversationForRequest(cleanQuestion, priorConversation);
2429:     try {
2430:       const remoteAnswer = await askRemoteAssistant(cleanQuestion, context, {
2431:         allowWebSearch: assistantQuestionAllowsPublicLookup(cleanQuestion, intent),
2432:         conversation,
2433:         intent
2434:       });
2435:       if (remoteAnswer?.answer) {
2436:         replaceAssistantMessage(pendingMessage, remoteAnswer.answer, sources);
2437:         assistantRemember("assistant", remoteAnswer.answer);
2438:         setAssistantStatus(remoteAnswer.model ? `AI answer ready: ${remoteAnswer.model}` : "AI answer
ready");
2439:         return;
2440:       }
2441:     } catch (error) {
2442:       setAssistantStatus(error?.message || "Local answer used", "error");
2443:     }
2444:
2445:     const fallbackAnswer = assistantLocalAnswer(cleanQuestion, sources, intent);
2446:     replaceAssistantMessage(pendingMessage, fallbackAnswer, sources);
2447:     assistantRemember("assistant", fallbackAnswer);
2448:     if (!assistantEndpoint()) setAssistantStatus("Local portfolio intelligence");
2449:   } catch {
2450:     const errorAnswer = "I had trouble processing that question. Try rephrasing it with a project name,
tool name, or electronics topic.";
2451:     replaceAssistantMessage(pendingMessage, errorAnswer, []);
2452:     assistantRemember("assistant", errorAnswer);
2453:     setAssistantStatus("Assistant error", "error");
2454:   } finally {
2455:     setAssistantBusy(false);
2456:     aiAssistantInput?.focus();
2457:   }
2458: }

```

linkAttributes - lines 2459-2462

Item	Explanation
Source location	Lines 2459-2462 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function linkAttributes(url, item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2459: function linkAttributes(url, item = {}) {
2460:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(item, url) });
2461:   return /^https?:\/\//i.test(target) ? ' target="_blank" rel="noreferrer"' : "";
2462: }

```

isLocalDownloadTarget - lines 2464-2467

Item	Explanation
Source location	Lines 2464-2467 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function isLocalDownloadTarget(target = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2464: function isLocalDownloadTarget(target = "") {
2465:   const value = normalizeLinkTarget(target, { assumeWeb: true });
2466:   return Boolean(value) && !/^(https?:)?\/\//i.test(value) && !/^(mailto:|tel:|#)/i.test(value);
2467: }

```

downloadAttribute - lines 2469-2472

Item	Explanation
Source location	Lines 2469-2472 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function downloadAttribute(target = "", item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem, isLocalDownloadTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2469: function downloadAttribute(target = "", item = {}) {
2470:   const value = normalizeLinkTarget(target, { assumeWeb: isWebsiteLinkItem(item, target) });
2471:   return isLocalDownloadTarget(value) ? " download" : "";
2472: }

```

fileNameFromUrl - lines 2474-2481

Item	Explanation
Source location	Lines 2474-2481 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function fileNameFromUrl(url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, pop
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2474: function fileNameFromUrl(url = "") {
2475:     const clean = String(url || "").split(/[?#]/)[0].split("/").pop() || "Download file";
2476:     try {
2477:         return decodeURIComponent(clean);
2478:     } catch {
2479:         return clean;
2480:     }
2481: }

```

resourceLink - lines 2483-2501

Item	Explanation
Source location	Lines 2483-2501 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function resourceLink(item = {}, label = item.label item.title item.name fileNameFromUrl(item.url item.artifact)) {
Touches	filesystem
Calls detected	fileNameFromUrl, normalizeLinkTarget, isWebsiteLinkItem, escapeHtml, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2483: function resourceLink(item = {}, label = item.label || item.title || item.name ||
fileNameFromUrl(item.url || item.artifact)) {
2484:     const rawTarget = item.url || item.artifact || item.href || item.file || item.path || item.src || "";
2485:     const target = normalizeLinkTarget(rawTarget, { assumeWeb: isWebsiteLinkItem(item, rawTarget) });
2486:
2487:     if (!target || item.status === "planned") {
2488:         return `<span class="resource-link muted">${escapeHtml(label || "Planned file")}</span>`;
2489:     }
2490:
2491:     return `
2492:     <a
2493:         class="resource-link"
2494:         href="${escapeHtml(target)}"
2495:         ${linkAttributes(target, item)}
2496:         ${downloadAttribute(target, item)}
2497:     >
2498:         ${escapeHtml(label || fileNameFromUrl(target))}
2499:     </a>
2500: `;
2501: }

```

pillList - lines 2503-2510

Item	Explanation
Source location	Lines 2503-2510 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function pillList(items, className = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2503: function pillList(items, className = "") {
2504:   return (items || [])
2505:     .map((item) => {
2506:       const label = typeof item === "string" ? item : item.name || item.title || item.label || "";
2507:       return label ? `

```

evidenceList - lines 2512-2518

Item	Explanation
Source location	Lines 2512-2518 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function evidenceList(items, renderItem, emptyMessage) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2512: function evidenceList(items, renderItem, emptyMessage) {
2513:   if (!items || !items.length) {
2514:     return `
```

detailBlock - lines 2520-2527

Item	Explanation
Source location	Lines 2520-2527 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function detailBlock(title, className, content) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2520: function detailBlock(title, className, content) {
2521:   return `
2522:     <details class="${className}">
2523:       <summary>${title}</summary>
2524:       ${content}
2525:     </details>
2526:   `;
2527: }

```

mediaGrid - lines 2529-2549

Item	Explanation
Source location	Lines 2529-2549 (21 source lines).
Purpose	Part of the public website runtime.
Declaration	function mediaGrid(items) {
Touches	filesystem
Calls detected	map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2529: function mediaGrid(items) {
2530:   if (!items || !items.length) {
2531:     return `<p class="evidence-empty">No project images have been added yet.</p>`;
2532:   }
2533: }
2534: return `
2535:   <div class="media-grid">
2536:     ${items.map((item) => `
2537:       <figure>
2538:         <a href="${escapeHtml(normalizeLinkTarget(item.url, { assumeweb: isWebsiteLinkItem(item,
item.url) })))}"${linkAttributes(item.url, item)}>
2539:         
2540:         </a>
2541:         <figcaption>
2542:           <strong>${escapeHtml(item.title)}</strong>
2543:           <span>${escapeHtml(item.caption || "")}</span>
2544:         </figcaption>
2545:       </figure>
2546:     `).join("")}
2547:   </div>
2548: `;
2549: }

```

siteSectionHasContent - lines 2551-2560

Item	Explanation
Source location	Lines 2551-2560 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function siteSectionHasContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	some

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2551: function siteSectionHasContent(section) {
2552:   return Boolean(
2553:     section?.description ||
2554:     section?.richDescription?.blocks?.length ||
2555:     section?.backgroundImage ||
2556:     (section?.links || []).some((link) => link.label || link.url) ||
2557:     (section?.assets || []).some((asset) => asset.title || asset.url) ||
2558:     (section?.subsections || []).some((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length)
2559:   );
2560: }

```

siteSectionRenderable - lines 2562-2564

Item	Explanation
Source location	Lines 2562-2564 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function siteSectionRenderable(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	siteSectionHasContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2562: function siteSectionRenderable(section) {
2563:   return section?.visible !== false && siteSectionHasContent(section);
2564: }

```

renderDynamicLinks - lines 2566-2570

Item	Explanation
Source location	Lines 2566-2570 (5 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDynamicLinks(items = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, map, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2566: function renderDynamicLinks(items = []) {
2567:   const links = items.filter((item) => item?.url && (item.label || item.title));
2568:   if (!links.length) return "";

```

```

2569:   return `<div class="resource-list dynamic-link-list">${links.map((item) => resourceLink(item,
item.label || item.title || "Open")).join("")}</div>`;
2570: }

```

renderSiteSections - lines 2572-2609

Item	Explanation
Source location	Lines 2572-2609 (38 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteSections() {
Touches	browser/Electron DOM
Calls detected	querySelector, filter, map, url, escapeHtml, renderRichContent, renderDynamicLinks, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2572: function renderSiteSections() {
2573:   const mount = document.querySelector("#dynamic-sections");
2574:   if (!mount) return;
2575:   const visibleSections = (siteSections || []).filter(siteSectionRenderable);
2576:   mount.innerHTML = visibleSections.map((section) => {
2577:     const style = section.backgroundImage ? ` style="--section-bg:
url('${escapeHtml(section.backgroundImage)}')` : "";
2578:     const links = [...(section.links || []), ...(section.assets || [])];
2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>
2582:         <div class="dynamic-section-surface">
2583:           <div class="section-heading">
2584:             <div>
2585:               <h2>${escapeHtml(section.title || "Untitled section")}</h2>
2586:             </div>
2587:           </div>
2588:           ${section.description || section.richDescription?.blocks?.length
? `<div class="dynamic-section-copy">${renderRichContent(section.richDescription,
section.description || "")}</div>`
2590:             : ""}
2591:           ${subsections.length ? `
2592:             <div class="dynamic-section-grid">
2593:               ${subsections.map((item) => `
2594:                 <article class="dynamic-section-card">
2595:                   <h3>${escapeHtml(item.title || "Untitled")}</h3>
2596:                   ${item.description || item.richDescription?.blocks?.length
? renderRichContent(item.richDescription, item.description || "")
2598:                     : ""}
2599:                   ${renderDynamicLinks(item.links || [])}
2600:                 </article>
2601:               `).join("")}
2602:             </div>
2603:             : ""}
2604:             ${renderDynamicLinks(links)}
2605:           </div>
2606:         </section>
2607:       `;
2608:   }).join("");
2609: }

```

subsections - lines 2579-2581

Item	Explanation
Source location	Lines 2579-2581 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	const subsections = (section.subsections []).filter((item) => item.title item.description item.richDescription?.blocks?.length (item.links []).length);
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	filter, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>

```

customSectionBlocks - lines 2611-2622

Item	Explanation
Source location	Lines 2611-2622 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function customSectionBlocks(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, detailBlock, evidenceList, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2611: function customSectionBlocks(project) {
2612:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
2613:     ${section.description ? `<p class="evidence-empty">${section.description}</p>` : ""}
2614:     ${evidenceList(section.items || [], (item) => `
2615:       <li>
2616:         ${item.url ? resourceLink(item, item.title) : `<strong>${item.title}</strong>`}
2617:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"}</span>
2618:         ${item.description ? `<p>${item.description}</p>` : ""}
2619:       </li>
2620:     `, "No content has been added yet.")]
2621:   `)).join("");
2622: }

```

itemUrl - lines 2623-2625

Item	Explanation
Source location	Lines 2623-2625 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function itemUrl(item = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2623: function itemUrl(item = {}) {  
2624:     return item.url || item.artifact || item.href || item.file || item.path || item.src || "";  
2625: }
```

itemLabel - lines 2627-2630

Item	Explanation
Source location	Lines 2627-2630 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function itemLabel(item = {}, fallback = "Download file") {
Touches	Mostly local calculations or local UI state.
Calls detected	itemUrl, fileNameFromUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2627: function itemLabel(item = {}, fallback = "Download file") {  
2628:     const url = itemUrl(item);  
2629:     return item.title || item.label || item.name || item.caption || fileNameFromUrl(url) || fallback;  
2630: }
```

normalizeDownloadAsset - lines 2632-2643

Item	Explanation
Source location	Lines 2632-2643 (12 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeDownloadAsset(item = {}, fallbackType = "File") {
Touches	filesystem
Calls detected	itemUrl, itemLabel
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2632: function normalizeDownloadAsset(item = {}, fallbackType = "File") {  
2633:     const url = itemUrl(item);  
2634:     if (!url || item.status === "planned") return null;  
2635:     return {  
2636:         title: itemLabel(item),  
2637:         url,  
2638:         status: item.status || "uploaded",  
2639:         type: item.type || item.kind || item.meta || fallbackType,  
2640:         description: item.description || item.caption || item.summary || ""  
2641:     };  
2642: }  
2643: }
```

collectRichDownloads - lines 2645-2659

Item	Explanation
Source location	Lines 2645-2659 (15 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function collectRichDownloads(rich, fallbackType = "File", output = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	forEach, push, cleanRichImageTitle, fileNameFromUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2645: function collectRichDownloads(rich, fallbackType = "File", output = []) {
2646:   (rich?.blocks || []).forEach((block) => {
2647:     if (!block?.url) return;
2648:
2649:     output.push({
2650:       title: cleanRichImageTitle(block) || block.title || block.caption ||
fileNameFromUrl(block.url),
2651:       url: block.url,
2652:       status: "uploaded",
2653:       type: block.type === "image" ? "Image" : fallbackType,
2654:       description: block.caption || ""
2655:     });
2656:   });
2657:
2658:   return output;
2659: }

```

collectDownloadsFromValue - lines 2661-2689

Item	Explanation
Source location	Lines 2661-2689 (29 source lines).
Purpose	Part of the public website runtime.
Declaration	function collectDownloadsFromValue(value, fallbackType = "File", output = []) {
Touches	filesystem
Calls detected	isArray, forEach, normalizeDownloadAsset, push, collectRichDownloads
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2661: function collectDownloadsFromValue(value, fallbackType = "File", output = []) {
2662:   if (!value) return output;
2663:
2664:   if (Array.isArray(value)) {
2665:     value.forEach((item) => collectDownloadsFromValue(item, fallbackType, output));
2666:     return output;
2667:   }
2668:
2669:   if (typeof value !== "object") return output;
2670:
2671:   const asset = normalizeDownloadAsset(value, fallbackType);
2672:   if (asset) output.push(asset);
2673:
2674:   collectRichDownloads(value.rich, fallbackType, output);
2675:   collectRichDownloads(value.summaryRich, fallbackType, output);
2676:
2677:   [
2678:     "files",
2679:     "assets",
2680:     "items",
2681:     "children",
2682:     "links",
2683:     "results",
2684:     "references",

```

```

2685:         "mathAnalysis"
2686:     ].forEach((key) => collectDownloadsFromValue(value[key], fallbackType, output));
2687:
2688:     return output;
2689: }

```

uniqueDownloads - lines 2691-2700

Item	Explanation
Source location	Lines 2691-2700 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function uniqueDownloads(items = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	Set, filter, has, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2691: function uniqueDownloads(items = []) {
2692:     const seen = new Set();
2693:
2694:     return items.filter((item) => {
2695:         const key = `${item.url}:${item.title}`;
2696:         if (!item.url || seen.has(key)) return false;
2697:         seen.add(key);
2698:         return true;
2699:     });
2700: }

```

renderDownloadBlock - lines 2702-2717

Item	Explanation
Source location	Lines 2702-2717 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDownloadBlock(title, items = []) {
Touches	filesystem
Calls detected	uniqueDownloads, detailBlock, escapeHtml, map, resourceLink, renderMultilineInlineText, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2702: function renderDownloadBlock(title, items = []) {
2703:     const downloads = uniqueDownloads(items);
2704:     if (!downloads.length) return "";
2705:
2706:     return detailBlock(escapeHtml(title), "evidence-block evidence-wide download-evidence-block", `
2707:     <ul class="download-list">
2708:         ${downloads.map((item) => `
2709:             <li>
2710:                 ${resourceLink(item, item.title)}
2711:                 <span>${escapeHtml(item.type || "File")} &middot; ${escapeHtml(item.status ||
"uploaded")}</span>
2712:                 ${item.description ? `<p>${renderMultilineInlineText(item.description)}</p>` : ""}
2713:             </li>
2714:         `).join("")}
2715:     </ul>
2716:     `);
2717: }

```


renderCollectedDownloadSections - lines 2719-2760

Item	Explanation
Source location	Lines 2719-2760 (42 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCollectedDownloadSections(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, push, renderDownloadBlock, addBlock, forEach, displayTitle, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2719: function renderCollectedDownloadSections(project) {
2720:   const blocks = [];
2721:
2722:   const addBlock = (title, source, fallbackType = "File") => {
2723:     const downloads = uniqueDownloads(collectDownloadsFromValue(source, fallbackType, []));
2724:     if (downloads.length) blocks.push(renderDownloadBlock(title, downloads));
2725:   };
2726:
2727:   addBlock("Overview downloads", [
2728:     { summaryRich: project.summaryRich },
2729:     project.design?.brief?.files,
2730:     project.design?.brief?.summaryRich
2731:   ], "Overview file");
2732:
2733:   addBlock("Design downloads", [
2734:     project.design?.documentation?.files,
2735:     project.design?.documentation?.references,
2736:     project.design?.documentation?.mathAnalysis
2737:   ], "Design file");
2738:
2739:   addBlock("Simulation downloads", [
2740:     project.design?.simulation?.files,
2741:     project.design?.simulation?.results
2742:   ], "Simulation file");
2743:
2744:   addBlock("Project documents", project.documents, "Document");
2745:   addBlock("Test artifacts", project.tests, "Test artifact");
2746:   addBlock("PCB artifacts", project.pcb, "PCB artifact");
2747:   addBlock("Images and media", project.media, "Image");
2748:
2749:   (project.sections || []).forEach((section) => {
2750:     addBlock(`${displayTitle(section.title, "Custom section")} downloads`, section.items ||
section.files || [], section.title || "Section file");
2751:   });
2752:
2753:   (project.portfolioView?.sections || []).
2754:     .filter((section) => section.id !== "brief")
2755:     .forEach((section) => {
2756:       addBlock(`${displayTitle(section.title, "Section")} downloads`, section.items || [],
section.title || "Section file");
2757:     });
2758:
2759:   return blocks.join("");
2760: }
```

renderParsedBriefBlock - lines 2761-2771

Item	Explanation
Source location	Lines 2761-2771 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderParsedBriefBlock(section, fallbackSummary = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	renderRichContent

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2761: function renderParsedBriefBlock(section, fallbackSummary = "") {
2762:   const briefItem = section?.items?.[0] || {};
2763:   const briefText = briefItem.description || fallbackSummary || "";
2764:   if (!briefItem.rich?.blocks?.length && !briefText) return "";
2765:   return
2766:     <details class="project-brief-default parsed-summary" open>
2767:       <summary>Overview</summary>
2768:       ${renderRichContent(briefItem.rich, briefText)}
2769:     </details>
2770:   `;
2771: }

```

pathToString - lines 2773-2775

Item	Explanation
Source location	Lines 2773-2775 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function pathToString(path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2773: function pathToString(path = []) {
2774:   return path.join(".");
2775: }

```

pathFromString - lines 2777-2780

Item	Explanation
Source location	Lines 2777-2780 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function pathFromString(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, filter, isInteger
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2777: function pathFromString(value = "") {
2778:   if (!value) return [];
2779:   return value.split(".").map((item) => Number(item)).filter((item) => Number.isInteger(item));

```

2780: }

sectionRouteState - lines 2782-2789

Item	Explanation
Source location	Lines 2782-2789 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteState(projectId, sectionIndex, resourcePath = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2782: function sectionRouteState(projectId, sectionIndex, resourcePath = "") {
2783:   return {
2784:     appwindow: sectionRouteKey,
2785:     projectId: String(projectId || ""),
2786:     resourcePath: String(resourcePath || ""),
2787:     sectionIndex: String(sectionIndex ?? "")
2788:   };
2789: }
```

canUseSectionHistory - lines 2791-2793

Item	Explanation
Source location	Lines 2791-2793 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function canUseSectionHistory() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2791: function canUseSectionHistory() {
2792:   return Boolean(window.history && typeof window.history.pushState === "function" && typeof
window.history.replaceState === "function");
2793: }
```

sectionBaseUrl - lines 2795-2799

Item	Explanation
Source location	Lines 2795-2799 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionBaseUrl() {
Touches	Mostly local calculations or local UI state.
Calls detected	URL, values, forEach, delete

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2795: function sectionBaseUrl() {
2796:   const url = new URL(window.location.href);
2797:   Object.values(sectionRouteParams).forEach((param) => url.searchParams.delete(param));
2798:   return url;
2799: }

```

sectionRouteUrl - lines 2801-2809

Item	Explanation
Source location	Lines 2801-2809 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionBaseUrl, set, delete
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2801: function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {
2802:   const url = sectionBaseUrl();
2803:   url.searchParams.set(sectionRouteParams.project, String(projectId || ""));
2804:   url.searchParams.set(sectionRouteParams.section, String(sectionIndex ?? ""));
2805:   if (resourcePath) url.searchParams.set(sectionRouteParams.path, String(resourcePath));
2806:   else url.searchParams.delete(sectionRouteParams.path);
2807:   url.hash = "projects";
2808:   return url;
2809: }

```

sectionRouteFromState - lines 2811-2819

Item	Explanation
Source location	Lines 2811-2819 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteFromState(state) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2811: function sectionRouteFromState(state) {
2812:   if (!state || state.appWindow !== sectionRouteKey) return null;
2813:   if (!state.projectId || state.sectionIndex === "") return null;

```

```

2814:   return {
2815:     projectId: String(state.projectId),
2816:     resourcePath: String(state.resourcePath || ""),
2817:     sectionIndex: String(state.sectionIndex)
2818:   };
2819: }

```

sectionRouteFromLocation - lines 2821-2831

Item	Explanation
Source location	Lines 2821-2831 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteFromLocation() {
Touches	Mostly local calculations or local UI state.
Calls detected	URLSearchParams, get
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2821: function sectionRouteFromLocation() {
2822:   const params = new URLSearchParams(window.location.search);
2823:   const projectId = params.get(sectionRouteParams.project);
2824:   const sectionIndex = params.get(sectionRouteParams.section);
2825:   if (!projectId || sectionIndex === null) return null;
2826:   return {
2827:     projectId,
2828:     resourcePath: params.get(sectionRouteParams.path) || "",
2829:     sectionIndex
2830:   };
2831: }

```

sameSectionRoute - lines 2833-2838

Item	Explanation
Source location	Lines 2833-2838 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function sameSectionRoute(left, right) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2833: function sameSectionRoute(left, right) {
2834:   return Boolean(left && right &&
2835:     left.projectId === right.projectId &&
2836:     String(left.sectionIndex) === String(right.sectionIndex) &&
2837:     String(left.resourcePath || "") === String(right.resourcePath || ""));
2838: }

```

syncSectionRouteToHistory - lines 2840-2851

Item	Explanation
Source location	Lines 2840-2851 (12 source lines).

Item	Explanation
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {
Touches	Mostly local calculations or local UI state.
Calls detected	canUseSectionHistory, sectionRouteState, sectionRouteUrl, sectionRouteFromState, sectionRouteFromLocation, sameSectionRoute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2840: function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {
2841:   if (isApplyingSectionRoute || mode === "none" || !canUseSectionHistory()) return;
2842:   const route = sectionRouteState(projectId, sectionIndex, resourcePath);
2843:   const url = sectionRouteUrl(projectId, sectionIndex, resourcePath);
2844:   const currentRoute = sectionRouteFromState(history.state) || sectionRouteFromLocation();
2845:   const method = mode === "replace" || sameSectionRoute(currentRoute, route) ? "replaceState" :
"pushState";
2846:   try {
2847:     history[method](route, "", url);
2848:   } catch {
2849:     // Preview iframes and restricted browser contexts may reject history writes.
2850:   }
2851: }

```

clearSectionRouteInHistory - lines 2853-2860

Item	Explanation
Source location	Lines 2853-2860 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearSectionRouteInHistory() {
Touches	Mostly local calculations or local UI state.
Calls detected	canUseSectionHistory, replaceState, sectionBaseUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2853: function clearSectionRouteInHistory() {
2854:   if (isApplyingSectionRoute || !canUseSectionHistory()) return;
2855:   try {
2856:     history.replaceState({ appWindow: "portfolio-base" }, "", sectionBaseUrl());
2857:   } catch {
2858:     // The visible dialog is already closed; URL cleanup is best effort.
2859:   }
2860: }

```

initializeSectionHistoryState - lines 2862-2871

Item	Explanation
Source location	Lines 2862-2871 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function initializeSectionHistoryState() {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	canUseSectionHistory, sectionRouteFromLocation, sectionRouteFromState, replaceState, sectionRouteState, sectionRouteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2862: function initializeSectionHistoryState() {
2863:   if (!canUseSectionHistory()) return;
2864:   const route = sectionRouteFromLocation() || sectionRouteFromState(history.state);
2865:   try {
2866:     if (route) history.replaceState(sectionRouteState(route.projectId, route.sectionIndex,
route.resourcePath), "", sectionRouteUrl(route.projectId, route.sectionIndex, route.resourcePath));
2867:     else if (!history.state?.appwindow) history.replaceState({ appwindow: "portfolio-base" }, "",
window.location.href);
2868:   } catch {
2869:     // Browser history state is progressive enhancement for the portfolio windows.
2870:   }
2871: }

```

nodeAtPath - lines 2873-2882

Item	Explanation
Source location	Lines 2873-2882 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeAtPath(section, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2873: function nodeAtPath(section, path = []) {
2874:   let node = section;
2875:   let children = section?.items || [];
2876:   for (const index of path) {
2877:     node = children[index];
2878:     if (!node) return null;
2879:     children = node.children || [];
2880:   }
2881:   return node;
2882: }

```

nodeChildren - lines 2884-2886

Item	Explanation
Source location	Lines 2884-2886 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2884: function nodeChildren(node) {
2885:   return node?.items || node?.children || [];
2886: }

```

richHasRenderableContent - lines 2888-2897

Item	Explanation
Source location	Lines 2888-2897 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function richHasRenderableContent(rich) {
Touches	filesystem
Calls detected	some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2888: function richHasRenderableContent(rich) {
2889:   return Boolean(rich?.blocks?.some((block) =>
2890:     block.type === "image" && block.url ||
2891:     block.type === "formula" && block.formula ||
2892:     block.type === "code" && block.code ||
2893:     block.text ||
2894:     block.title ||
2895:     block.caption
2896:   ));
2897: }

```

nodeHasRenderableContent - lines 2899-2910

Item	Explanation
Source location	Lines 2899-2910 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeHasRenderableContent(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasRenderableContent, nodeChildren, some, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2899: function nodeHasRenderableContent(node) {
2900:   if (!node) return false;
2901:   if (node.kind === "summary") return Boolean(node.description || richHasRenderableContent(node.rich));
2902:   const children = nodeChildren(node);
2903:   if (node.kind === "subsection") {
2904:     return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2905:   }

```



```

2906:   if (["tool", "language"].includes(node.kind)) {
2907:     return Boolean(node.title || node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2908:   }
2909:   return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2910: }

```

sectionHasRenderableContent - lines 2912-2914

Item	Explanation
Source location	Lines 2912-2914 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionHasRenderableContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasRenderableContent, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2912: function sectionHasRenderableContent(section) {
2913:   return Boolean(section?.description || richHasRenderableContent(section?.rich) || (section?.items ||
[])).some(nodeHasRenderableContent));
2914: }

```

responsiveChildren - lines 2916-2918

Item	Explanation
Source location	Lines 2916-2918 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2916: function responsiveChildren(node) {
2917:   return node?.items || node?.children || [];
2918: }

```

richContainsMedia - lines 2920-2922

Item	Explanation
Source location	Lines 2920-2922 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function richContainsMedia(rich) {
Touches	Mostly local calculations or local UI state.
Calls detected	some

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2920: function richContainsMedia(rich) {
2921:   return Boolean(rich?.blocks?.some((block) => block.type === "image" && block.url));
2922: }

```

responsiveNodeHasMedia - lines 2924-2930

Item	Explanation
Source location	Lines 2924-2930 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeHasMedia(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	richContainsMedia, responsiveChildren, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2924: function responsiveNodeHasMedia(node) {
2925:   if (!node) return false;
2926:   if (node.kind === "image" || richContainsMedia(node.rich)) return true;
2927:   const url = String(node.url || "");
2928:   if (/\. (apng|avif|gif|jpe?g|png|svg|webp)(\?|#|$)/i.test(url)) return true;
2929:   return responsiveChildren(node).some(responsiveNodeHasMedia);
2930: }

```

responsiveNodeCount - lines 2932-2935

Item	Explanation
Source location	Lines 2932-2935 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeCount(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeHasRenderableContent, responsiveChildren, reduce
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2932: function responsiveNodeCount(node) {
2933:   if (!node || !nodeHasRenderableContent(node)) return 0;
2934:   return 1 + responsiveChildren(node).reduce((count, child) => count + responsiveNodeCount(child), 0);
2935: }

```

responsiveNodeDepth - lines 2937-2942

Item	Explanation
Source location	Lines 2937-2942 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeDepth(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeHasRenderableContent, responsiveChildren, filter, max, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2937: function responsiveNodeDepth(node) {
2938:   if (!node || !nodeHasRenderableContent(node)) return 0;
2939:   const children = responsiveChildren(node).filter(nodeHasRenderableContent);
2940:   if (!children.length) return 1;
2941:   return 1 + Math.max(...children.map(responsiveNodeDepth));
2942: }

```

inferResponsiveProfile - lines 2944-2976

Item	Explanation
Source location	Lines 2944-2976 (33 source lines).
Purpose	Part of the public website runtime.
Declaration	function inferResponsiveProfile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, sectionHasRenderableContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2944: function inferResponsiveProfile(project) {
2945:   const sections = project?.portfolioView?.sections || [];
2946:   const visibleSections = sections.filter((section) => section?.id !== "brief" &&
sectionHasRenderableContent(section));
2947:   const itemCount = visibleSections.reduce(
2948:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
2949:     0
2950:   );
2951:   const maxDepth = visibleSections.reduce(
2952:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
2953:     1
2954:   );
2955:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
2956:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
2957:     ? "dense"
2958:     : itemCount <= 6 && visibleSections.length <= 2
2959:     ? "simple"
2960:     : "balanced";
2961:   return {
2962:     version: 1,
2963:     breakpoints: {
2964:       mobile: 640,
2965:       tablet: 900,
2966:       desktop: 1180
2967:     },
2968:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
2969:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
2970:     density,

```

```

2971:     mediaMax: hasMedia ? "860px" : "720px",
2972:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
2973:     touchTarget: "44px",
2974:     windowMode: "full-screen"
2975:   };
2976: }

```

projectResponsiveProfile - lines 2978-2982

Item	Explanation
Source location	Lines 2978-2982 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectResponsiveProfile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	inferResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2978: function projectResponsiveProfile(project) {
2979:   const savedProfile = project?.portfolioView?.responsive;
2980:   if (savedProfile?.version) return { ...inferResponsiveProfile(project), ...savedProfile };
2981:   return inferResponsiveProfile(project);
2982: }

```

nodeSummary - lines 2984-2992

Item	Explanation
Source location	Lines 2984-2992 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, renderRichContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2984: function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
2985:   if (!rich?.blocks?.length && !text) return "";
2986:   return `
2987:     <details class="parsed-summary" open>
2988:       <summary>${escapeHtml(title)}</summary>
2989:       ${renderRichContent(rich, text || "")}
2990:     </details>
2991:   `;
2992: }

```

nodesOverview - lines 2994-2997

Item	Explanation
Source location	Lines 2994-2997 (4 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function nodeIsOverview(item = {}) {
Touches	filesystem
Calls detected	normalize, displayTitle, endsWith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2994: function nodeIsOverview(item = {}) {
2995:   const title = normalize(displayTitle(item.title || item.label || ""));
2996:   return item.kind === "summary" || title === "overview" || title.endsWith(" overview");
2997: }

```

nodeOverviewDetails - lines 2999-3017

Item	Explanation
Source location	Lines 2999-3017 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeOverviewDetails(node, children = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, push, renderRichContent, forEach, renderInlineSectionFiles, nodeChildren, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2999: function nodeOverviewDetails(node, children = []) {
3000:   const overviewItems = children.filter(nodeIsOverview);
3001:   const blocks = [];
3002:   if (node?.rich?.blocks?.length || node?.description) {
3003:     blocks.push(renderRichContent(node.rich, node.description || ""));
3004:   }
3005:   overviewItems.forEach((item) => {
3006:     if (item.rich?.blocks?.length || item.description) blocks.push(renderRichContent(item.rich,
item.description || ""));
3007:     const itemFiles = renderInlineSectionFiles(nodeChildren(item));
3008:     if (itemFiles) blocks.push(itemFiles);
3009:   });
3010:   if (!blocks.length) return "";
3011:   return `
3012:     <details class="parsed-summary section-overview-default" open>
3013:       <summary>Overview</summary>
3014:       ${blocks.join("")}
3015:     </details>
3016:   `;
3017: }

```

parsedNodeCard - lines 3019-3032

Item	Explanation
Source location	Lines 3019-3032 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedNodeCard(node, projectId, sectionIndex, path) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	displayTitle, pathToString, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3019: function parsedNodeCard(node, projectId, sectionIndex, path) {
3020:   const title = displayTitle(node.title);
3021:   return `
3022:     <button
3023:       class="section-open-card subsection-open-card"
3024:       type="button"
3025:       data-section-project="${projectId}"
3026:       data-section-index="${sectionIndex}"
3027:       data-resource-path="${pathToString(path)}"
3028:     >
3029:       <span>${escapeHtml(title)}</span>
3030:     </button>
3031:   `;
3032: }

```

parsedChildCards - lines 3034-3042

Item	Explanation
Source location	Lines 3034-3042 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedChildCards(children, projectId, sectionIndex, basePath = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, nodeHasRenderableContent, nodeIsOverview, map, parsedNodeCard, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3034: function parsedChildCards(children, projectId, sectionIndex, basePath = []) {
3035:   const visibleChildren = children.filter((child) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child));
3036:   if (!visibleChildren.length) return "";
3037:   return `
3038:     <div class="subsection-grid section-content-grid">
3039:       ${children.map((child, index) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child) ? parsedNodeCard(child, projectId, sectionIndex, [...basePath, index]) : "").join("")}
3040:     </div>
3041:   `;
3042: }

```

parsedNodeContent - lines 3044-3058

Item	Explanation
Source location	Lines 3044-3058 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedNodeContent(node, projectId, sectionIndex, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	renderInlineSectionFiles, nodeChildren, filter, nodeIsOverview, nodeOverviewDetails, parsedChildCards
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3044: function parsedNodeContent(node, projectId, sectionIndex, path = []) {
3045:   const isContainer = !node.kind || node.kind === "subsection";
3046:   if (!isContainer && node.url) return renderInlineSectionFiles([node]);
3047:
3048:   const children = nodeChildren(node);
3049:   const contentChildren = children.filter((child) => !nodeIsOverview(child));
3050:
3051:   return `
3052:   <article class="parsed-window-panel section-directory">
3053:     ${nodeOverviewDetails(node, children)}
3054:     ${parsedChildCards(children, projectId, sectionIndex, path)}
3055:     ${renderInlineSectionFiles(contentChildren)}
3056:   </article>
3057: `;
3058: }

```

parsedSectionContent - lines 3060-3064

Item	Explanation
Source location	Lines 3060-3064 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedSectionContent(section, projectId, sectionIndex, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeAtPath, parsedNodeContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3060: function parsedSectionContent(section, projectId, sectionIndex, path = []) {
3061:   const node = path.length ? nodeAtPath(section, path) : section;
3062:   if (!node) return `<p class="evidence-empty">This section could not be found.</p>`;
3063:   return parsedNodeContent(node, projectId, sectionIndex, path);
3064: }

```

parsedSection - lines 3066-3080

Item	Explanation
Source location	Lines 3066-3080 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedSection(section, index, project) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, some, escapeHtml, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3066: function parsedSection(section, index, project) {
3067:   const visibleItems = (section.items || []).filter(nodeHasRenderableContent);
3068:   const hasImages = visibleItems.some((item) => item.kind === "image");
3069:
3070:   return `
3071:     <button
3072:       class="section-open-card evidence-block ${hasImages ? "evidence-wide" : ""}"
3073:       type="button"
3074:       data-section-project="${project.id}"
3075:       data-section-index="${index}"
3076:     >
3077:       <span>${escapeHtml(displayTitle(section.title, "Section"))}</span>
3078:     </button>
3079:   `;
3080: }

```

projectCard - lines 3082-3164

Item	Explanation
Source location	Lines 3082-3164 (83 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectCard(project) {
Touches	filesystem
Calls detected	find, filter, sectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedSection, join, renderMultilineInlineText, detailBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3082: function projectCard(project) {
3083:   if (project.portfolioView) {
3084:     const category = categories.find((item) => item.id === project.category) || {};
3085:     const accent = category.accent || "#117c7a";
3086:     const sections = project.portfolioView.sections || [];
3087:     const briefSection = sections.find((section) => section.id === "brief");
3088:     const otherSections = sections.filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3089:
3090:     return `
3091:       <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3092:         <div class="project-body">
3093:           <h3>${project.portfolioView.title || project.title}</h3>
3094:           ${renderParsedBriefBlock(briefSection, project.summary)}
3095:           <div class="evidence-grid" aria-label="${project.title} parsed project content">
3096:             ${otherSections.map((section, index) => parsedSection(section, index, project)).join("")}
3097:           </div>
3098:         </div>
3099:       </article>
3100:     `;
3101:   }
3102:
3103:   const category = categories.find((item) => item.id === project.category) || {};
3104:   const accent = category.accent || "#117c7a";
3105:
3106:   return `
3107:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3108:       <div class="project-body">
3109:         <h3>${project.title}</h3>
3110:         <p class="rich-paragraph">${renderMultilineInlineText(project.summary)}</p>
3111:
3112:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
3113:           <ul class="highlight-list">
3114:             ${project.highlights.map((item) => `<li>${item}</li>`).join("")}
3115:           </ul>
3116:         `) : ""}
3117:
3118:         <div class="evidence-grid" aria-label="${project.title} evidence blocks">
3119:           ${detailBlock("Documents", "evidence-block", evidenceList(project.documents, (item) => `
3120:             <li>
3121:               ${resourceLink(item, item.title)}
3122:               <span>${item.type || "Document"} &mdot; ${item.status || "tracked"}</span>
3123:             </li>

```



```

3124:         ` , "No document artifact has been added yet."))}
3125:
3126:     ${detailBlock("Tests and results", "evidence-block", evidenceList(project.tests, (item) => `
3127:         <li>
3128:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3129:             <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
3130:             ${item.result ? `<p>${item.result}</p>` : ""}
3131:         </li>
3132:     ` , "No test artifact has been added yet."))}
3133:
3134:     ${detailBlock("PCBs built", "evidence-block", evidenceList(project.pcb, (item) => `
3135:         <li>
3136:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3137:             <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
3138:         </li>
3139:     ` , "No PCB build has been added yet."))}
3140:
3141:     ${detailBlock("Images", "evidence-block evidence-wide", mediaGrid(project.media))}
3142:     ${customSectionBlocks(project)}
3143: </div>
3144:
3145:     ${detailBlock("Tools and implementation files", "project-drawer", `
3146:         <div class="project-tooling">
3147:             <div>
3148:                 <h4>Tools Used</h4>
3149:                 <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
3150:             </div>
3151:             <div>
3152:                 <h4>Languages</h4>
3153:                 <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
3154:             </div>
3155:         </div>
3156:     `)}
3157:
3158:     <div class="resource-list">
3159:         ${project.links || []}.map((item) => resourceLink(item)).join("")
3160:     </div>
3161: </div>
3162: </article>
3163: `;
3164: }

```

categorySection - lines 3166-3181

Item	Explanation
Source location	Lines 3166-3181 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function categorySection(category, visibleProjects) {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3166: function categorySection(category, visibleProjects) {
3167:     return `
3168:         <section class="category-section ${visibleProjects.length ? "" : "empty-category-section"}" aria-
3169:             labelledby="${escapeHtml(category.id)}-title">
3170:             <div class="category-heading">
3171:                 <div>
3172:                     <h3 id="${escapeHtml(category.id)}-title">${escapeHtml(category.label)}</h3>
3173:                 </div>
3174:                 ${visibleProjects.length ? `<span>${visibleProjects.length} project${visibleProjects.length === 1
3175: ? "" : "s"}</span>` : ""}
3176:             </div>
3177:             ${visibleProjects.length && category.description ? `<p class="category-
3178: description">${escapeHtml(category.description)}</p>` : ""}
3179:             <div class="category-projects">
3180:                 ${visibleProjects.map(projectCard).join("")}
3181:             </div>
3182:         </section>
3183:     `;
3184: }

```

escapeRegExp - lines 3183-3185

Item	Explanation
Source location	Lines 3183-3185 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function escapeRegExp(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3183: function escapeRegExp(value = "") {
3184:   return String(value).replace(/[*+?^${}()|[\]\\]/g, "\\$&");
3185: }
```

highlightQueryHtml - lines 3187-3193

Item	Explanation
Source location	Lines 3187-3193 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function highlightQueryHtml(value = "", query = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, trim, escapeRegExp, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3187: function highlightQueryHtml(value = "", query = "") {
3188:   const escaped = escapeHtml(value);
3189:   const cleanQuery = String(query || "").trim();
3190:   if (!cleanQuery) return escaped;
3191:   const pattern = new RegExp(`${escapeRegExp(escapeHtml(cleanQuery))}`, "ig");
3192:   return escaped.replace(pattern, '<mark class="search-result-mark">$1</mark>');
3193: }
```

clearSearchHighlights - lines 3195-3197

Item	Explanation
Source location	Lines 3195-3197 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearSearchHighlights() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3195: function clearSearchHighlights() {
3196:   window.CSS?.highlights?.delete?.(searchHighlightName);
3197: }

```

textNodeSearchTargets - lines 3199-3204

Item	Explanation
Source location	Lines 3199-3204 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function textNodeSearchTargets() {
Touches	browser/Electron DOM
Calls detected	querySelector, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3199: function textNodeSearchTargets() {
3200:   return [
3201:     document.querySelector("main"),
3202:     document.querySelector("#section-view-dialog[open] .section-view-content")
3203:   ].filter(Boolean);
3204: }

```

applySearchHighlights - lines 3206-3238

Item	Explanation
Source location	Lines 3206-3238 (33 source lines).
Purpose	Part of the public website runtime.
Declaration	function applySearchHighlights(query) {
Touches	browser/Electron DOM, filesystem
Calls detected	clearSearchHighlights, trim, normalize, textNodeSearchTargets, forEach, createTreeWalker, acceptNode, includes, closest, nextNode, indexOf, createRange
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3206: function applySearchHighlights(query) {
3207:   clearSearchHighlights();
3208:   const cleanQuery = String(query || "").trim();
3209:   if (!cleanQuery || !window.CSS?.highlights || typeof window.Highlight !== "function") return 0;
3210:   const ranges = [];
3211:   const needle = normalize(cleanQuery);
3212:
3213:   textNodeSearchTargets().forEach((root) => {
3214:     const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
3215:       acceptNode(node) {
3216:         if (!node.textContent || !normalize(node.textContent).includes(needle)) return
NodeFilter.FILTER_REJECT;
3217:         if (node.parentElement?.closest("script, style, input, textarea, select, button, .search-engine,

```

```

.site-header")) return NodeFilter.FILTER_REJECT;
3218:     return NodeFilter.FILTER_ACCEPT;
3219:   }
3220: });
3221: let node = walker.nextNode();
3222: while (node) {
3223:   const haystack = normalize(node.textContent);
3224:   let index = haystack.indexOf(needle);
3225:   while (index >= 0) {
3226:     const range = document.createRange();
3227:     range.setStart(node, index);
3228:     range.setEnd(node, index + cleanQuery.length);
3229:     ranges.push(range);
3230:     index = haystack.indexOf(needle, index + Math.max(1, cleanQuery.length));
3231:   }
3232:   node = walker.nextNode();
3233: }
3234: });
3235:
3236: if (ranges.length) window.CSS.highlights.set(searchHighlightName, new window.Highlight(...ranges));
3237: return ranges.length;
3238: }

```

queueSearchHighlights - lines 3240-3243

Item	Explanation
Source location	Lines 3240-3243 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function queueSearchHighlights() {
Touches	Mostly local calculations or local UI state.
Calls detected	applySearchHighlights
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3240: function queueSearchHighlights() {
3241:   clearTimeout(searchHighlightTimer);
3242:   searchHighlightTimer = setTimeout(() => applySearchHighlights(searchInput?.value || ""), 30);
3243: }

```

ensureSearchPanel - lines 3245-3283

Item	Explanation
Source location	Lines 3245-3283 (39 source lines).
Purpose	Part of the public website runtime.
Declaration	function ensureSearchPanel() {
Touches	browser/Electron DOM
Calls detected	createElement, replaceWith, append, querySelector, addEventListener, updateSearchDropdown, preventDefault, closest, goToSearchResult
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3245: function ensureSearchPanel() {
3246:   if (!searchInput || !searchWrap || searchPanel) return;
3247:   const shell = document.createElement("div");
3248:   shell.className = "search-engine";
3249:   searchWrap.replaceWith(shell);

```

```

3250: shell.append(searchWrap);
3251: searchPanel = document.createElement("div");
3252: searchPanel.className = "search-results-panel";
3253: searchPanel.hidden = true;
3254: searchPanel.innerHTML = `
3255:   <div class="search-results-tools">
3256:     <span id="search-status">Type to search projects, files, sections, and phrases.</span>
3257:     <label>
3258:       <span>Show</span>
3259:       <select id="search-result-limit" aria-label="Number of search results">
3260:         <option value="5">5</option>
3261:         <option value="10" selected>10</option>
3262:         <option value="15">15</option>
3263:         <option value="20">20</option>
3264:       </select>
3265:     </label>
3266:   </div>
3267:   <div class="search-results-list" role="listbox" aria-label="Search results"></div>
3268: `;
3269: shell.append(searchPanel);
3270: searchStatus = searchPanel.querySelector("#search-status");
3271: searchLimitSelect = searchPanel.querySelector("#search-result-limit");
3272: searchLimitSelect.addEventListener("change", () => {
3273:   searchResultLimit = Number(searchLimitSelect.value) || 10;
3274:   updateSearchDropdown();
3275: });
3276: searchPanel.addEventListener("mousedown", (event) => event.preventDefault());
3277: searchPanel.addEventListener("click", (event) => {
3278:   const button = event.target.closest("[data-search-result-index]");
3279:   if (!button) return;
3280:   const result = currentSearchResults[Number(button.dataset.searchResultIndex)];
3281:   if (result) goToSearchResult(result);
3282: });
3283: }

```

setSearchMessage - lines 3285-3289

Item	Explanation
Source location	Lines 3285-3289 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setSearchMessage(message, state = "neutral") {
Touches	browser/Electron DOM
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3285: function setSearchMessage(message, state = "neutral") {
3286:   if (!searchStatus) return;
3287:   searchStatus.textContent = message;
3288:   searchStatus.dataset.state = state;
3289: }

```

renderSearchResults - lines 3291-3327

Item	Explanation
Source location	Lines 3291-3327 (37 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSearchResults(query, visibleCount) {
Touches	browser/Electron DOM
Calls detected	ensureSearchPanel, querySelector, trim, searchResultsFor, slice, setAttribute, setSearchMessage, removeAttribute, map, highlightQueryHtml, escapeHtml, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3291: function renderSearchResults(query, visibleCount) {
3292:   ensureSearchPanel();
3293:   if (!searchPanel) return;
3294:   const list = searchPanel.querySelector(".search-results-list");
3295:   const cleanQuery = String(query || "").trim();
3296:   currentSearchResults = searchResultsFor(cleanQuery);
3297:   const limitedResults = currentSearchResults.slice(0, searchResultLimit);
3298:   searchPanel.hidden = !cleanQuery;
3299:   searchInput.setAttribute("aria-expanded", String(Boolean(cleanQuery)));
3300:
3301:   if (!cleanQuery) {
3302:     list.innerHTML = "";
3303:     setSearchMessage("Type to search projects, files, sections, and phrases.");
3304:     searchInput.removeAttribute("aria-invalid");
3305:     return;
3306:   }
3307:
3308:   if (!currentSearchResults.length) {
3309:     list.innerHTML = "";
3310:     setSearchMessage(`"${cleanQuery}" was not found on this portfolio.`, "not-found");
3311:     searchInput.setAttribute("aria-invalid", "true");
3312:     return;
3313:   }
3314:
3315:   searchInput.removeAttribute("aria-invalid");
3316:   const visibleText = visibleCount
3317:     ? `${visibleCount} visible match${visibleCount === 1 ? "" : "es"} highlighted.`
3318:     : "Matches found elsewhere. Choose a result below.";
3319:   setSearchMessage(`${visibleText} Showing ${limitedResults.length} of ${currentSearchResults.length}.`,
3320: "found");
3321:   list.innerHTML = limitedResults.map((result, index) => `
3322:     <button class="search-result-item" type="button" role="option" data-search-result-index="${index}">
3323:       <span class="search-result-title">${highlightQueryHtml(result.title || "Untitled result",
3324: cleanQuery)}</span>
3325:       <span class="search-result-meta">${escapeHtml([result.type, result.context].filter(Boolean).join("
3326: / ")).join("<br>")}</span>
3327:       ${searchSnippet(result.text, cleanQuery)} ? <span class="search-result-
3328: snippet">${highlightQueryHtml(searchSnippet(result.text, cleanQuery), cleanQuery)}</span>` : ""}
3329:     </button>
3330:   `).join("");
3331: }

```

updateSearchDropdown - lines 3329-3333

Item	Explanation
Source location	Lines 3329-3333 (5 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSearchDropdown() {
Touches	Mostly local calculations or local UI state.
Calls detected	applySearchHighlights, renderSearchResults
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3329: function updateSearchDropdown() {
3330:   const query = searchInput?.value || "";
3331:   const visibleCount = applySearchHighlights(query);
3332:   renderSearchResults(query, visibleCount);
3333: }

```

showSearchPanelIfNeeded - lines 3335-3338

Item	Explanation
Source location	Lines 3335-3338 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showSearchPanelIfNeeded() {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3335: function showSearchPanelIfNeeded() {
3336:   if (!searchInput?.value?.trim()) return;
3337:   updateSearchDropdown();
3338: }

```

scrollToDomTarget - lines 3340-3352

Item	Explanation
Source location	Lines 3340-3352 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	function scrollToDomTarget(id) {
Touches	browser/Electron DOM
Calls detected	getElementById, closeSectionDialogForRoute, scrollIntoView, replaceState, queueSearchHighlights
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3340: function scrollToDomTarget(id) {
3341:   const target = id ? document.getElementById(id) : null;
3342:   if (!target) return false;
3343:   closeSectionDialogForRoute();
3344:   target.scrollIntoView({ behavior: "smooth", block: "start" });
3345:   try {
3346:     history.replaceState(history.state || { appWindow: "portfolio-base" }, "", `#${id}`);
3347:   } catch {
3348:     // Scrolling is still the important behavior.
3349:   }
3350:   queueSearchHighlights();
3351:   return true;
3352: }

```

goToSearchResult - lines 3354-3377

Item	Explanation
Source location	Lines 3354-3377 (24 source lines).
Purpose	Part of the public website runtime.
Declaration	function goToSearchResult(result) {
Touches	browser/Electron DOM
Calls detected	scrollToDomTarget, openParsedSection, queueSearchHighlights, getElementById, setActiveFilter, renderProjects, scrollIntoView, open

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3354: function goToSearchResult(result) {
3355:   if (!result) return;
3356:   searchPanel.hidden = true;
3357:   if (result.domTarget && scrollToDomTarget(result.domTarget)) return;
3358:
3359:   if (result.projectId && result.sectionIndex !== undefined) {
3360:     openParsedSection(result.projectId, result.sectionIndex, result.resourcePath || "");
3361:     queueSearchHighlights();
3362:     return;
3363:   }
3364:
3365:   if (result.projectId) {
3366:     if (!document.getElementById(result.projectId)) {
3367:       setActiveFilter("all");
3368:       renderProjects();
3369:     }
3370:     const target = document.getElementById(result.projectId);
3371:     target?.scrollIntoView({ behavior: "smooth", block: "center" });
3372:     queueSearchHighlights();
3373:     return;
3374:   }
3375:
3376:   if (result.url) window.open(result.url, "_blank", "noopener");
3377: }

```

handleSearchInput - lines 3379-3382

Item	Explanation
Source location	Lines 3379-3382 (4 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleSearchInput() {
Touches	Mostly local calculations or local UI state.
Calls detected	renderProjects, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3379: function handleSearchInput() {
3380:   renderProjects();
3381:   updateSearchDropdown();
3382: }

```

renderProjects - lines 3384-3409

Item	Explanation
Source location	Lines 3384-3409 (26 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProjects() {
Touches	filesystem
Calls detected	normalize, trim, filter, projectVisible, queueSearchHighlights, map, categorySection, join
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3384: function renderProjects() {
3385:   const query = normalize(searchInput.value).trim();
3386:   const visible = projects.filter((project) => projectVisible(project, query));
3387:
3388:   projectCount.textContent = projects.length;
3389:
3390:   if (!visible.length) {
3391:     grid.innerHTML = `
3392:       <div class="empty-state">
3393:         <h3>No projects match that view.</h3>
3394:         <p>Search covers categories, project names, documents, tests, PCB builds, tools, languages, and
links.</p>
3395:       </div>
3396:     `;
3397:     queueSearchHighlights();
3398:     return;
3399:   }
3400:
3401:   grid.innerHTML = categories
3402:     .map((category) => {
3403:       const visibleProjects = visible.filter((project) => project.category === category.id);
3404:       const shouldShowEmptyCategory = !query && (activeFilter === "all" || activeFilter === category.id);
3405:       return visibleProjects.length || shouldShowEmptyCategory ? categorySection(category,
visibleProjects) : "";
3406:     })
3407:     .join("");
3408:   queueSearchHighlights();
3409: }

```

ensureSectionDialog - lines 3411-3470

Item	Explanation
Source location	Lines 3411-3470 (60 source lines).
Purpose	Part of the public website runtime.
Declaration	function ensureSectionDialog() {
Touches	browser/Electron DOM
Calls detected	querySelector, createElement, append, addEventListener, closeSectionDialog, clearSectionRouteInHistory, remove, updateSectionDialogMinimize, focus, pathFromString, sectionForwardStack, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3411: function ensureSectionDialog() {
3412:   let dialog = document.querySelector("#section-view-dialog");
3413:   if (dialog) return dialog;
3414:
3415:   dialog = document.createElement("dialog");
3416:   dialog.id = "section-view-dialog";
3417:   dialog.className = "section-view-dialog";
3418:   dialog.innerHTML = `
3419:     <div class="section-view-shell">
3420:       <div class="section-view-heading">
3421:         <div class="section-view-titlebar">
3422:           <button class="section-view-back" type="button" title="Previous view" aria-label="Previous
view" hidden>&larr;</button>
3423:           <button class="section-view-forward" type="button" title="Next view" aria-label="Next view"
hidden>&rarr;</button>
3424:           <h2 id="section-view-title">Section</h2>
3425:         </div>
3426:         <div class="section-view-actions">
3427:           <button class="section-view-close" type="button" aria-label="Close section">&times;</button>
3428:         </div>

```

```

3429:         </div>
3430:         <div class="section-view-content" id="section-view-content"></div>
3431:     </div>
3432: `;
3433:     document.body.append(dialog);
3434:     dialog.querySelector(".section-view-close").addEventListener("click", () =>
closeSectionDialog(dialog));
3435:     dialog.addEventListener("cancel", () => clearSectionRouteInHistory());
3436:     dialog.addEventListener("close", () => {
3437:         document.body.classList.remove("full-window-open");
3438:         dialog.classList.remove("is-minimized-dialog");
3439:         updateSectionDialogMinimize(dialog);
3440:         const returnTarget = document.querySelector(
3441:             `[data-section-project="${dialog.dataset.projectId}"][data-section-
index="${dialog.dataset.sectionIndex}"]`
3442:         );
3443:         returnTarget?.focus({ preventScroll: true });
3444:     });
3445:     dialog.querySelector(".section-view-back").addEventListener("click", () => {
3446:         const path = pathFromString(dialog.dataset.resourcePath || "");
3447:         if (!path.length) {
3448:             closeSectionDialog(dialog);
3449:             return;
3450:         }
3451:         const stack = sectionForwardStack(dialog);
3452:         stack.push(pathToString(path));
3453:         dialog.dataset.forwardStack = JSON.stringify(stack);
3454:         path.pop();
3455:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3456:     });
3457:     dialog.querySelector(".section-view-forward").addEventListener("click", () => {
3458:         const stack = sectionForwardStack(dialog);
3459:         const nextPath = stack.pop();
3460:         if (!nextPath) return;
3461:         dialog.dataset.forwardStack = JSON.stringify(stack);
3462:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, nextPath, { historyMode:
"replace", preserveForward: true });
3463:     });
3464:     dialog.querySelector("#section-view-content").addEventListener("click", (event) => {
3465:         const nodeButton = event.target.closest("[data-resource-path]");
3466:         if (!nodeButton) return;
3467:         openParsedSection(nodeButton.dataset.sectionProject, nodeButton.dataset.sectionIndex,
nodeButton.dataset.resourcePath);
3468:     });
3469:     return dialog;
3470: }

```

sectionForwardStack - lines 3472-3478

Item	Explanation
Source location	Lines 3472-3478 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionForwardStack(dialog) {
Touches	browser/Electron DOM
Calls detected	parse
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3472: function sectionForwardStack(dialog) {
3473:     try {
3474:         return JSON.parse(dialog.dataset.forwardStack || "[]");
3475:     } catch {
3476:         return [];
3477:     }
3478: }

```

updateSectionNavigation - lines 3480-3486

Item	Explanation
Source location	Lines 3480-3486 (7 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSectionNavigation(dialog, path) {
Touches	browser/Electron DOM
Calls detected	querySelector, setAttribute, sectionForwardStack
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3480: function updateSectionNavigation(dialog, path) {
3481:   const back = dialog.querySelector(".section-view-back");
3482:   back.hidden = false;
3483:   back.title = path.length ? "Previous view" : "Back to projects";
3484:   back.setAttribute("aria-label", path.length ? "Previous view" : "Back to projects");
3485:   dialog.querySelector(".section-view-forward").hidden = !sectionForwardStack(dialog).length;
3486: }

```

closeSectionDialog - lines 3488-3491

Item	Explanation
Source location	Lines 3488-3491 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeSectionDialog(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	close, clearSectionRouteInHistory
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3488: function closeSectionDialog(dialog) {
3489:   dialog.close();
3490:   clearSectionRouteInHistory();
3491: }

```

closeOrStepBackSectionDialog - lines 3493-3504

Item	Explanation
Source location	Lines 3493-3504 (12 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeOrStepBackSectionDialog(dialog) {
Touches	browser/Electron DOM
Calls detected	pathFromString, sectionForwardStack, push, pathToString, stringify, pop, openParsedSection, closeSectionDialog
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3493: function closeOrStepBackSectionDialog(dialog) {
3494:   const path = pathFromString(dialog.dataset.resourcePath || "");
3495:   if (path.length) {
3496:     const stack = sectionForwardStack(dialog);
3497:     stack.push(pathToString(path));
3498:     dialog.dataset.forwardStack = JSON.stringify(stack);
3499:     path.pop();
3500:     openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3501:     return;
3502:   }
3503:   closeSectionDialog(dialog);
3504: }

```

openParsedSection - lines 3506-3532

Item	Explanation
Source location	Lines 3506-3532 (27 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {
Touches	browser/Electron DOM
Calls detected	find, filter, sectionHasRenderableContent, pathFromString, nodeAtPath, nodeHasRenderableContent, ensureSectionDialog, applyProjectTemplateToElement, pathToString, syncSectionRouteToHistory, querySelector, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3506: function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {
3507:   const project = projects.find((item) => item.id === projectId);
3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;
3515:
3516:   const dialog = ensureSectionDialog();
3517:   const category = categories.find((item) => item.id === project.category) || {};
3518:   applyProjectTemplateToElement(dialog, project, category.accent || "#117c7a");
3519:   dialog.dataset.projectId = projectId;
3520:   dialog.dataset.sectionIndex = String(sectionIndex);
3521:   dialog.dataset.resourcePath = pathToString(path);
3522:   if (!options.preserveForward) dialog.dataset.forwardStack = "[]";
3523:   syncSectionRouteToHistory(projectId, sectionIndex, pathToString(path), options.historyMode || "push");
3524:   dialog.querySelector("#section-view-title").textContent = displayTitle(node.title || section.title,
"Section");
3525:   updateSectionNavigation(dialog, path);
3526:   dialog.querySelector("#section-view-content").innerHTML = parsedSectionContent(section, projectId,
Number(sectionIndex), path);
3527:   document.body.classList.add("full-window-open");
3528:   if (!dialog.open) dialog.showModal();
3529:   dialog.scrollTop = 0;
3530:   queueSearchHighlights();
3531:   return true;
3532: }

```

sections - lines 3508-3514

Item	Explanation
Source location	Lines 3508-3514 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>const sections = (project?.portfolioView?.sections []).filter((section) => section.id !== "brief" && sectionHasRenderableContent(section));</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>filter</code> , <code>sectionHasRenderableContent</code> , <code>pathFromString</code> , <code>nodeAtPath</code> , <code>nodeHasRenderableContent</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 3 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;

```

closeSectionDialogForRoute - lines 3534-3537

Item	Explanation
Source location	Lines 3534-3537 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	<code>function closeSectionDialogForRoute() {</code>
Touches	browser/Electron DOM
Calls detected	<code>querySelector</code> , <code>close</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 0 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3534: function closeSectionDialogForRoute() {
3535:   const dialog = document.querySelector("#section-view-dialog");
3536:   if (dialog?.open) dialog.close();
3537: }

```

applySectionRoute - lines 3539-3555

Item	Explanation
Source location	Lines 3539-3555 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function applySectionRoute(route) {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>closeSectionDialogForRoute</code> , <code>openParsedSection</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3539: function applySectionRoute(route) {
3540:   isApplyingSectionRoute = true;
3541:   try {
3542:     if (!route) {
3543:       closeSectionDialogForRoute();
3544:       return;
3545:     }
3546:
3547:     const opened = openParsedSection(route.projectId, route.sectionIndex, route.resourcePath, {
3548:       historyMode: "none",
3549:       preserveForward: true
3550:     });
3551:     if (!opened) closeSectionDialogForRoute();
3552:   } finally {
3553:     isApplyingSectionRoute = false;
3554:   }
3555: }

```

restoreSectionRouteAfterCatalogLoad - lines 3557-3560

Item	Explanation
Source location	Lines 3557-3560 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function restoreSectionRouteAfterCatalogLoad() {
Touches	Mostly local calculations or local UI state.
Calls detected	initializeSectionHistoryState, applySectionRoute, sectionRouteFromLocation, sectionRouteFromState
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3557: function restoreSectionRouteAfterCatalogLoad() {
3558:   initializeSectionHistoryState();
3559:   applySectionRoute(sectionRouteFromLocation() || sectionRouteFromState(history.state));
3560: }

```

renderInlineSectionFiles - lines 3621-3639

Item	Explanation
Source location	Lines 3621-3639 (19 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderInlineSectionFiles(items = []) {
Touches	filesystem
Calls detected	uniqueDownloads, map, normalizeDownloadAsset, filter, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute, fileNameFromUrl, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3621: function renderInlineSectionFiles(items = []) {
3622:   const downloads = uniqueDownloads(items.map((item) => normalizeDownloadAsset(item,
"File")).filter(Boolean));
3623:   if (!downloads.length) return "";
3624:
3625:   return `
3626:     <div class="section-file-list">
3627:       ${downloads.map((item) => `
3628:         <a
3629:           class="section-file-link"
3630:           href="${escapeHtml(normalizeLinkTarget(item.url, { assumeWeb: isWebsiteLinkItem(item, item.url)
3631:         })}"}"
3632:           ${linkAttributes(item.url, item)}
3633:           ${downloadAttribute(item.url, item)}
3634:         >
3635:           ${escapeHtml(item.title || fileNameFromUrl(item.url))}
3636:         </a>
3637:       `).join("")}
3638:     </div>
3639:   `;
3640: }

```

normalizeTextColor - lines 3640-3647

Item	Explanation
Source location	Lines 3640-3647 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeTextColor(value = "") {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3640: function normalizeTextColor(value = "") {
3641:   const color = String(value || "").trim();
3642:   return /^[0-9a-f]{3}|[0-9a-f]{6}$/i.test(color) ||
3643:     /^rgba?\(/i.test(color) ||
3644:     /^hsla?\(/i.test(color)
3645:     ? color
3646:     : "";
3647: }

```

loadProjectCatalog - lines 3648-3708

Item	Explanation
Source location	Lines 3648-3708 (61 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	function loadProjectCatalog() {
Touches	filesystem, network/API requests
Calls detected	hydrateProjectCategories, normalizeFieldStyles, normalizeSiteContent, normalizeProfile, normalizeFunFacts, renderProfile, renderSiteContent, renderFunFacts, renderSiteSections, renderCategoryFilters, renderProjects, rebuildSearchIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3648: function loadProjectCatalog() {

```

```

3649:   if (window.__PORTFOLIO_CATALOG__) {
3650:     projects = window.__PORTFOLIO_CATALOG__.projects || [];
3651:     categories = hydrateProjectCategories(window.__PORTFOLIO_CATALOG__.categories || [], projects);
3652:     siteSections = window.__PORTFOLIO_CATALOG__.siteSections || [];
3653:     fieldStyles = normalizeFieldStyles(window.__PORTFOLIO_CATALOG__.fieldStyles || {});
3654:     siteContent = normalizeSiteContent(window.__PORTFOLIO_CATALOG__.siteContent || {});
3655:     siteContentRich = window.__PORTFOLIO_CATALOG__.siteContentRich || {};
3656:     profile = normalizeProfile(window.__PORTFOLIO_CATALOG__.profile || {});
3657:     profileRich = window.__PORTFOLIO_CATALOG__.profileRich || {};
3658:     funFacts = normalizeFunFacts(window.__PORTFOLIO_CATALOG__.funFacts || []);
3659:     funFactsRich = window.__PORTFOLIO_CATALOG__.funFactsRich || null;
3660:     renderProfile();
3661:     renderSiteContent();
3662:     renderFunFacts();
3663:     renderSiteSections();
3664:     renderCategoryFilters();
3665:     renderProjects();
3666:     rebuildSearchIndex();
3667:     updateSearchDropdown();
3668:     restoreSectionRouteAfterCatalogLoad();
3669:     return;
3670:   }
3671:
3672:   const catalogUrl = new URL("projects.json", window.location.href);
3673:   catalogUrl.searchParams.set("v", String(Date.now()));
3674:   fetch(catalogUrl.href, { cache: "no-store" })
3675:     .then((response) => {
3676:       if (!response.ok) throw new Error("Could not load projects.json");
3677:       return response.json();
3678:     })
3679:     .then((data) => {
3680:       projects = data.projects || [];
3681:       categories = hydrateProjectCategories(data.categories || [], projects);
3682:       siteSections = data.siteSections || [];
3683:       fieldStyles = normalizeFieldStyles(data.fieldStyles || {});
3684:       siteContent = normalizeSiteContent(data.siteContent || {});
3685:       siteContentRich = data.siteContentRich || {};
3686:       profile = normalizeProfile(data.profile || {});
3687:       profileRich = data.profileRich || {};
3688:       funFacts = normalizeFunFacts(data.funFacts || []);
3689:       funFactsRich = data.funFactsRich || null;
3690:       renderProfile();
3691:       renderSiteContent();
3692:       renderFunFacts();
3693:       renderSiteSections();
3694:       renderCategoryFilters();
3695:       renderProjects();
3696:       rebuildSearchIndex();
3697:       updateSearchDropdown();
3698:       restoreSectionRouteAfterCatalogLoad();
3699:     })
3700:     .catch(() => {
3701:       grid.innerHTML = `
3702:         <div class="empty-state">
3703:           <h3>Project data did not load.</h3>
3704:           <p>The project catalog is temporarily unavailable.</p>
3705:         </div>
3706:       `;
3707:     });
3708: }

```

Representative opening snippet

```

1: const grid = document.querySelector("#project-grid");
2: const searchInput = document.querySelector("#project-search");
3: const projectFilters = document.querySelector("#project-filters");
4: let filterButtons = [...document.querySelectorAll(".filter-button")];
5: const projectCount = document.querySelector("#project-count");
6: const projectTrackCount = document.querySelector("#project-track-count");
7: const projectTrackLabels = document.querySelector("#project-track-labels");
8: const year = document.querySelector("#year");
9: const funFactsCallout = document.querySelector("#fun-facts-callout");
10: const heroEyebrow = document.querySelector(".hero-content .eyebrow");
11: const heroTitle = document.querySelector("#hero-title");
12: const heroCopy = document.querySelector(".hero-copy");
13: const brandName = document.querySelector(".brand strong");
14: const brandSubtitle = document.querySelector(".brand small");
15: const brandIcon = document.querySelector(".brand-icon");
16: const brandText = document.querySelector(".omb-engraving");
17: const headerAvatar = document.querySelector(".header-avatar");
18: const headerAvatarImage = document.querySelector(".header-avatar img");
19: const heroImage = document.querySelector(".hero-image");
20: const resumeSection = document.querySelector("#resume");
21: const contactBand = document.querySelector("#contact");
22: const profilePhoto = document.querySelector(".profile-photo");
23: const contactTitle = document.querySelector("#contact-title");
24: const contactIntro = document.querySelector(".contact-intro");
25: const contactDetails = document.querySelector(".contact-details");
26: const contactLinks = document.querySelector(".contact-links");
27: const footerOwner = document.querySelector(".site-footer span");

```



```

28: const searchWrap = searchInput?.closest(".search-wrap");
29: const aiAssistantForm = document.querySelector("#ai-assistant-form");
30: const aiAssistantPanel = document.querySelector("#ask-ai");
31: const aiAssistantInput = document.querySelector("#ai-assistant-input");
32: const aiAssistantLog = document.querySelector("#ai-assistant-log");
33: const aiAssistantStatus = document.querySelector("#ai-assistant-status");
34: const aiClearChatButton = document.querySelector("#ai-clear-chat");
35:
36: let categories = [];
37: let projects = [];
38: let siteSections = [];
39: let funFacts = [];
40: let funFactsRich = null;
41: let fieldStyles = {};
42: let siteContent = null;
43: let siteContentRich = {};
44: let profile = null;
45: let profileRich = {};
46: let activeFilter = "all";
47: let activeSectionDialogDrag = null;
48: let activeSectionDialogResize = null;
49: let sectionDialogDragEnabled = false;
50: let isApplyingSectionRoute = false;
51: let searchPanel = null;
52: let searchStatus = null;
53: let searchLimitSelect = null;
54: let currentSearchResults = [];
55: let assistantSourceCounter = 0;
56: let assistantSourceMap = new Map();
57: let assistantChatHistory = [];
58: let searchableEntries = [];
59: let searchResultLimit = 10;
60: let searchHighlightTimer = 0;
61: const searchHighlightName = "portfolio-search-highlight";
62:
63: const sectionRouteKey = "portfolio-section";
64: const sectionRouteParams = {
65:   path: "portfolioPath",
66:   project: "portfolioProject",
67:   section: "portfolioSection"
68: };
69:
70: const supportedCodeLanguages = [
71:   { id: "c", label: "C", aliases: ["c"] },
72:   { id: "cpp", label: "C++", aliases: ["cpp", "c++", "cplusplus"] },

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.